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On the Renormalizability of Quasi Parton Distribution Functions

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Quasi-parton distribution functions have received a lot of attentions in both perturbative QCD and lattice QCD communities in recent years because they not only carry good information on the parton distribution functions, but also could be evaluated by lattice QCD simulations. However, unlike the parton distribution functions, the quasi-parton distribution functions have perturbative ultraviolet power divergences because they are not defined by twist-2 operators. In this paper, we identify all sources of ultraviolet divergences for the quasi-parton distribution functions in coordinate-space, and demonstrate that power divergences, as well as all logarithmic divergences can be renormalized multiplicatively to all orders in QCD perturbation theory.

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I. INTRODUCTION

Parton distribution functions (PDFs), $f_{i/h}(x,\mu^2)$, are defined as the probability distributions to find a quark, an antiquark, or a gluon ($i = q, \bar{q}, g$, respectively) in a fast moving hadron h to carry the hadron's momentum fraction between x and $x+dx$, probed at the factorization scale μ [1]. They are fundamental and important nonperturbative functions in QCD quantifying the relation between a hadron and the quarks and gluons within it, and playing an essential role to connect colliding hadron(s) to short-distance QCD dynamics in high energy scattering processes [2]. However, calculation of PDFs from the first principle, both analytically or from lattice QCD (LQCD), is a challenge, if it is not impossible, due to the fact that PDFs contain the dynamics at long-distance scales and of nonperturbative in nature, and are defined in terms of time-dependent operators. Traditionally, PDFs have been extracted from high energy scattering data by QCD global analysis in the framework of QCD factorization [3–7].

Recently, Ji introduced a set of quasi-PDFs for a hadron of momentum p_z, say along z-direction, and argued that they are equal to corresponding PDFs when the hadron momentum p_z goes to infinity [8]. Without setting $p_z \to \infty$, the quasi-PDFs could be factorized to PDFs to all orders in QCD perturbation theory so long as quasi-PDFs can be multiplicatively renormalized [9]. Like PDFs, the quasi-PDFs are defined by hadronic matrix elements of two-field correlators with a straight line gauge link between the two fields to ensure the gauge invariance,

$$\tilde{f}_{q/p}(\tilde{x},\tilde{\mu}^2,p_z) \equiv \int \frac{d\xi_z}{2\pi} e^{i\tilde{x}p_z\xi_z} \langle h(p)|\overline{\psi}_q(\xi_z)\,\frac{\gamma_z}{2} \times \Phi_{n_z}^{(f)}(\{\xi_z,0\})\,\psi_q(0)|h(p)\rangle \qquad (1)$$

for quasi-quark distribution with $\xi_0 = \xi_\perp = 0$, and

$$\tilde{f}_{g/p}(\tilde{x},\tilde{\mu}^2,p_z) \equiv \frac{1}{\tilde{x}p_z}\int \frac{d\xi_z}{2\pi} e^{i\tilde{x}p_z\xi_z} \langle h(p)|F_z{}^{\nu}(\xi_z) \times \Phi_{n_z}^{(a)}(\{\xi_z,0\})\,F_{z\nu}(0)|h(p)\rangle \qquad (2)$$

for quasi-gluon distribution with ν summing over transverse directions. In Eqs. (1) and (2), $\Phi_{n_z}^{(f,a)}(\{\xi_z,0\}) = \exp[-ig\int_0^{\xi_z} d\eta_z\, A_z^{(f,a)}(\eta_z)]$ are the gauge links with "f" and "a" representing fundamental and adjoint representation, respectively. Unlike PDFs, the two-field correlators of the quasi-PDFs are defined to be off the light-cone and have an equal time separation, which makes it possible to calculate the quasi-PDFs in LQCD [10–14]. However, since the hadron momentum in LQCD calculation is effectively bounded by the lattice spacing, the $p_z \to \infty$ limit is hard to achieve in LQCD calculation, and it is a challenge to control the corrections due to the finite p_z [15, 16]. Nevertheless, with the potential to calculate the PDFs from the first principle in QCD by using LQCD and the challenges in doing so, the concept of the quasi-PDFs has generated a lot of interests and activities in both perturbative QCD (PQCD) and LQCD community [17–33]. Besides of the corrections due to the limited range of p_z, the key to derive PDFs from the LQCD calculated quasi-PDFs is of two-folds: (1) being able to renormalize all ultraviolet (UV) divergences of quasi-PDFs nonperturbatively, and (2) ensuring the renormalized quasi-PDFs and PDFs to share the same collinear (CO) divergences.

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It was demonstrated in Ref. [9] by two of the present authors that quasi-PDFs and corresponding PDFs share the same leading logarithmic CO perturbative divergences to all orders in QCD perturbation theory, if the quasi-PDFs can be multiplicatively renormalized. With this fact, instead of deriving the PDFs from quasi-PDFs by taking the hadron momentum $p_z \to \infty$, two of the present authors proposed in Ref. [9] to extract PDFs by using the PQCD factorization approach from LQCD calculated quasi-PDFs or other LQCD calculable single-hadron matrix elements, so long as these matrix elements could be factorized into the desired PDFs with perturbatively calculable coefficients. In this PQCD factorization approach, the hard scale is of $\tilde{x}p_z \sim 1/\xi_z \sim$ GeV, and the corrections are power suppressed by the hard scale and characterized by the hadronic matrix elements of high twist operators, just like how experimentally measurable and perturbative factorizable hadronic cross sections are connected to PDFs via PQCD factorization.

Therefore, understanding the renormalizability of the operators defining the quasi-PDFs is the most critically important challenge for extracting PDFs from LQCD calculated quasi-PDFs, reliably. Since the quasi-PDFs are not defined by twist-2 operators, PDFs and quasi-PDFs have different perturbative UV behavior. Instead of the logarithmic perturbative UV divergence of PDFs, quasi-PDFs have power UV divergences [9, 34]. Although LQCD calculations of quasi-PDFs are naturally regularized by the lattice spacing a, the perturbative power divergence in $1/a$ makes it difficult to take the continuous limit of lattice results to extract the correct PDFs [35–39]. Under the UV renormalization, it is also possible that the operators defining the quasi-PDFs might mix with other operators, for example, the quark PDFs can mix with gluon PDF. Therefore, it is very important to find out that not only if the operators defining quasi-PDFs are renormalizable, and but also if the operator mixing under the renormalization, if there is any, could be limited to a closed set of operators. In this paper, we address the issues concerning the renormalizability of the operators defining quasi-PDFs.

The rest of this paper is organized as following. In Sec. II we will show that quasi-PDFs defined in Eqs. (1) and (2) have bad short-distance behavior, while coordinate-space quasi-PDFs are better candidates for extracting PDFs. We also define quasi-PDFs in coordinate-space and explain why its renormalization is difficult. We then study the one-loop expansion of quasi-PDFs in coordinate-space in Sec. III. Using the UV power counting derived in Sec. IV, we identify all perturbative UV divergent regions for the quasi-PDFs. We find that, once subdivergences are subtracted off, all UV divergences are originated from the integration regions where all loop momenta are large, which is significantly different from the behavior of PDFs. Based these observations and findings, we prove in Sec. V that quasi-PDFs in coordinate-space can be multiplicatively renormalized. Most importantly, we found that quasi-PDFs do not mix with other quantities under the running of the renormalization scale, which completes our proof of the renormalizability of coordinate-space quasi-PDFs. Finally, we give our summary in Sec. VI.

II. COORDINATE-SPACE QUASI-PDFS

The interest and excitement of studying quasi-PDFs is based on the potential of extracting PDFs and hadron structure from the first principle calculation of LQCD. Consequently, the value for studying quasi-PDFs depends on the reliability and accuracy of the matching between quasi-PDFs and PDFs, as proposed in Eq. (11) of Ref. [8],

$$\tilde{q}(x,\mu^2,P^z) = \int_x^1 \frac{dy}{y} Z\left(\frac{x}{y},\frac{\mu}{P^z}\right) q(y,\mu^2)\,, \qquad (3)$$

with $Z(x,\mu/P^z) = \delta(x-1) + (\alpha_s/2\pi)Z^{(1)}(x,\mu/P^z) + \cdots \to \delta(x-1)$ as $P^z \to \infty$ for quark distribution. This is equivalent to require the reliability and accuracy of the PQCD factorization of the quasi-PDFs,

$$\tilde{f}_{i/p}(\tilde{x},\tilde{\mu}^2,p_z) \;\approx\; \sum_j \int_0^1 \frac{dx}{x}\mathcal{C}_{ij}(\frac{\tilde{x}}{x},\tilde{\mu}^2,\mu^2,p_z) \times f_{j/p}(x,\mu^2) + \mathcal{O}\left(\frac{1}{\tilde{\mu}^2}\right), \quad (4)$$

where $i,j = q,\bar{q},g$, $\mathcal{C}$'s are IR safe and perturbatively calculated matching coefficients, $\tilde{\mu}$ is the renormalization scale of quasi-PDFs, $\mu(\sim\tilde{\mu})$ is the factorization scale of PDFs, and the power correction, $\mathcal{O}(1/\tilde{\mu}^2)$, is characterized by the size of high-twist quark-gluon correlation functions, like in all collinear PQCD factorization formalisms. We found that if the perturbative UV divergences of quasi-PDFs in the left-hand-side (LHS) of Eq. (4) are regularized by a momentum cutoff, the factorized convolution in the right-hand-side (RHS) is well behaved. However, the momentum cutoff regulator is hard to implement consistently at higher order calculations in perturbative expansion. On the other hand, we noticed that if we regularize the perturbative UV divergences of quasi-PDFs by dimensional regularization (DR), the integration over the momentum fraction x on the RHS of the Eq. (4) will be divergent. This is because sea-quark and gluon PDFs, $f_{j/p}(x,\mu^2) \to x^{-\alpha}$ as $x \to 0$, with $1 < \alpha < 2$, and the calculated matching coefficient, $\mathcal{C}^{(1)}_{ij}(\tilde{x}/x,\tilde{\mu}^2,\mu^2,p_z)$, evaluated in using DR, has a term proportional to $1/(\tilde{x}/x)$ as $x \to 0$ [9, 34]. As a result, the lower end of the x-integration in Eq. (4) leads to the divergence, $\int_0 dx/x\,(x/\tilde{x})\,x^{-\alpha} \to \infty$.

By applying the factorization formula in Eq. (4) to an asymptotic parton state, "j", and expanding the both sides of the factorized equation to the first order in power of α_s, we have $\mathcal{C}^{(1)}_{ij}(y,\tilde{\mu}^2,\mu^2,p_z) = \tilde{f}^{(1)}_{i/j}(y,\tilde{\mu}^2,p_z) -$

$f^{(1)}_{i/j}(y,\mu^2)$. As $f^{(1)}_{i/j}(y,\mu^2)$ vanishes as $y\to\infty$, the asymptotic behavior of $\mathcal{C}^{(1)}_{ij}(\tilde{x}/x,\tilde{\mu}^2,\mu^2,p_z)$ as $x\to 0$ is fully determined by the large $\tilde{x}$ behavior of $\tilde{f}^{(1)}_{i/j}(\tilde{x},\tilde{\mu}^2,p_z)$.

As $\tilde{x}\to\infty$, the large momentum behavior of quasi-PDFs, defined in Eqs. (1) and (2), is closely related to the asymptotic behavior of the hadronic matrix elements as the separation of the two fields, $\xi_z\to 0$. The divergence in the x-convolution of the factorized formalism in Eq. (4), found above, indicates that hadronic matrix elements could be ill-defined when $\xi_z\to 0$, which is indeed the case as we will demonstrate in the next section. As pointed out in Ref. [9], the coordinate-space quasi-PDFs are better quantities for the calculation in LQCD, for the discussion of renormalization, and for the extraction of PDFs.

We define coordinate-space quasi-PDFs as following,

$$\begin{aligned}&\tilde{F}_{q/p}(\xi_z,\tilde{\mu}^2,p_z)\\ =&\frac{e^{ip_z\xi_z}}{p_z}\langle h(p)|\overline{\psi}_q(\xi_z)\,\frac{\gamma_z}{2}\,\Phi^{(f)}_{n_z}(\{\xi_z,0\})\,\psi_q(0)|h(p)\rangle,\end{aligned}\tag{5}$$

and

$$\begin{aligned}&\tilde{F}_{g/p}(\xi_z,\tilde{\mu}^2,p_z)\\ =&\frac{e^{ip_z\xi_z}}{p_z^2}\langle h(p)|F_z{}^{\nu}(\xi_z)\,\Phi^{(a)}_{n_z}(\{\xi_z,0\})\,F_{z\nu}(0)|h(p)\rangle,\end{aligned}\tag{6}$$

where $\tilde{\mu}$ is a renormalization scale, and $\Phi^{(f,a)}_{n_z}(\{\xi_z,0\})$ are gauge links, defined in Sec. I. We also define $n_z^\mu=(0,0_\perp,1)$ with $g^{\mu\nu}=\text{diag}(1,-1,-1,-1)$, and $v\cdot n_z=-v_z$ for any vector v^μ, and we have $n_z^2=-1$.

We will demonstrate in the next section that coordinate-space quasi-PDFs are well defined for finite ξ_z, while they are divergent when $\xi_z\to 0$. if one wants to have a well defined momentum-space quasi-PDFs, one has to properly and consistently subtract off the divergence at $\xi_z\to 0$. From the definitions of quasi-PDFs, we have

$$\tilde{F}_{i/p}(-\xi_z,\tilde{\mu}^2,p_z)=\tilde{F}^*_{i/p}(\xi_z,\tilde{\mu}^2,p_z).\tag{7}$$

To take the advantage of this relation and simplify the discussion, we assume $\xi_z>0$ in the following calculation. For the final result, we will then express it in the form that is correct for arbitrary value of ξ_z.

Before going into the details of proving the renormalizability of quasi-PDFs, we first briefly explain the complexity of quasi-PDFs' renormalization:

1) Lorentz symmetry is broken. Because of the explicitly "z"-direction dependence of quasi-PDFs, to identify all possible UV divergences, we need to study both four-dimensional loop momentum integration and three-dimensional loop momentum integration (with "z"-direction fixed)[1] for each individual loop. This amounts to 2^n different cases for a n-loop Feynman diagram, which is hard to handle.

2) Renormalizaiton of composite operators is needed. As an example, let's choose an $A_z=0$ axial gauge, and the quasi-quark PDFs become

$$\tilde{F}_{q/p}(\xi_z,\tilde{\mu}^2,p_z)=\frac{e^{ip_z\xi_z}}{p_z}\,\langle h(p)|\overline{\psi}_q(\xi_z)\,\frac{\gamma_z}{2}\,\psi_q(0)|h(p)\rangle\,.\tag{8}$$

For the renormalization of the quasi-quark PDFs, we need to renormalize the individual quark field, $\overline{\psi}_q(\xi_z)$ and $\psi_q(0)$ in Eq. (8), as well as the bi-local composite operator as a whole if it generates UV divergence. The renormalization of the fields can be naturally taken care of by using the renormalized QCD Lagrangian in $A_z=0$ gauge. It is the renormalization of the bi-local composite operators that is not covered by the renormalization of QCD. This is effectively the same procedure for proving the renormalizability of PDFs, for example, for quark PDFs in $A^+=0$ gauge, one has to verify the multiplicative renormalization of the bi-local composite operators defining the quark-PDFs to prove their renormalizability. It is the renormalization of the bi-local composite operators that mixes quark PDFs with gluon PDF [40]. That is, the renormalizability of QCD Lagrangian itself is not enough to guarantee the renormalizability of quasi-PDFs, which are defined by composite operators.

In our explicit calculation and discussion in the rest of this paper, we choose the Feynman gauge since the renormalization of QCD Lagrangian in Feynman gauge is well known. Since the renormalization of individual quark and gluon field is well defined, we will focus on the renormalization of the composite operators defining the quasi-PDFs. We will mainly discuss coordinate-space quasi-PDFs, and thus whenever we mention quasi-PDFs, we mean the coordinate-space quasi-PDFs. We will first concentrate on quasi-quark PDFs, and then generalize our study to quasi-gluon PDF at the end.

III. QUASI-QUARK PDFS AT ONE-LOOP ORDER

In this section, we study the quasi-quark PDFs at one-loop and its renormalization. Feynman diagrams for quasi-quark PDFs of an asymptotic quark of momentum p at one-loop order in the Feynman gauge are shown in

[1] Note that these are the only two cases that we need to study. Other cases of loop momentum integration cannot generate new UV divergences.

Fig. 1. The diagram (a) in Fig. 1 gives

$$
\begin{aligned}
M_{1a} =& \frac{e^{ip_z\xi_z}}{p_z}\frac{1}{N_c}\mathrm{Tr}_c[T^aT^a]\int_a^{\xi_z-2a}dr_1\int_{r_1+a}^{\xi_z-a}dr_2 \\
&\times\int\frac{d^4l}{(2\pi)^4}\,e^{-ip_z\xi_z}\,e^{il_z(r_2-r_1)}\left(\frac{-ig_{\mu\nu}}{l^2}\right) \\
&\times(-ig_s n_z^\mu)(-ig_s n_z^\nu)\mathrm{Tr}\left[\frac{1}{2}\not p\,\frac{1}{2}\gamma_z\right] \\
=& \frac{\alpha_s C_F}{4i\pi^3}\int_a^{\xi_z-2a}dr_1\int_{r_1+a}^{\xi_z-a}dr_2\int d^4l\frac{e^{il_z(r_2-r_1)}}{l^2}
\end{aligned}
\tag{9}
$$

where we introduced a cutoff "a" between fields along the gauge link to regularize potential linear UV divergence[2]. The upper limit of r_1-integration in Eq. (9) is $\xi_z - 2a$ because it needs a separation a from the upper limit of r_2-integration. We will show that this cutoff is enough to regularize all UV divergence in this diagram, and thus we do not introduce DR here. For the following calculation, we introduce a vector $\bar{l}^\mu$, which is the same as l^μ except that it does not have the z-direction component: $l^\mu = \bar{l}^\mu + l_z n_z^\mu$ and $l^2 = \bar{l}^2 - l_z^2$. We will refer the integration of $\bar{l}^\mu$ as three-dimensional (3-D) integration and the integration of l^μ as four-dimensional (4-D) integration. With $d^4l = d^3\bar{l}\,dl_z$, let's consider the following phase space integration,

$$
\begin{aligned}
\int\frac{d^3\bar{l}}{l^2} &= \int\frac{d^3\bar{l}}{\bar{l}^2 - l_z^2} \\
&= \int d^3\bar{l}\left(\frac{1}{\bar{l}^2} + \frac{l_z^2}{(\bar{l}^2 - l_z^2)\bar{l}^2}\right).
\end{aligned}
\tag{10}
$$

The first term in Eq. (10) is linear divergent, but its coefficient is proportional to $\int dl_z e^{il_z(r_2-r_1)} = 2\pi\delta(r_2 - r_1)$, which vanishes because r_2 and r_1 cannot be at the same point. Thus, effectively, the 3-D integration above is finite. On the other hand, it is easy to see that, if keeping $\bar{l}^\mu$ finite, the integration of l_z is also finite even if $a \to 0$. Therefore, the Eq. (9) can be UV divergent only in the region where all components of l^μ go to infinity. As a result, the spacing "a" can regularize all UV divergences, which gives

$$
M_{1a} \overset{\text{div}}{=} -\frac{\alpha_s C_F}{\pi}\frac{|\xi_z|}{a} + \frac{\alpha_s C_F}{\pi}\ln\frac{|\xi_z|}{a},
\tag{11}
$$

where we have included the situation when $\xi_z < 0$.

From Fig. 1, we have the contribution from the diagram (b) as

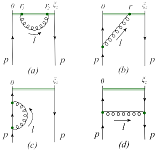

FIG. 1: Feynman diagrams for quasi-quark PDFs of an asymptotic quark of momentum p at one-loop order.

$$
\begin{aligned}
M_{1b} =& g_s^2 C_F\int_a^{\xi_z-a}dr\int\frac{d^4l}{(2\pi)^4}\frac{e^{il_z r}}{l^2(p-l)^2} \\
&\times\frac{1}{p_z}\mathrm{Tr}\left[\frac{1}{2}\not p\,\frac{1}{2}\gamma_z(\not p - \not l)\gamma_z\right],
\end{aligned}
\tag{12}
$$

where we again use the spacing "a" as a regulator. For the 3-D integration, it has potential logarithmic divergence from the term proportional to $\not l$,

$$
\int\frac{l^\mu\,d^3\bar{l}}{l^2(p-l)^2}.
\tag{13}
$$

However, the above integration is finite because, to extract the potential logarithmic divergence, we can set the external physical scale $p \to 0$ in the above integration and thus it becomes an odd function in $\bar{l}^\mu$, which vanishes under integration. This explains why the spacing a can regularize the UV divergence of Eq. (12). The UV divergence of the diagram (b) is

$$
M_{1b} \overset{\text{div}}{=} -\frac{\alpha_s C_F}{2\pi}\ln\frac{|\xi_z|}{a},
\tag{14}
$$

where we again included the situation when $\xi_z < 0$.

Similarly, we have from the diagram (c) in Fig. 1,

$$
\begin{aligned}
M_{1c} =& \frac{ig_s^2\mu_r^{2\epsilon}}{2p_z}(1-\epsilon)C_F\int\frac{d^dl}{(2\pi)^d}\frac{\mathrm{Tr}[\not p\gamma_z\not p(\not p - \not l)]}{p^2l^2(p-l)^2} \\
=& ig_s^2\mu_r^{2\epsilon}C_F(1-\epsilon)\int\frac{d^dl}{(2\pi)^d}\frac{1}{l^2(p-l)^2},
\end{aligned}
\tag{15}
$$

where space-time dimension is defined as $d = 4 - 2\epsilon$. It is clear that the integral in Eq. (15) vanishes in DR when $p^2 = 0$, due to the apparent cancelation between the UV and IR poles in $1/\epsilon$. The UV divergence from

[2] Although explicit result of UV divergence depends on UV regulators, our conclusions, like power counting rules and renormalization structure, are independent of them.

this diagram is logarithmic only if all components of l^μ go to infinity, and is given in DR by

$$M_{1c} \overset{\text{div}}{=} -\frac{\alpha_s C_F}{4\pi}\frac{1}{\epsilon}. \tag{16}$$

It is well-known that, at this order and higher orders, UV divergence of massless quark self-energy diagrams can be removed by the renormalization of quark field.

The diagram (d) in Fig. 1 gives

$$\begin{aligned} M_{1d} =& -ig_s^2 C_F e^{ip_z\xi_z}\int \frac{d^4k}{(2\pi)^4} e^{-ik_z\xi_z}\frac{1}{k^4(p-k)^2} \\ &\times \frac{1}{4p_z}\mathrm{Tr}[\not p\gamma_\mu \not k\gamma_z \not k\gamma^\mu] \end{aligned} \tag{17}$$

where $k = p - l$. Because of the oscillation factor $e^{-ik_z\xi_z}$, contribution from the region where k_z goes to infinity is highly suppressed. Thus, it is UV finite at finite ξ_z, although it is divergent as $\xi_z \to 0$,

$$M_{1d} \overset{\text{div}}{=} -\frac{\alpha_s C_F}{2\pi}\ln(|\xi_z p_z|). \tag{18}$$

In summary, the total one-loop contribution to the UV divergence of quasi-quark PDFs of a quark of momentum p at an arbitrary ξ_z is

$$\begin{aligned} M^{(1)} &\overset{\text{div}}{=} M_{1a} + 2\times M_{1b} + 2\times\frac{1}{2}M_{1c} + M_{1d} \\ &= \frac{\alpha_s C_F}{\pi}\left(-\frac{|\xi_z|}{a} - \frac{1}{4\epsilon} - \frac{1}{2}\ln(|\xi_z p_z|)\right), \end{aligned} \tag{19}$$

where the $\ln(|\xi_z p_z|)$ term is not UV divergent, but divergent as $\xi_z \to 0$ [3]. We find that, at this order, UV divergences only come from the region where all loop momenta go to infinity, thus UV divergences are localized in coordinate space. We will show in the next section that this behavior remains true up to all orders in QCD perturbation theory.

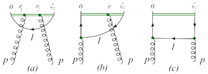

FIG. 2: Feynman diagram for quasi-quark PDFs of an asymptotic gluon of momentum p at one-loop order.

Feynman diagrams for quasi-quark PDFs of an asymptotic gluon of momentum p at one-loop order in the Feynman gauge are shown in Fig. 2, plus the complex conjugate diagram of (b). A general argument in the next section will show that UV divergences of all diagrams of quasi-PDFs can only come from the 4-D integration, and thus localized in coordinate space. If we keep ξ_z to be finite, all one-loop diagrams in Fig. 2 cannot be local, just like the diagram (d) in Fig. 1, and therefore, all these one-loop diagrams in Fig. 2 must be UV-finite. But, they can be divergent as $\xi_z \to 0$. To demonstrate this feature, let's take the diagram (a) in Fig. 2 as an example. The Fig. 2(a) gives

$$\begin{aligned} M_{2a} \propto& \int_0^{\xi_z} dr_1 \int_{r_1}^{\xi_z} dr_2 \int d^4l\, e^{-il_z\xi_z}\frac{l_z}{l^2} \\ =& \frac{\xi_z^2}{2}\int dl_z\, e^{-il_z\xi_z}\, l_z \int d^3\bar{l}\left(\frac{1}{\bar{l}^2} + \frac{l_z^2}{(\bar{l}^2 - l_z^2)\bar{l}^2}\right), \end{aligned} \tag{20}$$

which seems to be linearly UV divergent for the 3-D integration. However, similar to the case of diagram (a) in Fig. 1, the linear divergent term is proportional to $\delta'(\xi_z)$, which vanishes for finite ξ_z. The second term is finite under integration of $\bar{l}$, which gives

$$\begin{aligned} &\frac{\xi_z^2}{2}\int dl_z\, e^{-il_z\xi_z}\, l_z \int d^3\bar{l}\frac{l_z^2}{(\bar{l}^2 - l_z^2)\bar{l}^2} \\ \propto& \frac{\xi_z^2}{2}\int dl_z\, e^{-il_z\xi_z}\,\frac{l_z^3}{|l_z|} \\ =& \frac{2i}{\xi_z}, \end{aligned} \tag{21}$$

where the proportional relation can be simply obtained by dimensional counting. We therefore find that Fig. 2(a) is UV finite, but behaves as $1/\xi_z$ when $\xi_z \to 0$. Similarly, we found that Figs. 2(b,c) are also UV finite, which indicates that at one-loop, the quasi-quark PDFs get no mixing from a gluon under the UV renormalization.

Before continue, we want to note that the UV behavior found above is significantly different from that of PDFs. UV divergences of PDFs come from the region of loop momentum $(l_+, l_-, \vec{l}_\perp) \sim (1, \lambda^2, \lambda)$ when $\lambda \to \infty$. Thus, with the momentum component $l_+ \sim \mathcal{O}(1)$, the UV divergence for the normal PDFs is nonlocal in the "$-$" direction in coordinate space. It is this fact that the renormalization of PDFs is a convolution, mixed with all twist-2 PDFs with operators along the "$-$" direction, while the renormalizaiton of quasi-PDFs is a multiplicative factor as we will show.

We further note that, even if UV divergences ($1/a$, $\ln(a)$ or $1/\epsilon$) are renormalized, the $\ln(|\xi_z|)$ dependence (see Eq. 18 as an example) and $1/\xi_z$ dependence (see Eq. 21 as an example) of one-loop diagrams signal the

[3] If we take $\xi_z \to 0$, what we calculated is an one-loop correction to a local vector current. The finite ξ_z in Eq. (19) effectively regularizes the UV divergence of the one-loop vertex diagram Fig. 1(d), while UV divergence of the self-energy diagram in Fig. 1(c) is regularized by the dimensional regularization. From Fig. 1(d), we obtained the $\ln(|\xi_z p_z|)$ term after we took $\epsilon \to 0$ with ξ_z fixed. If we need to take $\xi_z \to 0$ to mimic the local current, we should keep ϵ finite to regularize the one-loop UV divergence, which changes $\ln(|\xi_z p_z|)$ to $-\frac{1}{2\epsilon}$. As a result, we find from Eq. (19) that UV divergences of the vector current vanish at one-loop level if we take $\xi_z \to 0$ with fixed ϵ and a, which is consistent with the expectation of current conservation.

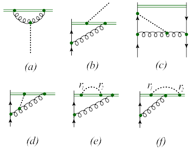

FIG. 3: Some loop diagrams, with an additional gluon (denoted as dotted curves) attachments, that could contribute to the quasi-PDFs at higher loop orders.

divergence of quasi-PDFs as $\xi_z \to 0$, which causes the difficulty in getting a well-defined momentum-space quasi-PDFs if one simply Fourier transforms the coordinate-space quasi-PDFs, as mentioned in the last section. This behaviour will certainly remain true to higher orders in QCD perturbation theory.

IV. POWER COUNTING

A. Superficial UV divergences

In order to identify all UV divergences from any Feynman diagram contributing to the quasi-PDFs, we start with a set of general diagrams constructed from the lower order Feynman diagrams by adding one more gluon to them. We will then examine how this additional gluon could change the UV divergence of the original diagrams. Specifically, we introduce and study the change of divergence index $\Delta\omega_3$ and $\Delta\omega_4$ corresponding to the 3-D integration and 4-D integration of corresponding loop, respectively. A sufficient condition for quasi-PDFs to be renormalizable is that $\Delta\omega_3 \le 0$ and $\Delta\omega_4 \le 0$ are satisfied for all cases, but it is not necessary as we will argue. We divide our discussions into five distinctive cases:

Case I: Only one end of the gluon is attached to parton lines of a loop, like Fig. 3(a). In this case we have one more propagator and one more QCD vertex in the loop, which results in $\Delta\omega_3 = -1$ and $\Delta\omega_4 = -1$. Thus a linear divergence can be changed to a logarithmic divergence, and a logarithmic divergent diagram is changed to be finite.

Case II: Only one end of the gluon is attached to gauge link of a loop, like Fig. 3(b). In this case, we have one more $z-$direction integration in coordinate space, which does not change the degree of divergence of 3-D integration but reduce the degree of divergence of 4-D integration by 1, such that $\Delta\omega_3 = 0$ and $\Delta\omega_4 = -1$.

Case III: Both ends of the gluon are attached to parton lines of a loop, like Fig. 3(c). In this case we have three more propagators, two more QCD vertexes, and four more momentum integrations for the loop, which results in $\Delta\omega_3 = -1$ and $\Delta\omega_4 = 0$.

Case IV: One end of the gluon is attached to parton lines of a loop, and the other end of the gluon is attached to gauge link of the loop, like Fig. 3(d). In this case we have two more propagators, one more QCD vertex, one more $z-$direction integration in coordinate space, and four more momentum integrations for the loop, which results in $\Delta\omega_3 = 0$ and $\Delta\omega_4 = 0$.

Case V: Both ends of the gluon are attached to gauge link of a loop, like Figs. 3(e,f). In this case we have one more propagator, two more $z-$direction integration in coordinate space, and four more momentum integrations for the loop. This results in $\Delta\omega_3 = 1$ and $\Delta\omega_4 = 0$.

A few comments for the above power counting rules are in order. 1) We only concentrated on overall divergences but not on subdivergences, because subdivergences can be taken care by forest subtraction in the step of renormalization. 2) The change of divergence indexes discussed here are only for superficial divergence. It means that even if the power counting indicates that a diagram is divergent, it may not be really divergent due to other considerations. But, on the other hand, if the power counting indicates that a diagram is finite, it must be UV finite.

The **Case V** is dangerous for renormalization, because $\Delta\omega_3 > 0$ means that the number of potentially UV divergent topologies is not finite, and thus a finite number of renormalization constants may not be enough to remove all UV divergences. However, as we pointed out, the above power counting rules are only for superficial divergence. We will show in the next subsection that all Feynman diagrams of quasi-PDFs do not have overall UV divergence if only the 3-D integration is considered for any of its loop momenta. As a result, only $\Delta\omega_4$ is relevant for identifying real UV divergent topologies.

B. Finiteness of the 3-D integration

To demonstrate that the 3-D integration of Feynman diagrams contributing to the quasi-PDFs cannot generate a real UV divergence, we consider the "gauge-link-irreducible" (GLI) diagrams. Similar to the concept of one-particle-irreducible (1PI) diagrams, a GLI diagram is a diagram that is still connected even if all gauge links are cut (or removed). For example, the diagrams (a), (b) and (c) in Fig. 1 are not the GLI diagrams, while the diagram (d) is. Similarly, the diagrams (a), (c) and (d) in Fig. 3 are the GLI diagrams, while the diagrams (b), (e) and (f) are not. For studying the renormalization of quasi-PDFs, since the renormalizaiton of QCD Lagrangian is well known, we only need to study the UV properties of the general GLI diagrams as shown in Fig. 4, where the dashed lines connecting to gauge link can be either gluon or quark (if it is at the end point of the gauge link).

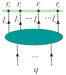

FIG. 4: A general "gauge-link-irreducible" (GLI) diagram, where $l_0 = q - l_1 - \cdots - l_n$.

In Fig. 4, we assume that the momentum flows into the blob is q, $l_1, \cdots, l_n$ are n loop momenta, and $l_0 = q - l_1 - \cdots - l_n$ is determined by the momentum conservation. Note that this assignment of momentum is always possible for a GLI diagram. Then, the general diagram in Fig. 4 can be expressed as

$$e^{iq_z r_0} \prod_{j=1}^{n} \int_{r_{j-1}+a}^{r_{max}-a} dr_j \int \frac{d^4 l_j}{(2\pi)^4} e^{il_{jz}(r_j - r_0)} \mathcal{M}(q, l_1, \cdots, l_n), \tag{22}$$

where $\mathcal{M}(q, l_1, \cdots, l_n)$ denotes the blob combined with propagators of the $n+1$ lines connecting to the gauge link. Since the GLI diagrams can be constructed from one-loop diagrams in Figs. 1 or 2 combined with insertions in **Cases I, III** and **IV** in the last subsection, their overall superficial UV divergence index $\omega \leq 1$, no matter we apply the 3-D integration or 4-D integration to each loop momentum.

Now let us study the case in which only the 3-D integration is performed for l_j, while carrying out either 3-D or 4-D integration for other loop momenta. The $\mathcal{M}(q, l_1, \cdots, l_n)$ has two kinds of dependence on l_j. One is the polynomial dependence in the numerator, for which we can simply decompose l_j to $\bar{l}_j$ and l_{jz}. The other is in the propagator like $1/(l_j + k)^2$ where k can be vanishing or depending on other loop momenta. We can decompose the propagator as

$$\begin{aligned} \frac{1}{(l_j+k)^2} =& \frac{1}{\Delta - 2k_z l_{jz} - l_{jz}^2} \\ =& \frac{1}{\Delta} + \frac{2k_z l_{jz}}{\Delta^2} + \frac{(\Delta + 4k_z^2 + 2k_z l_{jz}) l_{jz}^2}{(\Delta - 2k_z l_{jz} - l_{jz}^2)\Delta^2}, \end{aligned} \tag{23}$$

where $\Delta = (\bar{l}_j + \bar{k})^2 - k_z^2$ independent of l_{jz}. Based on dimensional counting, each additional l_{jz} in the numerator will suppress the divergence index ω of the 3-D integration by one unit. Thus the last term in Eq. (23) can be safely ignored as we are considering UV divergence from the 3-D integration of l_j. Since the other terms factorize out the l_{jz} dependence from $\mathcal{M}(q, l_1, \cdots, l_n)$, potential UV divergences of these terms are proportional to

$$\int dl_{jz} e^{il_{jz}(r_j - r_0)} l_z^m \propto \delta^{(m)}(r_j - r_0), \tag{24}$$

with m being non-negative integer. As r_j cannot equal to r_0 as defined in Eq. (22), $\delta^{(m)}(r_j - r_0)$ in Eq. (24) vanishes before we take $a \to 0$ limit. Therefore, UV divergences of Fig. 4, obtained by integrating out $\bar{l}_j$ and other loop momenta but fixing l_{jz}, eventually vanish after the integration of l_{jz}.

In summary, the overall UV divergences of GLI diagrams only come from the region where all loop momenta are large. Furthermore, the third term in Eq. (23) is responsible for real UV divergences from the 4-D integration of the GLI diagrams.

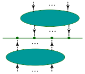

FIG. 5: A diagram made of two "gauge-link-irreducible" (GLI) sub-diagrams.

Now let us study a diagram made of two GLI sub-diagrams as shown in Fig. 5. This non-GLI diagram can be generated either from the diagram in Fig. 6 or by the insertion of **Cases II** and **V** in the last subsection. Based on the power counting rules, this diagram has overall superficial UV divergence index $\omega \leq 2$. When extracting overall UV divergences, we can follow the above discussion for each GLI sub-diagram, and found that the overall UV divergence of the combined diagram vanishes if we fix the z-component of any loop momentum. This conclusion can be easily generalized to any diagram made of more GLI sub-diagrams.

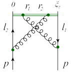

FIG. 6: A two-loop diagram that might give UV divergent contribution to quasi-quark PDF based on the power counting rules and building blocks in Figs. 1 and 2.

We thus conclude that overall UV divergences of any diagram that contributes to quasi-PDFs can only come from the region where all loop momenta are large. An immediate consequence of this finding is that to identify UV divergent topologies, we only need to consider the value of $\Delta\omega_4$ when we are evaluating the five case insertions discussed in the last subsection. Since we found that $\Delta\omega_4 \leq 0$ for all cases, there should be only finite number of topologies of UV divergent Feynman diagrams

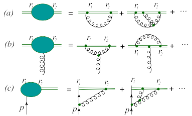

FIG. 7: Three topologies of diagrams that could give UV divergent contributions to the quasi-quark PDFs.

that could contribute to the quasi-PDFs, which we will present in the next subsection.

Let us conclude this subsection by explaining the key difference in UV behavior between quasi-PDFs and PDFs, which have UV divergence from the 3-D integration. As pointed out at the end of Sec. III, UV divergences of PDFs are obtained from a different 3-D integration of loop momentum, say l_- and $\vec{l}_\perp$. We can certainly do a similar decomposition of propagators as that in Eq. (23),

$$\begin{aligned}\frac{1}{(l+k)^2} =& \frac{1}{\hat{\Delta}+2l_+(l_-+k_-)} \\ =& \frac{1}{\hat{\Delta}} - \frac{2(l_-+k_-)l_+}{\hat{\Delta}^2} + \frac{4(l_-+k_-)^2 l_+^2}{(\hat{\Delta}+2l_+(l_-+k_-))\hat{\Delta}^2},\end{aligned} \tag{25}$$

where $\hat{\Delta} = 2k_+(l_-+k_-) - (\vec{l}_\perp + \vec{k}_\perp)^2$, which is independent of l_+. We can also argue that, because l_+ is factorized out, the first two terms do not contribute after the integration of l_+. However, the last term in Eq. (25) can still generate the UV divergence from the 3-D integration. This is because the UV divergence for PDFs comes from the region where any loop momentum l behaves as $(l_+, l_-, \vec{l}_\perp) \sim (1, \lambda^2, \lambda)$ as $\lambda \to \infty$, and thus $l_- l_+ \sim l_\perp^2 \sim \lambda^2$. The boost invariance ensures that each l_+ in the numerator will be either accompanied with a "$-$" momentum component in the numerator or a "$+$" momentum component in the denominator, and neither of these cases suppresses UV divergence index of the loop momentum integration.

C. Divergent diagrams for quasi-quark PDFs

Based on the above strategy and power counting rules, we can find out all UV divergent Feynman diagrams begin with a complete set of building blocks. To identify all UV divergent Feynman diagrams for quasi-quark PDFs, we need one more diagram in Fig. 6 to form the set of building blocks in addition to the one-loop diagrams in Figs. 1 and 2, because the superficial UV divergence of this two-loop diagram does not agree with that obtained from Fig. 1(b) by inserting one more gluon like in **Case IV**. Because we only need to consider 4-D integrations, non-vanishing ξ_z in Fig. 6 ensures that this diagram is UV finite.

Using the above building blocks, we can generate all possible higher order Feynman diagrams. Among them, there are only three types of topologies that could give UV divergent contribution to the quasi-quark PDFs as shown in Fig. 7, in addition to those from the renormalizaiton of QCD Lagrangian. From the above discussion, we know that all of these three topologies can be UV divergent only in the region where all loop momenta go to infinity, thus UV divergent contributions are localized in coordinate space. An immediate consequence is that to make the diagrams in Fig. 7 divergent, $r_2 - r_1$ must go to 0. This finding also explains the result of the two-loop calculation in Ref. [41], where one finds that UV divergence of quasi-quark PDF under DR is proportional to $\delta(1-z)$ in momentum space.

In addition to these three UV divergent topologies in Fig. 7, we also show some topologies that are UV finite in Fig. 8. Especially, the last diagram in Fig. 8 indicates that quasi-quark PDFs do not mix with quasi-gluon PDF under the renormalization to all orders in QCD perturbation theory.

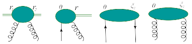

FIG. 8: Sample topologies of diagrams that give UV finite contributions to the quasi-quark PDFs.

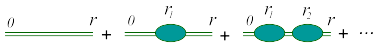

FIG. 9: Contributions to the general diagrams in Fig. 7(a) with all loop diagrams reorganized in terms of 1PI diagrams.

V. RENORMALIZATION

In this section, we present our general arguments for the renormalization of the three topologies of diagrams that could give UV divergent contribution to the quasi-quark PDFs, as identified in the last section.

Let us first consider the power UV divergence from the general diagram in Fig. 7(a), which could be expressed in terms of the sum of diagrams made of 1PI diagrams, as shown in Fig. 9. Since the UV divergence is local in coordinate space, we assign the "i-th" blob in Fig. 9 a specific point r_i. We express the linear UV power divergence of the "1st" blob in Fig. 9 as $c\int_0^r dr_1$, where perturbative coefficient $c = c^{(1)} + c^{(2)} + \cdots$ where $c^{(1)} = -\frac{\alpha_s C_F}{\pi}\frac{1}{a}$ is the first order expansion in α_s, as derived in Eq. (11).

With the geometric sum of the 1PI diagrams, as shown in Fig. 9, we can derive all linear UV power divergent contribution to the general diagram in Fig. 7(a) by summing over contributions of 1PI diagrams located between 0 and r ($r > 0$) to all orders as,

$$1 + c\int_0^r dr_1 + c^2\int_0^r dr_1 \int_{r_1}^r dr_2 + \cdots \\ = \mathcal{P} e^{c\int_0^r dr'} = e^{c\,r}\,, \tag{26}$$

with $\mathcal{P}$ indicating the path ordering. Therefore, one could introduce an overall factor $e^{-c|\xi_z|}$ to remove all linear power UV divergences of the quasi-quark PDFs. This overall factor could be thought as the mass renormalization of a test particle moving along the gauge link. Renormalizing power divergence in this way was first proposed in Ref. [42].

Besides power UV divergence, there are also logarithmic UV divergences from Fig. 7(a). It is well known [43] that these divergences can be removed by a "wave function" renormalization of the test particle, Z_{wq}^{-1}.

The general diagrams in Fig. 7(b) have only the logarithmic UV divergences, effectively from the loop corrections to the gluon coupling to the gauge link. it was proven in Ref. [43] that these logarithmic UV divergences can be absorbed by the coupling constant renormalization of QCD. Therefore, we do not need to worry about them if we use the renormalized QCD Lagrangian.

Finally, let us examine UV divergence from the general diagrams shown in Fig. 7(c). Unlike the general diagrams in Figs. 7(a) and (b), the loop momentum of diagrams in Fig. 7(c) go through active quark (or gluon for the case of quasi-gluon PDF). Since the UV divergence from the diagrams in Fig. 7(c) effectively comes from high order loop corrections to the quark-gauge-link vertex, which is not a fundamental coupling in QCD Lagrangian, using the renormalized QCD Lagrangian to do the calculation does not help remove this kind of UV divergences. That is, the renormalziation of the operators defining quasi-PDFs is required to remove this kind of UV divergences, if we want to have renormalizable quasi-PDFs.

The key question is then if the operators defining quasi-PDFs will mix with other operators under renormalization? If it does, there could be a danger that the operators defining quasi-PDFs may not form a closed set under the renormalization.

The quark-gauge-link vertex at the lowest order is a simple gamma matrix γ_z for the quasi-quark PDFs (is the same for the $A_z = 0$ case). As we demonstrated in the last section, UV divergence comes only from the region where all loop momenta are very large, thus we can set $p = 0$ if we are only interested in leading UV divergence, which is logarithmic as demonstrated in Sec. IV. In addition, the logarithmic UV divergence, $\ln(a)$ as $a \to 0$, is local in coordinate space, and does not have a direct dependence on ξ_z. That is, we find that the UV divergent term of Fig. 7(c) only depends on the vector n_z^μ, and consequently, it is proportional to γ_z with a constant coefficient, which is proportional to the quark-gauge-link vertex at the lowest order. Therefore, a constant counterterm is sufficient to remove this kind of UV divergences. Using bookkeeping forests subtraction method, it is straight forward to remove the high order divergences and to show that the net effect is to introduce a constant multiplicative renormalizaton factor Z_{vq}^{-1} for the quark-gauge-link vertex.

In summary, by using renormalized QCD Lagrangian in Feynman gauge, we find that all remained perturbative UV divergences of the quasi-quark PDFs can be removed by introducing a multiplicative renormalization factor, with the multiplicative renormalization factor calculated order by order in QCD perturbation theory. We expect that similar arguments should also apply for quasi-gluon PDF. We thus can define the renormalized coordinate-space quasi-PDFs as

$$\tilde{F}^R_{i/p}(\xi_z, \tilde{\mu}^2, p_z) = e^{-C_i|\xi_z|} Z_{wi}^{-1} Z_{vi}^{-1} \tilde{F}^b_{i/p}(\xi_z, \tilde{\mu}^2, p_z), \tag{27}$$

where C_i, Z_{wi} and Z_{vi} are renormalization constant depending on parton flavor "i" but independent of ξ_z, and perturbatively calculable order-by-order in powers of α_s. We conclude that the coordinate-space quasi-PDFs are renormalizable and they do not mix with each other or with any other operators under renormalization group equation.

VI. SUMMARY

We demonstrated that the behavior of UV divergences of quasi-PDFs is very different from that of PDFs. While the renormalization of quasi-PDFs is a simple multiplicative factor, the renormalization of PDFs is of a convolution form, due to the fact that the UV divergences of PDFs are not completely local, and consequently, PDFs of different flavors mix with each other under the renormalization.

We show that the locality in space-time of the perturbative UV divergences of the coordinate-space quasi-PDFs makes it possible to have the coordinate-space quasi-PDFs multiplicatively renormalizable, as shown in Eq. (27), to all orders in QCD perturbation theory. With the all-order arguments for factorizing all leading power CO divergences of quasi-PDFs into PDFs, given in Ref. [9], we conclude that the renormalized quasi-PDFs could be good candidates for extracting PDFs from LQCD calculations.

For the LQCD calculation of quasi-PDFs, however, the introduction of the overall factor, $e^{-C_i|\xi_z|}$, with perturbatively calculated C_i, is not sufficient to remove all power divergences, since we cannot calculate the C_j to all orders. That is, we need to renormalize the power UV divergence of quasi-PDFs, nonperturbatively. As shown in Ref. [35], for example, we could introduce an overall non-perturbative factor to remove this kind of power divergences to all orders, for which we will not get into the details here. In principle, the choice to renormalize the power divergences of quasi-PDFs nonperturbatively is not unique. But, so long as the renormalization is multiplicative, the renormalization procedure does not alter the CO properties of quasi-PDFs (so that the quasi-PDFs could be factorized into PDFs), the difference in renormalization procedures/choices corresponds to different renormalization schemes, which should lead to different matching coefficients between quasi-PDFs and PDFs.

In addition, we show that the coordinate-space quasi-PDFs are well behaved for all values of ξ_z, except when $\xi_z = 0$. That is, if we want to Fourier transform the coordinate-space quasi-PDFs to derive momentum-space quasi-PDFs, we will have to define a consistent subtraction scheme to remove all divergent terms as $\xi_z \to 0$, before the Fourier transformation. Without this subtraction, the momentum-space quasi-PDFs are ill-defined at large $\tilde{x}$ region.

Note added: Two independent studies of the renormalization of quasi-PDFs appeared recently [44, 45], in which the same conclusion is reached although approaches are very different from ours.

Acknowledgments

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P20–p011

P21：威尔逊线重正化改进准分布

主题归属	非局域算符重整化与 hybrid 方案
底稿	arXiv:1609.08102
arXiv / DOI	1609.08102
完整原始 PDF	9 页；来源 PyQCD；SHA-256 722b1a13447f...
转排索引	EN 8 页；ZH 7 页（只作书目对照）
备注	索引未记录额外备注。

阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

页码范围：P21-p001 – P21-p009

Improved quasi parton distribution through Wilson line renormalization

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Abstract

Recent developments showed that hadron light-cone parton distributions could be directly extracted from spacelike correlators, known as quasi parton distributions, in the large hadron momentum limit. Unlike the normal light-cone parton distribution, a quasi parton distribution contains ultraviolet (UV) power divergence associated with the Wilson line self energy. We show that to all orders in the coupling expansion, the power divergence can be removed by a "mass" counterterm in the auxiliary z-field formalism, in the same way as the renormalization of power divergence for an open Wilson line. After adding this counterterm, the quasi quark distribution is improved such that it contains at most logarithmic divergences. Based on a simple version of discretized gauge action, we present the one-loop matching kernel between the improved non-singlet quasi quark distribution with a lattice regulator and the corresponding quark distribution in dimensional regularization.

P21-p001

I. INTRODUCTION

One of the most important goals of QCD is to understand the hadron structure from its fundamental degrees of freedom - quarks and gluons. This necessarily goes to the nonperturbative regime of QCD, and it is difficult in general to obtain first principle results directly from the QCD Lagrangian. A powerful tool of obtaining such results is Lattice QCD, which is an approach defined on Euclidean spacetime, and has been used to calculate hadron masses, charges etc. to a remarkable accuracy. However, it cannot be used to directly access intrinsically Minkowskian quantities such as the parton distribution functions (PDFs). PDFs are defined as the forward hadronic matrix elements of light-cone correlations, and describe the momentum distribution of quarks and gluons inside the hadron. They play a crucial role in understanding the experimental data at high energy hadron colliders such as the LHC. Owing to their real time dependence, Lattice QCD can only be used to indirectly extract the information on the PDFs by calculating their moments, which is limited by technical complications such as operator mixing. Another commonly used approach to determine PDFs is to assume a suitable parametrized form and fit to a large variety of experimental data. Most PDF groups obtained their PDF sets in this way [1–6]. A main drawback of this approach is that it suffers from parametrization uncertainty, and different groups usually produce different results for the same PDF.

Recent developments [7–21] showed that the light-cone observables such as the PDFs can be directly extracted from the large momentum limit of the hadronic matrix element of a spacelike correlator, which is known as the quasi observable, using a large momentum effective theory [15] (for other approaches to extract light-cone quantities see e.g. [22–28]). The quasi PDF does not have a real time dependence, and thus can be simulated on the lattice. The infrared (IR) behaviors between the flavor non-singlet quasi-PDF and its corresponding light-cone PDF are shown to be the same at one loop by direct computations, and argued to be the same at all loops in Ref. [9], based on which a factorization formula was also presented.

The factorization in Ref. [9] was given for the bare quasi quark distribution, where all fields and couplings entering the quasi distribution are bare ones. However, as the light-cone distribution, the quasi distribution also contains ultraviolet (UV) divergences and therefore needs renormalization. Ref. [17] explores the renormalization property of the quasi distribution, and shows that it is multiplicatively renormalizable at two-loop order. Also an equivalence was established between the virtual correction of the quasi quark distribution and the correction to the heavy-light quark vector current in heavy quark effective theory, so that the UV divergences in the former can be renormalized as the renormalization of the latter. Dimensional regularization was used in Ref. [17], and the linear divergence present in a cutoff or lattice regularization was ignored. In realistic lattice calculations, one needs to know how to deal with such power divergences. This is one goal of the present paper. We will show that the power divergence in the quasi quark distribution can be removed to all-loop orders by a mass counterterm, which is the same as the renormalization of an open Wilson line. After such a mass renormalization, the quasi quark distribution is improved such that it contains at most logarithmic divergences.

The rest of this paper is organized as follows. In Sec. 2, we discuss the renormalization of power divergences arising from the Wilson line self energy in the quasi quark distribution. In Sec. 3, we present a lattice perturbation theory matching for the quasi quark distribution. We then conclude in Sec. 4.

P21-p002

II. IMPROVED QUASI QUARK DISTRIBUTION THROUGH WILSON LINE RENORMALIZATION

Let us start by recalling the definition of the quasi quark distribution. For the unpolarized quark density, it is given as [8]

$$\tilde{q}(x,\Lambda,p^z) = \int_{-\infty}^{\infty} \frac{dz}{4\pi} e^{izk^z} \langle p|\overline{\psi}(0,0_\perp,z)\gamma^z L(z,0)\psi(0)|p\rangle \ , \tag{1}$$

where the quark fields are separated along the spatial z-direction, $L(z,0)$ is the Wilson line gauge link inserted to ensure gauge invariance, and Λ denotes the UV cutoff.

The one-loop correction to the above quark density has been computed both in the axial gauge [9] and in the Feynman gauge [17]. In Ref. [17], the Wilson line $L(z,0)$ was separated into two Wilson lines along the z-axis $L(\infty,0)$ and $L(z,\infty)$, and associated respectively to the quark field to form gauge invariant combinations. Here we keep the gauge link as $L(z,0)$. The one-loop diagrams are then given in Fig. 1. It is straightforward to check that these diagrams yield the same one-loop result as that in Ref. [17]. The linear divergence comes from the last diagram, which is a Wilson line self energy. It is known that such a linear divergence can be removed by a mass renormalization with the help of an auxiliary z-field formalism [29–31], where the z-field is defined in a one-dimensional parameter space, and the non-local Wilson line can be interpreted as a two-point function of the z-field. In a sense, the auxiliary field $Z(z)$ can be understood as a Wilson line extending between $[z,\infty]$, i.e.

$$Z(z) = L(z,\infty), \tag{2}$$

which satisfies an equation of motion

$$\left[\partial_z - igA_z(z)\right] Z(z) = 0, \tag{3}$$

analogous to a heavy quark field (in the heavy quark limit). The Wilson line

$$L(z,0) = Z(z)Z^\dagger(0) \tag{4}$$

also renormalizes analogously to a heavy quark two point function

$$L^{\rm ren}(z,0) = \mathcal{Z}_Z^{-1} e^{-\delta m|z|} L(z,0), \tag{5}$$

where δm is analogous to the dimension one heavy quark mass counterterm and is linearly divergent, while the dimension zero wave function renormalization constant $\mathcal{Z}_Z$ is logarithmically divergent [29].

To illustrate the cancellation of linear divergence from the mass counterterm, let us determine δm at one-loop level. Recall that the last diagram in Fig. 1 leads to the following linear divergence

$$\lim_{\epsilon\to 0}\int dk_z \frac{\alpha_s C_F \Lambda}{2\pi} \frac{[\delta(k_z - \bar{x}p_z) - \delta(\bar{x}p_z)]p_z}{k_z^2+\epsilon^2} \tag{6}$$

with $\bar{x} = 1-x$. On the other hand, plugging Eq. (5) into Eq. (1), one finds that the mass

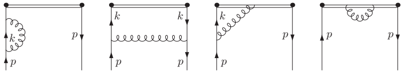

FIG. 1: One-loop diagrams for quasi quark distribution in Feynman gauge, the conjugate diagrams are not shown.

counterterm yields the following contribution

$$
\begin{aligned}
&-\int \frac{dz}{2\pi} p_z\, e^{i(x-1)p_z z}|z|\,\delta m = -\lim_{\epsilon\to 0}\int \frac{dz}{2\pi} p_z\, e^{-i\bar{x}p_z z}\frac{1-e^{-\epsilon|z|}}{\epsilon}\delta m \\
&= -\lim_{\epsilon\to 0}\int \frac{dz}{2\pi} p_z\, e^{-i\bar{x}p_z z}\int_0^\infty \frac{d\alpha}{\sqrt{\pi\alpha}}(1-e^{-\frac{z^2}{4\alpha}})e^{-\alpha\epsilon^2}\delta m \\
&= -\lim_{\epsilon\to 0}\int \frac{dz}{2\pi} p_z\, e^{-i\bar{x}p_z z}\int_0^\infty d\alpha \int dk_z \frac{e^{-\alpha(k_z^2+\epsilon^2)}(1-e^{ik_z z})}{\pi}\delta m \\
&= -\lim_{\epsilon\to 0}\int \frac{dz}{2\pi} p_z\, e^{-i\bar{x}p_z z}\int dk_z \frac{1}{\pi}\frac{(1-e^{ik_z z})}{k_z^2+\epsilon^2}\delta m \\
&= -\lim_{\epsilon\to 0}\int \frac{dk_z}{\pi} p_z \frac{\delta(\bar{x}p_z)-\delta(k_z-\bar{x}p_z)}{k_z^2+\epsilon^2}\delta m. \qquad (7)
\end{aligned}
$$

We therefore have

$$\delta m = -\frac{\alpha_s C_F}{2\pi}(\pi\Lambda) \qquad (8)$$

at one-loop.

In coordinate space, by expanding the Wilson line to $\mathcal{O}(g^2)$, we have

$$g^2 C_F \int_0^z dz_1 \int_0^{z_1} dz_2 \frac{1}{4\pi^2}\frac{1}{(z_1-z_2)^2+a^2} = \frac{g^2 C_F}{4\pi^2}\Big[\frac{z}{a}\tan^{-1}\Big(\frac{z}{a}\Big) - \frac{1}{2}\ln\Big(1+\frac{z^2}{a^2}\Big)\Big], \qquad (9)$$

where we have introduced a cutoff a to regularize the short distance singularity in the coordinate space gluon propagator when $z_1 \to z_2$. By Fourier transforming to momentum space, the above result leads to

$$\frac{g^2 C_F}{8\pi^2}\Big[\frac{1}{ap_z(1-x)^2} - \frac{1}{|1-x|} + C\delta(1-x)\Big], \qquad (10)$$

where C is a constant that can be determined by demanding the x-integration of the above expression vanish. We can therefore identify

$$\Lambda = \frac{1}{a}. \qquad (11)$$

The mass counterterm diagram of order α_s is shown in Fig. 2. The gauge independence of δm can already be seen from that the one-loop correction in the axial gauge [9] and Feynman gauge [17] contains identical linear divergences. By using the R_ξ gauge, one easily

P21-p004

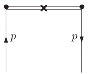

FIG. 2: One-loop mass counterterm diagram for quasi quark distribution in Feynman gauge.

finds that the extra piece $k^\mu k^\nu/k^2$ (where k is the gluon momentum) in the numerator of gluon propagator has no impact on the linear divergence, therefore δm remains the same in different gauges.

One can add more gluons and quarks to the one-loop diagrams in Fig. 1 to form higher loop diagrams. Power counting tells us that power divergence appears only in those diagrams which contain the Wilson line self energy as a subdiagram. As shown in the appendix, beyond one-loop the mass counterterm also removes all the power divergence from the Wilson line. Therefore, after including the mass counterterm contribution, there is at most logarithmic divergence in the quasi quark distribution. This indicates that, as far as the power divergence is concerned, the renormalization of a quark bilinear operator like the quasi quark distribution is the same as the renormalization of an open Wilson line. In the discussions above, we showed that the linear divergence is indeed cancelled by the mass counterterm at perturbative one-loop level. On the lattice, the mass counterterm δm has to be determined nonperturbatively, which can be done e.g. as in Ref. [32].

III. MATCHING BETWEEN LATTICE AND CONTINUUM THROUGH LATTICE PERTURBATION THEORY

In this section, we aim at establishing the one-loop matching connecting the quasi quark distribution on the lattice to the normal quark distribution in the continuum. On the lattice, besides the diagrams in Fig. 1, there exists an extra one-loop diagram (shown in Fig. 3).

The first diagram in Fig. 1 and the diagram in Fig. 3 are nothing but the Feynman gauge quark field renormalization, and have been well understood in lattice QCD. Their contribution can also be obtained from the other diagrams in Fig. 1 through quark number conservation. As shown in Refs. [9] and [17], in the continuum the second and third diagrams in Fig. 1 contain a logarithmic dependence on the hadron momentum instead of a logarithmic divergence. Their contribution is independent of the regularization and thus will be the same in the continuum and on the lattice. One only needs to match the linearly divergent part of the last diagram in Fig. 1 between lattice and continuum. This gives a connection between the lattice cutoff and the linear divergence in the one-loop kernel in the continuum [9] (in general they will differ by a finite term). To obtain this connection, we compute the Wilson line self energy using lattice perturbation theory based on a simple version of discretized gauge action.

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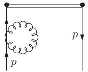

FIG. 3: The extra one-loop diagram for quasi quark distribution on the lattice, the conjugate diagram is not shown.

The contribution of the Wilson line self energy diagram can be written as

$$\begin{aligned}\delta W &= \frac{\alpha_s C_F}{8\pi^3}\int d^4(ka_L)\frac{1-\cos(k_z z)}{(1-\cos(k_z a_L))4\sum_\lambda \sin^2\frac{k_\lambda a_L}{2}} \\ &= \alpha_s C_F[\frac{\pi}{2}\frac{|z|}{a_L}-\frac{1}{\pi}\ln\frac{\pi|z|}{a_L}-\frac{1}{\pi}]+\mathcal{O}(\frac{a_L}{z}),\end{aligned} \tag{12}$$

where a_L denotes the lattice spacing. Comparing this with the expansion of Eq. (9) leads to the following matching between the linear divergence in the continuum and on the lattice

$$\Lambda = \frac{\pi}{a_L}. \tag{13}$$

By replacing Λ in the one-loop kernel of Ref. [9] with the above matching condition, we then obtain the one-loop kernel relating the quasi PDF on the lattice to the normal PDF. By adding a factor of $e^{-\delta m|z|}$ to the integrand of Eq. (1), we obtain the improved quasi-PDF $\tilde{q}_{\rm imp}$ free of power divergence

$$\tilde{q}_{\rm imp}(x,\Lambda,p^z) = \int_{-\infty}^{\infty}\frac{dz}{4\pi}e^{izk^z-\delta m|z|}\langle p|\overline{\psi}(0,0_\perp,z)\gamma^z L(z,0)\psi(0)|p\rangle\ . \tag{14}$$

The matching kernel Z between the improved quasi-PDF $\tilde{q}_{\rm imp}$ and the PDF q is defined as

$$\tilde{q}_{\rm imp}(x,a_L,p^z) = \int_{-1}^{1}\frac{dy}{|y|}Z\left(\frac{x}{y},p^z a_L,\frac{\mu}{p^z}\right)q(y,\mu)+\mathcal{O}\left(\Lambda^2_{\rm QCD}/(p^z)^2,M^2/(p^z)^2\right)\ . \tag{15}$$

At one-loop the Z factor can be extracted from Eqs. (15)-(19) of Ref. [9] with the $1/a_L$ contribution subtracted by the counterterm diagram in Fig. 2.

$$Z\left(\xi,p^z a_L,\frac{\mu}{p^z}\right) = \delta(\xi-1)+\frac{\alpha_s}{2\pi}Z^{(1)}\left(\xi,p^z a_L,\frac{\mu}{p^z}\right)+\ldots, \tag{16}$$

where

$$Z^{(1)}/C_F = \begin{cases}\left(\frac{1+\xi^2}{1-\xi}\right)\ln\frac{\xi}{\xi-1}+1\ , & \xi>1\ , \\ \left(\frac{1+\xi^2}{1-\xi}\right)\ln\frac{(p^z)^2}{\mu^2}+\left(\frac{1+\xi^2}{1-\xi}\right)\ln\left[4\xi(1-\xi)\right]-\frac{2\xi}{1-\xi}+1\ , & 0<\xi<1\ , \\ \left(\frac{1+\xi^2}{1-\xi}\right)\ln\frac{\xi-1}{\xi}-1\ , & \xi<0\ ,\end{cases} \tag{17}$$

and the extra contribution near $\xi = 1$ is

$$\delta Z^{(1)} = \frac{\alpha_S C_F}{2\pi} \int dy \begin{cases} -\frac{1+y^2}{1-y} \ln \frac{y}{y-1} - 1 \,, & y > 1 \,, \\ -\frac{1+y^2}{1-y} \ln \frac{(p^z)^2}{\mu^2} - \frac{1+y^2}{1-y} \ln \left[4y(1-y)\right] + \frac{2y(2y-1)}{1-y} + 1 \,, & 0 < y < 1 \,, \\ -\frac{1+y^2}{1-y} \ln \frac{y-1}{y} + 1 \,, & y < 0 \,. \end{cases} \tag{18}$$

Given that the power divergence has been removed, the above one-loop matching kernel facilitates a reliable extraction of normal PDFs from lattice data.

IV. CONCLUSION

In this paper, we have shown that the power divergence present in the lattice regularization of the quasi PDFs can be removed to all-loop orders by a mass counterterm, which can be interpreted as the mass renormalization for a test particle moving on the Wilson line. After such a mass renormalization, the quasi distribution is improved such that it contains at most logarithmic divergences. We also present the one-loop matching for the quasi quark distribution between lattice and continuum using lattice perturbation theory based on a simple version of discretized gauge action. Our results facilitate a reliable extraction of physical PDFs from lattice data.

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Note added

While this work is being finalized, a preprint by Jian-Wei Qiu et al. [33] dealing with the same topic has appeared, where they obtain similar conclusions as ours on the renormalization of power divergence.

V. APPENDIX

In this appendix, we present the arguments that the mass counterterm removes the power divergence to all-loop orders.

For simplicity, we start with the one-loop diagrams for the quasi quark distribution in Ref. [17], which can be equivalently drawn as the diagrams in Fig. 4 with a vertex insertion (1 for virtual diagrams, and $\gamma^z \delta(x - k^z/p^z)$ for real diagrams to impose momentum constraint). At one-loop, it has been shown [17] that the virtual diagrams are respectively equivalent to the wave function renormalization of a light quark field, the renormalization of a heavy-light vector current, and the wave function renormalization of a heavy quark field. All these

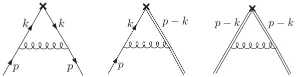

FIG. 4: One-loop diagrams to illustrate the real contribution of quasi quark distribution in Feynman gauge.

can be done to all-loop orders. Moreover, the mass counterterm can be chosen to exactly cancel the linearly divergent contribution from the third diagram in Fig. 4 (both virtual contribution with vertex insertion 1 and real contribution with $\gamma^z\delta(x - k^z/p^z)$). Following Ref. [30], this also holds to all-loop orders. After this mass renormalization, as the real diagrams contain an extra δ-function which effectively reduces the UV degree of divergence by one, they will produce at most logarithmic divergences at higher-loop orders. Therefore, as far as power divergence is concerned, the mass renormalization is sufficient to remove them to all-loop orders.

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P21-p009

P22：大动量有效理论的重正化

主题归属	非局域算符重整化与 hybrid 方案
底稿	arXiv:1706.08962
arXiv / DOI	1706.08962
完整原始 PDF	5 页；来源 course-cache；SHA-256 7a288164a4f2...
转排索引	EN 7 页；ZH 6 页（只作书目对照）
备注	索引未记录额外备注。

阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

页码范围：P22-p001 – P22-p005

Renormalization in Large Momentum Effective Theory of Parton Physics

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In the large-momentum effective field theory approach to parton physics, the matrix elements of non-local operators of quark and gluon fields, linked by straight Wilson lines in a spatial direction, are calculated in lattice quantum chromodynamics as a function of hadron momentum. Using the heavy-quark effective theory formalism, we show a multiplicative renormalization of these operators at all orders in perturbation theory, both in dimensional and lattice regularizations. The result provides a theoretical basis for extracting parton properties through properly renormalized observables in Monte Carlo simulations.

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I. INTRODUCTION

One of the most important goals of quantum chromodynamics (QCD) is to understand the hadron structure from its fundamental degrees of freedom — quarks and gluons. A powerful tool of obtaining such results is lattice QCD, which has been used to study the static properties of the hadron, such as mass, charge radius etc., to good accuracy (see e.g. [1, 2]). However, in general it cannot be used to directly access quantities with real-time dependence. An example is the parton distribution functions (PDFs), which are defined as the quark and gluon matrix elements of light-cone correlations and describe the momentum distribution of quarks and gluons inside the hadron. They play a crucial role in understanding the experimental data at high energy hadron colliders such as the LHC. Lattice QCD can only be used to indirectly extract the information of the PDFs by calculating their moments [3, 4], which is limited by technical difficulties.

In the past few years, a new approach has been proposed to study parton physics from lattice QCD, which is now known as the large-momentum effective theory (LaMET) [5, 6]. Here one starts from a time-independent quasi-PDF, which is a matrix element of non-local quark and gluon operators that can be directly calculated on the lattice, and then uses it to extract the light-cone PDF through a perturbative factorization up to power corrections suppressed by the nucleon momentum. The factorization in Refs. [5, 7] was given for the bare quark quasi-PDF, where all fields and couplings entering the quasi-distribution are bare ones. The ultraviolet (UV) physics of the quasi-PDF is all factorized perturbatively into the matching coefficients. However, lattice perturbation theory is known to converge slowly, and non-perturbative renormalization is often necessary to define a continuum limit where the matching to parton physics is under better control. Thus to make LaMET more practical, a complete understanding of UV divergences of the relevant non-local operators is essential.

Renormalization of non-local operators in QCD has not been well studied in the literature. A distinct feature of the LaMET non-local operators is the appearance of Wilson lines connecting the parton fields at different spacetime points to ensure gauge invariance. The renormalization of an open Wilson line as well as a closed Wilson loop has been studied decades ago [8–10], where it was shown that the Wilson line with a smooth simple contour can be renormalized, and in particular, the power divergence associated with the Wilson line can be absorbed into an exponential mass renormalization. In an auxiliary field formalism [10], it was also shown that the Wilson line can be replaced by a Green's function of the auxiliary field and studied as such. On the other hand, the renormalization of non-local operators such as the quasi-PDF has extra complications due to the parton fields attached to the endpoints. We have shown previously that the quasi-PDF is multiplicatively renormalizable in dimensional regularization up to two-loop order [11]. The full renormalization properties of the quasi-PDF has been conjectured in recent studies [12–17] (for an earlier work on the renormalization of transverse-momentum-dependent distributions with a similar conjecture, see [18]), but not proven.

In this paper, we perform a systematic study of the renormalization of gauge-invariant non-local operators for the quasi-PDF. We first present a general framework by introducing an auxiliary "heavy quark" field into the QCD Lagrangian, where the Wilson line can be replaced by a product of the auxiliary fields. In analogy to heavy-quark effective theory (HQET), we argue that this theory is renormalizable to all orders in perturbation theory. By taking the unpolarized quark quasi-PDF as an example, we then show that its renormalization reduces to that of two heavy-light quark currents, which can also be done to all orders in perturbation theory. This removes the obstacle to define a continuous quasi-PDF through lattice simulations. The same conclusion is not limited to the quark quasi-PDF, but can also be generalized to the gluon ones, as well as other non-local correlators, such as the Euclidean observables defined in the LaMET framework to extract the generalized parton distributions, transverse-momentum-dependent distributions, hadron distribution amplitudes, etc.

II. EFFECTIVE QCD WITH AUXILIARY "HEAVY-QUARK" FIELD

To be concrete, consider the following non-local operator

$$O(x,y) = \overline{\psi}(x)\Gamma L(x,y)\psi(y), \tag{1}$$

where $\psi, \bar{\psi}$ denote the bare quark fields and Γ a Dirac structure. $L(x,y)$ is a path-ordered gauge link from y to x,

$$L(x,y) = \mathcal{P}\exp\left(-ig\int_0^1 d\lambda \frac{dz^\mu}{d\lambda}A_\mu(z(\lambda))\right) \tag{2}$$

carrying a $(3,\bar{3})$ representation in color SU(3) group, where the path is parametrized such that $z(0) = y$ and $z(1) = x$. The gauge link ensures gauge invariance of the operator $O(x,y)$. For a straight-line gauge link which is of our main interest in the present paper, we can choose $z^\mu(\lambda) = y^\mu + \eta\lambda\, n^\mu$ with n^μ a unit vector specifying the direction and η the length of the gauge link.

To study the renormalization property of the above operator, we introduce an "heavy quark" auxiliary field (color source without spin degrees of freedom) Q with color 3, and its conjugate $\overline{Q}$, with color $\bar{3}$. We extend QCD to include this "heavy quark" interaction with the gluon field, and introduce the following Lagrangian

$$\mathcal{L} = \mathcal{L}_{\rm QCD} + \overline{Q}(x) in\cdot DQ(x), \tag{3}$$

where $D_\mu = \partial_\mu + igt^a A^a_\mu$ is the covariant derivative in fundamental representation. For a real heavy quark or timelike Wilson line, n^μ is a timelike vector $n^\mu = (1,0,0,0)$, whereas for a spacelike Wilson line n^μ can be chosen as $n^\mu = (0,0,0,1)$. We will focus on the latter case in the following, although the discussion may go through for any n. So long as there is a non-zero time component n^0, the "heavy quark" is dynamical, but follows a designated world line.

In the above theory, we can replace the bilocal operator $O(x,y)$ by a new operator,

$$O(x,y) = \overline{\psi}(x)\Gamma Q(x)\overline{Q}(y)\psi(y). \tag{4}$$

This can be seen as following. After integrating out the "heavy quark" field, we have

$$\int \mathcal{D}\overline{Q}\mathcal{D}Q\, Q(x)\overline{Q}(y) e^{i\int d^4x\mathcal{L}} = S_Q(x,y)e^{i\int d^4x\mathcal{L}_{\rm QCD}} \tag{5}$$

up to a determinant $\det(in\cdot D)$ which can be shown to be a constant and absorbed into the normalization of the generating functional [19], where $S_Q(x,y)$ satisfies

$$n\cdot D\, S_Q(x,y) = \delta^{(4)}(x-y), \tag{6}$$

with the solution in the case of $n^\mu = (0,0,0,1)$

$$\begin{aligned} S_Q(x,y) &= \theta(x^z-y^z)\delta(x^0-y^0)\delta^{(2)}(\vec{x}_\perp - \vec{y}_\perp)L(x,y) \\ &= \theta(x^z-y^z)\delta(x^0-y^0)\delta^{(2)}(\vec{x}_\perp - \vec{y}_\perp)L(x^z,y^z), \end{aligned} \tag{7}$$

under an appropriate boundary condition. In the following we will restrict to the case $x^z > y^z$ without loss of generality. The δ-functions ensure that the time and transverse components of x and y are equal, and therefore generate a spacelike Wilson line along the longitudinal direction. The above result allows us to replace the Wilson line $L(x^z,y^z)$, which is the one appearing in the quasi-PDF, by the product of two auxiliary heavy quark fields $Q(x^z)\overline{Q}(y^z)$. The non-local operator in the quasi-PDF then becomes the product of two composite operators in Eq. (4).

III. RENORMALIZATION OF EFFECTIVE QCD WITH "HEAVY QUARK" IN DIMENSIONAL REGULARIZATION

Let us consider renormalization of the above effective theory with a "heavy quark". We will show that such a theory can be renormalized perturbatively to all orders in perturbation theory, first in dimensional regularization where power divergences do not exist. The basic argument is the same as the all-order proof of the renormalization for HQET, first presented in Ref. [20].

The standard proof of the renormalization of QCD chooses the covariant gauge $\partial_\mu A^\mu = 0$, implemented in the functional integrals with a gauge parameter ξ. To recover a local theory after gauge fixing, one has to introduce the color-octet ghost fields c^a and $\bar{c}^a$. The resulting theory has a residual global symmetry called BRST symmetry. The Ward-Takahashi identities from this symmetry are a key ingredient for showing the theory is renormalizable to all-orders in perturbation theory in the following sense: After absorbing all the infinities into wave function renormalization factors of quark (Z_1), gluon (Z_3), and ghost fields (Z_c), and the multiplicative renormalization of quark masses (Z_m), gauge coupling (Z_g) and gauge fixing parameter (Z_ξ), Green's functions of the renormalized elementary fields are UV finite.

In HQET where the Lagrangian has a similar form as Eq. (3) except that the vector n^μ is timelike instead of spacelike, it has been shown that to leading order in the heavy quark mass the Green's functions can be renormalized to all-orders in perturbation theory [20], where a BRST invariance of the HQET Lagrangian was used. In HQET, heavy quarks do not couple to heavy antiquarks, hence no heavy quark loops can occur. This implies that Green's functions without external heavy quark legs can be renormalized in the same way as in QCD. Only those involving external heavy quark legs need to be considered separately, which yield a new wave function renormalization constant (Z_Q). These Green's functions include the heavy-quark two-point function, the heavy-quark-gluon vertex and the heavy-quark-ghost vertex with one insertion of composite operators coming from the BRST transformation of the heavy quark field. All of them can be shown to be UV finite with the help of Ward-Takahashi identities under the given finite number of renormalization factors in the HQET Lagrangian [20].

Note that the above arguments are valid independent of whether the heavy quark field in the HQET Lagrangian is defined by a timelike or spacelike vector. Therefore, the statement about renormalizability of the HQET in Ref. [20] shall also be true for our auxiliary heavy quark Lagrangian with $n^\mu = (0,0,0,1)$ in Eq. (3). In the case of light-cone vector, however, new divergences appear due to light-cone singularities.

Local composite operators in this effective theory can be renormalized using the standard approach. In particular, the heavy-light composite operators, such as

$$\overline{j}(x) = \overline{\psi}(x)\Gamma Q(x), \tag{8}$$

acquires a divergent factor Z_j (Note that $\overline{j}(x)$ now carries a spinor index). In analogy to the Green's functions, one can show that the overall UV divergence of the insertion of $\bar{j}(x)$ into Green's functions is subtracted by Z_j to all orders of perturbation theory. Therefore, we can write $\overline{j} = Z_j \overline{j}_R$ where $\overline{j}_R$ is a renormalized operator. The calculation of its anomalous dimension has been worked out in the HQET [21, 22]. For the auxiliary heavy quark field, actually one can explicitly verify that at one-loop in dimensional regularization (with $D = 4 - 2\epsilon$)

$$Z_j = 1 + \frac{\alpha_s}{2\pi\epsilon}\,, \tag{9}$$

which is the same as in HQET.

IV. RENORMALIZATION OF NON-LOCAL OPERATORS IN DIMENSIONAL REGULARIZATION

Now let us consider the renormalization of the non-local operator, such as in Eq. (1), which appears in the LaMET approach to parton physics. In the effective theory, the non-local operator becomes a product of local composite operators,

$$O(z_2, z_1) = \overline{j}(z_2) j(z_1), \tag{10}$$

after we replace the non-local Wilson line by the product of two auxiliary "heavy quark" fields. This operator can be multiplicatively renormalized by

$$O(z_2, z_1) = Z_{\bar{j}} Z_j O_R(z_2, z_1) \tag{11}$$

to all orders in perturbation theory with $Z_j = Z_q^{1/2} Z_Q^{1/2} Z_{V_j}$, where Z_q, Z_Q, Z_{V_j} are the renormalization constant for the light-quark, heavy-quark and vertex, respectively.

Thus the renormalized operator becomes $O_R = Z_{\bar{j}}^{-1} Z_j^{-1} O(z_2, z_1)$. After integration over the "heavy quark" field, we have

$$O_R = Z_{\bar{j}}^{-1} Z_j^{-1} \overline{\psi}(z_2) \Gamma L(z_2, z_1) \psi(z_1), \tag{12}$$

where all fields on the r.h.s. are bare fields. In Ref. [11], it has been shown that the quasi-PDF operator requires the renormalization of virtual diagrams only, which is equivalent to the result in Eq. (12) that we only need renormalization of the heavy-light quark currents for the operator O. In particular, the one-loop renormalization factor for the quasi-PDF

$$Z_O = 1 + \frac{\alpha_s}{\pi\epsilon} \tag{13}$$

is consistent with the anomalous dimension of the heavy-light quark current in Eq. (9). The same is true at the two-loop order [11].

V. RENORMALIZATION OF NON-LOCAL OPERATORS IN LATTICE REGULARIZATION

The HQET Lagrangian in Eq. (3) takes the infinite heavy quark mass limit, therefore does not contain any mass term. The mass correction to the heavy quark can only receive power divergent contributions [23], which are absent in dimensional regularization.

However, in UV cut-off regularizations such as the lattice regularization, when going beyond leading-order perturbation theory, the self-energy of the heavy quark introduces a linear divergence which has to be absorbed into an effective mass counterterm,

$$\delta\mathcal{L}_m = -\delta m \overline{Q} Q \tag{14}$$

with δm of $\mathcal{O}(1/a)$ with a the lattice spacing [23]. Although lattice regularization breaks Lorentz symmetry and introduces operator mixing, it is not a concern for $O(z_2, z_1)$ because there is no lower-dimension operator that can mix with it in lattice QCD. In the HQET framework, the appearance of the linear divergence is easy to understand. Given that the heavy quark is infinitely heavy, it behaves like a static color source, whose energy will have a Coulomb-like form $1/r$, and therefore will diverge linearly if the source is a pointlike particle. In our auxiliary heavy quark Lagrangian, the physical picture as a static color source is lost, whereas the removal of linear divergence can be done essentially in the same way. Since the linear divergence appears only in the self energy of the auxiliary heavy quark, it can be absorbed into the renormalization of the heavy quark mass as in Ref. [23].

Thus, the total heavy-quark Lagrangian now becomes

$$\mathcal{L}_Q = \bar{Q}(in \cdot D - \delta m)Q. \tag{15}$$

Note that the BRST invariance of the Lagrangian requires a dependence of δm on the signature of n [10], which yields a vanishing δm for a lightlike n^μ. In the case of a spacelike n^μ, δm in Eq. (15) is imaginary, we can therefore write $\delta m = i\delta\bar{m}$. After integrating over the heavy fields, the renormalized operator becomes,

$$O_R = Z_{\bar{j}}^{-1} Z_j^{-1} e^{\delta\bar{m}|z_2-z_1|}\overline{\psi}(z_2)\Gamma L(z_2,z_1)\psi(z_1). \quad (16)$$

Thus there is a mass counterterm in the exponent which has a z-dependence. After the mass renormalization, there are at most logarithmic divergences which will be canceled by counterterms from the Lagrangian or from the renormalization of the heavy-light composite operator. The above renormaliztion form has been proposed in a number of papers before [12–16, 18], but not proven.

It is worthwhile to point out that the explicit one-loop calculation [13] in lattice regularization indeed shows that the exponential mass renormalization in the form of Eq. (16) cancels the linear divergence in the quasi-PDF in Ref. [7] after a Fourier transform to momentum space, where

$$\delta\bar{m} = \frac{2\pi\alpha_s}{3a}. \quad (17)$$

The renormalization presented above works the same when $z_2 < z_1$.

VI. CONCLUSION

In this paper, we have shown the non-local operator involving a spacelike Wilson line calculated in LaMET can be renormalized multiplicatively to all-orders in perturbation theory. Apart from the mass counterterm which yields a z-dependent exponential, all the other renormalization constants are independent of z and related to the renormalization of the heavy-light current in the HQET. This provides an important theoretical basis for defining the continuum limit of the quasi-PDFs in lattice QCD, which is necessary for precision calculations of parton physics using the LaMET approach.

Note: When this article is being finalized, we learnt that a similar consideration has been made by the DESY group [24].

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P22–p005

P23：非定域夸克双线性的非微扰重正化

主题归属	非局域算符重整化与 hybrid 方案
底稿	arXiv:1707.07152
arXiv / DOI	1707.07152
完整原始 PDF	6 页；来源 course-cache；SHA-256 71cc84a9fecb...
转排索引	EN 11 页；ZH 10 页（只作书目对照）
备注	索引未记录额外备注。

阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

页码范围：P23-p001 – P23-p006

Nonperturbative renormalization of nonlocal quark bilinears for quasi-PDFs on the lattice using an auxiliary field

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(Dated: January 23, 2018)

Quasi-PDFs provide a path toward an *ab initio* calculation of parton distribution functions (PDFs) using lattice QCD. One of the problems faced in calculations of quasi-PDFs is the renormalization of a nonlocal operator. By introducing an auxiliary field, we can replace the nonlocal operator with a pair of local operators in an extended theory. On the lattice, this is closely related to the static quark theory. In this approach, we show how to understand the pattern of mixing that is allowed by chiral symmetry breaking, and obtain a master formula for renormalizing the nonlocal operator that depends on three parameters. We present an approach for nonperturbatively determining these parameters and use perturbation theory to convert to the $\overline{\text{MS}}$ scheme. Renormalization parameters are obtained for two lattice spacings using Wilson twisted mass fermions and for different discretizations of the Wilson line in the nonlocal operator. Using these parameters we show the effect of renormalization on nucleon matrix elements with pion mass approximately 370 MeV, and compare renormalized results for the two lattice spacings. The renormalized matrix elements are consistent among the different Wilson line discretizations and lattice spacings.

Parton distribution functions (PDFs) describe the distribution of quarks and gluons inside a proton with respect to its longitudinal momentum. They are universal: the same PDFs appear in many different scattering processes, and they are phenomenologically determined from global fits to collider data. Except for their lowest Mellin moments, PDFs have resisted *ab initio* calculation. Lattice QCD can be used to calculate many properties of protons, but it is an inherently Euclidean space method, whereas PDFs are defined in Minkowski space via the matrix elements of operators with quark[1] creation and annihilation separated along the light cone. A possible solution to this problem was proposed by Ji [1]: compute *quasi-PDFs* using matrix elements of equal-time operators

$$\mathcal{O}_\Gamma(x,\xi,n) \equiv \bar\psi(x+\xi n)\Gamma W(x+\xi n,x)\psi(x), \qquad (1)$$

where the ψ and $\bar\psi$ are spatially separated by distance ξ in direction n and connected by a straight Wilson line W. From quasi-PDFs we can obtain PDFs via a matching formula [2, 3], in the limit where the proton's momentum component $p\cdot n$ goes to infinity.

One of the challenges in lattice calculations of quasi-PDFs is renormalization of $\mathcal{O}_\Gamma$, which is a nonlocal operator and is known to be power-law divergent [4, 5]. This operator also appears in the related approach of pseudo-PDFs [6, 7], and similar nonlocal operators are used to study transverse momentum-dependent PDFs [8–10]. The initial lattice studies of quasi-PDFs [3, 11–13] did not include a complete renormalization, but they did build the Wilson line using smeared gauge links, which has been shown in perturbation theory to reduce the power divergence [4].

Renormalization of $\mathcal{O}_\Gamma$ was studied in one-loop lattice perturbation theory in Ref. [14], where it was found that chiral symmetry breaking allows $\mathcal{O}_\Gamma$ to mix with $\mathcal{O}_{\{\not n,\Gamma\}}$. Numerical evidence for this mixing was also found in Ref. [10]. The study of nonperturbative renormalization was pioneered in Refs. [15, 16], which used the Rome-Southampton method [17] to obtain a complex ξ-dependent renormalization factor (or matrix, when there is mixing) $Z(\xi)$ in the RI-MOM scheme. This was then supplemented by a perturbative conversion [14] to the $\overline{\text{MS}}$ scheme used in phenomenology. The problems with this method are that a whole function, rather than a handful of parameters, must be determined, and that conversion at large ξ may occur outside the regime where perturbation theory is valid. Since the intermediate scheme fixes more than the minimal number of parameters, this means that nonperturbative information in correlation functions is lost, only to be recovered perturbatively in the conversion to $\overline{\text{MS}}$.

In this work, we study the use of an auxiliary scalar, color triplet field $\zeta(\xi)$ defined only on the line $x+\xi n$ to simplify the renormalization of $\mathcal{O}_\Gamma$. In this approach, we replace correlation functions in QCD involving $\mathcal{O}_\Gamma$ with correlation functions in the extended theory QCD+ζ involving the local color singlet bilinear $\phi \equiv \bar\zeta\psi$. This approach has been used long ago in the continuum [18, 19].

On the lattice, for now we restrict n to point along one of the axes, $n=\pm\hat\mu$, and use the action

$$S_\zeta = a\sum_\xi \frac{1}{1+am_0}\bar\zeta(x+\xi n)\left[\nabla_n + m_0\right]\zeta(x+\xi n),$$
$$\nabla_n \equiv \begin{cases} n\cdot\nabla^* = \nabla^*_\mu & \text{if } n=+\hat\mu \\ n\cdot\nabla = -\nabla_\mu & \text{if } n=-\hat\mu, \end{cases} \qquad (2)$$

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[1] Here we focus on quark and antiquark PDFs rather than gluon PDFs.

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where ∇ and ∇^* are the forward and backward lattice covariant derivatives and a is the lattice spacing. This yields the bare propagator in fixed gauge background

$$\left\langle \zeta(x+\xi n)\bar{\zeta}(x)\right\rangle_\zeta = \theta(\xi)e^{-m\xi}W(x+\xi n, x), \qquad (3)$$

where $m = a^{-1}\log(1+am_0)$ and W is the simple product of lattice gauge links connecting x and $x+\xi n$. Smeared gauge links can be used to construct W by using the same gauge links to define ∇_n. The mass term can not be forbidden by any symmetry, and corresponds to an $O(a^{-1})$ counterterm. Using this propagator, we obtain for $m=0$ and $\xi > 0$ the relation

$$\mathcal{O}_\Gamma(x,\xi,n) = \left\langle \bar{\phi}(x+\xi n)\Gamma\phi(x)\right\rangle_\zeta . \qquad (4)$$

To obtain $\mathcal{O}_\Gamma$ for $\xi < 0$, we reverse the direction in which ζ propagates, and use $\mathcal{O}_\Gamma(x,\xi,n) = \mathcal{O}_\Gamma(x,-\xi,-n)$.

In addition to determining the counterterm m_0, we must also renormalize the local composite operator ϕ. The lattice quark action may break chiral symmetry, in which case ϕ can mix with $\not{n}\phi$. We thus obtain the renormalization pattern

$$\phi_R = Z_\phi\left(\phi + r_{\rm mix}\not{n}\phi\right), \quad \bar{\phi}_R = Z_\phi\left(\bar{\phi} + r_{\rm mix}\bar{\phi}\not{n}\right). \qquad (5)$$

We can also use projectors to form operators $\phi^\pm \equiv \frac{1}{2}(1 \pm \not{n})\phi$ that renormalize diagonally with $Z_\phi^\pm \equiv Z_\phi(1 \pm r_{\rm mix})$. This pattern leads to the form of the renormalized $\mathcal{O}_\Gamma$, for $\xi \neq 0$:

$$\begin{aligned} \mathcal{O}_\Gamma^R(x,\xi,n) &= Z_\phi^2 e^{-m|\xi|}\mathcal{O}_{\Gamma'}(x,\xi,n), \\ \Gamma' &= \Gamma + r_{\rm mix}\,\mathrm{sgn}(\xi)\{\not{n},\Gamma\} + r_{\rm mix}^2\not{n}\Gamma\not{n}. \end{aligned} \qquad (6)$$

$\mathcal{O}_\Gamma$ can therefore be renormalized by determining three parameters: the linearly divergent m, the log-divergent Z_ϕ, and the finite $r_{\rm mix}$. Since $r_{\rm mix}$ is $O(g^2)$, this is the same pattern at one-loop order as in Ref. [14]. At $\xi = 0$, $\mathcal{O}_\Gamma$ is a local quark bilinear with different divergence structure, and should have a separate renormalization factor [20], which can be computed in the usual way; we use results from Ref. [21]. We also note that since the local operator ϕ is not flavor singlet, there is no mixing between quark and gluon quasi-PDFs even when $\mathcal{O}_\Gamma$ is flavor singlet. In the latter case mixing will occur in the matching to PDFs.

If we choose $n = \hat{t}$ and give ζ spin degrees of freedom (which do not couple in the action), then S_ζ becomes the action for a static quark on the lattice [22, 23]. In particular, ϕ is related to the static-light bilinears and we find relations between renormalization factors: $Z_V^{\rm stat} = Z_\phi^+$ and $Z_A^{\rm stat} = Z_\phi^-$. The static quark theory also tells us how to remove $O(a)$ lattice artifacts [24], which are present even if chiral symmetry is preserved on the lattice. In the continuum, the relation between $\mathcal{O}_\Gamma$ and the static quark theory was previously discussed in Ref. [25].

We determine the renormalization parameters nonperturbatively using a variant of the Rome-Southampton method. In Landau gauge on $N_f = 4$ twisted mass lattice ensembles [21], we compute the position-space ζ propagator,

$$S_\zeta(\xi) \equiv \left\langle \zeta(x+\xi n)\bar{\zeta}(x)\right\rangle_{{\rm QCD}+\zeta} = \left\langle W(x+\xi n, x)\right\rangle_{\rm QCD}, \qquad (7)$$

the momentum-space quark propagator $S_\psi(p)$, and the mixed-space Green's function for $\phi^\pm$:

$$G^\pm(\xi,p) \equiv \int d^4x\, e^{ip\cdot x}\left\langle \zeta(\xi n)\phi^\pm(0)\bar{\psi}(x)\right\rangle_{{\rm QCD}+\zeta}. \qquad (8)$$

The renormalization parameters, as well as the ζ and ψ field renormalizations can be determined by the conditions

$$-\frac{d}{d\xi}\log \mathrm{Tr}\, S_\zeta(\xi)\Big|_{\xi=\xi_0} + m = 0, \qquad (9)$$

$$\left[\frac{Z_\zeta}{3}\mathrm{Tr}\, S_\zeta(\xi_0)\right]^2 = \frac{Z_\zeta}{3}\mathrm{Tr}\, S_\zeta(2\xi_0), \qquad (10)$$

$$\frac{1}{6}\frac{Z_\phi^\pm}{\sqrt{Z_\zeta Z_\psi}}\Re\,\mathrm{Tr}\left[S_\zeta^{-1}(\xi_0)G^\pm(\xi_0,p_0)S_\psi^{-1}(p_0)\right] = 1, \qquad (11)$$

and the RI′-MOM/RI-SMOM condition for S_ψ [17, 26]. Eq. (9) is sensitive to m, whereas the others are constructed to eliminate dependence on it. These conditions define a two-parameter family of renormalization schemes at scale $\mu^2 = p_0^2$, that depend on the dimensionless quantities $y \equiv |p_0|\xi_0$ and $z \equiv p_0 \cdot n/|p_0|$. We call this family of schemes RI-xMOM. Restricting to the kinematics $p_0 \propto n$ (i.e. $z = 1$), we have computed the conversion to the $\overline{\rm MS}$ scheme at one-loop order, using dimensionally regularized perturbation theory [27]. We obtain in Landau gauge:

$$\begin{aligned} C_\phi &\equiv \frac{Z_\phi^{\overline{\rm MS}}(\mu)}{Z_\phi^{\text{RI-xMOM}}(\mu,y,z=1)} \\ &= 1 + \frac{\alpha_s C_F}{8\pi}\Bigg[6\log\frac{y}{4} + 6\gamma_E - 8\log 2 + 7 \\ &\qquad - \cos y - \left(8\cos\frac{y}{2} - y\sin\frac{y}{2}\right)\mathrm{Ci}\left(\frac{y}{2}\right) \\ &\qquad + 8\,\mathrm{Ci}(y)\Bigg] + O(\alpha_s^2), \end{aligned} \qquad (12)$$

where γ_E is the Euler-Mascheroni constant and Ci is the cosine integral function, $\mathrm{Ci}(z) \equiv -\int_z^\infty \frac{\cos(t)}{t}dt$. For converting m to the $\overline{\rm MS}$ scheme, we use the three-loop results for the static quark propagator from Refs. [28, 29].

In Fig. 1, we show the quantity[2] in Eq. (9), for two different lattice spacings: $a = 0.082$ fm ($\beta = 1.95$) and

[2] This is the "effective energy" of the auxiliary field propagator. It was previously studied in Ref. [30], where it was denoted $Y_{\rm line}$.

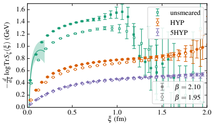

FIG. 1. Effective energy of the bare auxiliary field propagator, for two lattice spacings and three different link discretizations. Solid symbols show the finer lattice spacing and open symbols show the coarser one. The curve shows the three-loop perturbative result, shifted vertically by $-m$ to match it to the unsmeared data on the finer lattice spacing. Its error band indicates the size of the $O(\alpha_s^3)$ contribution.

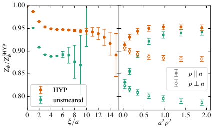

FIG. 2. Z_ϕ for $\beta = 2.10$, relative to the 5HYP case. Left: versus ξ, for $p \parallel n$ and $a^2p^2 \approx 0.35$. Right: versus p^2, for $\xi = 6a$ (unsmeared) and $\xi = 10a$ (HYP). For each link discretization, we take the average of the two values at the smallest p^2.

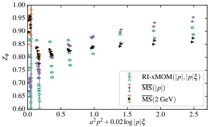

FIG. 3. Z_ϕ for $\beta = 2.10$, using unsmeared gauge links. Data are shown for a range of p^2 and $y \equiv |p|\xi$; the horizontal axis is a^2p^2, with a small displacement for different y at the same p^2. The green open squares are given in our family of schemes, the blue filled diamonds show the result from conversion to $\overline{\mathrm{MS}}$ at scale $|p|$ using Eq. (12), and the orange filled triangles with black outlines show the $\overline{\mathrm{MS}}$ results evolved to the scale 2 GeV, using the two-loop anomalous dimension of the static-light current [28, 33, 34].

$a = 0.064$ fm ($\beta = 2.10$). This is renormalized by adding m. Without smearing there is a significant difference between the two lattice spacings due to the linear divergence, but this is greatly reduced by applying one or five steps of HYP smearing [31], which also reduces the statistical uncertainty at large ξ. At small ξ/a, smearing distorts the shape and therefore we choose to impose our condition at $\xi_0 \approx 0.6$ fm, where the shapes are similar. We then convert to the $\overline{\mathrm{MS}}$ scheme at short distance by matching the result using unsmeared links on the fine ensemble to the perturbative result.

From Eq. (11) an estimator for r_{mix} can be isolated. In our data we see indications that this suffers from significant $O(a)$ lattice artifacts. These could be remedied through Symanzik improvement [24], but instead we choose to focus on the helicity quasi-PDF, which is unaffected by mixing[3] and depends only on r^2_{mix}. We find that the latter is not much greater than 1% and can be neglected at the current level of precision.

The remaining factor Z_ϕ depends on the kinematics that define our scheme, but the ratio between Z_ϕ for different discretizations is scheme independent. We evaluate this ratio in the plateau region at large ξ and at small p (Fig. 2) to reduce lattice artifacts. Finally, we determine Z_ϕ using unsmeared gauge links, converting to the $\overline{\mathrm{MS}}$ scheme and evolving to the scale 2 GeV, as shown in Fig. 3. We find that the one-loop conversion factor is effective at removing much of the dependence on the scheme parameter $|p|\xi$, and the two-loop evolution removes most of the dependence on the scale $|p|$.

We apply these renormalization parameters to nucleon matrix elements computed on one $N_f = 2+1+1$ twisted mass ensemble at each lattice spacing. The physical parameters are matched on the two ensembles: $m_\pi \approx 370$ MeV and $p \cdot n \approx 1.85$ GeV. The coarser ensemble was previously used in Refs. [3, 13] and our methodology is similar to Ref. [13], including the use of momentum smearing [35] in the nucleon interpolating operator to obtain a good signal at large momentum.

Figure 4 shows the effect of renormalization on the isovector helicity matrix element Δh_{u-d},

$$\langle p, \lambda' | \mathcal{O}_{\not{n}\gamma_5} | p, \lambda \rangle \equiv \bar{u}(p,\lambda')\not{n}\gamma_5 u(p,\lambda) \Delta h(\xi, p\cdot n), \quad (13)$$

computed on the fine ensemble, for different link dis-

[3] Following the same logic used to derive automatic $O(a)$ improvement [32], it can also be argued that for twisted mass lattice QCD at maximal twist, the contribution from the mixing operator to nucleon matrix elements will vanish at $O(a)$.

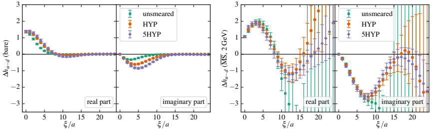

FIG. 4. Matrix element for the helicity quasi-PDF versus ξ on the $\beta = 2.10$ ensemble, for three different link discretizations, bare (left) and renormalized (right). Only $\xi \geq 0$ is shown, since the real part is even in ξ and the imaginary part is odd.

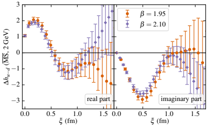

FIG. 5. Renormalized matrix element for the helicity quasi-PDF versus ξ on the two ensembles, using five steps of HYP smearing.

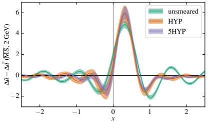

FIG. 6. Isovector helicity quasi-PDF on the $\beta = 2.10$ ensemble, for three different link discretizations, computed from renormalized matrix elements.

cretizations. Without renormalization there is a significant disagreement between the different link types, and renormalization brings them into good agreement. In the renormalized matrix elements, we also see the benefit of smearing: without it, the statistical errors grow rapidly at large ξ and there is no useful signal for $\xi \gtrsim 10a$. With smearing, we are able to see that the matrix elements return toward zero at large ξ. Five steps of HYP smearing also yields more precise data than one step, at large ξ. In Fig. 5, we compare the renormalized data with five steps of HYP smearing on the two ensembles. They are in excellent agreement, which suggests that the linear divergence is under control and discretization effects are not large.

Finally, we examine the helicity quasi-PDF, which is given by a Fourier transform of the matrix element:

$$\Delta\tilde{q}(x, p\cdot n) \equiv \frac{p\cdot n}{2\pi} \int d\xi\, e^{-ix\xi p\cdot n} \Delta h_q(\xi, p\cdot n). \qquad (14)$$

Without renormalization, unsmeared links lead to a much broader distribution, as shown in Fig. 4 of Ref. [3]. We show our renormalized results on the fine ensemble in Fig. 6. Because the data become noisy at large ξ, we restrict the integral to $|\xi| \leq 16a$ ($|\xi| \leq 10a$ for the unsmeared case). When this restriction removes part of the signal, it leads to oscillations, which are clearly visible in the unsmeared case. In future studies improved results could be obtained by replacing the hard cutoff with a model for the large-ξ behavior of the matrix elements, or by applying one of the methods proposed in Refs. [36, 37] for suppressing the contributions at large ξ. Ignoring the oscillations, we see that renormalization brings the data with different link discretizations into reasonably good agreement.

In this work, we have shown that the nonlocal problem of renormalizing lattice quasi-PDFs can be turned into a local problem by introducing an auxiliary field. The auxiliary field approach can also be applied to operators with staple-shaped gauge connection used for lattice studies of transverse momentum-dependent PDFs [8–10], where the mixing pattern will be different. Because this approach is closely connected with the static quark theory, it is possible to make use of existing results from that theory such as the three-loop continuum calculation in Ref. [28]. We have contributed to the evidence that link smearing is a very beneficial technique for these calculations, as it leads to greatly reduced uncertainty at large ξ; this was also explained some time ago in the static quark theory [38]. There are several steps from the results shown here to a full calculation of PDFs: matching from quasi-PDFs to PDFs, control over the $p \cdot n \to \infty$ limit, use of a physical pion mass, as well as control over finite-volume and excited-state effects.

Note: after one of us presented a preliminary version of this work at a conference [39], we were made aware of an independent parallel effort based on the auxiliary field approach [40]. We also note another very recent article discussing renormalizability of quasi-PDFs, without using the auxiliary field approach [41].

We thank our colleagues in ETMC for a pleasant collaboration. Gauge fixing was performed using Fourier-accelerated conjugate gradient [42], implemented in GLU [43]. Calculations were performed using the Grid library [44] and the DD-αAMG solver [45] with twisted mass support [46]. The authors gratefully acknowledge the computing time granted by the John von Neumann Institute for Computing (NIC) and provided on the supercomputer JURECA [47] at Jülich Supercomputing Centre (JSC).

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P23–p006

P24：准 PDF 完整非微扰重正化方案

主题归属	非局域算符重整化与 hybrid 方案
底稿	arXiv:1706.00265
arXiv / DOI	1706.00265
完整原始 PDF	24 页；来源 PyQCD；SHA-256 aa8daa6632b2...
转排索引	EN 22 页；ZH 19 页（只作书目对照）
备注	索引未记录额外备注。

阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

页码范围：P24-p001 – P24-p024

A complete non-perturbative renormalization prescription for quasi-PDFs

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Abstract

In this work we present, for the first time, the non-perturbative renormalization for the unpolarized, helicity and transversity quasi-PDFs, in an RI′ scheme. The proposed prescription addresses simultaneously all aspects of renormalization: logarithmic divergences, finite renormalization as well as the linear divergence which is present in the matrix elements of fermion operators with Wilson lines. Furthermore, for the case of the unpolarized quasi-PDF, we describe how to eliminate the unwanted mixing with the twist-3 scalar operator.

We utilize perturbation theory for the one-loop conversion factor that brings the renormalization functions to the $\overline{\text{MS}}$-scheme at a scale of 2 GeV. We also explain how to improve the estimates on the renormalization functions by eliminating lattice artifacts. The latter can be computed in one-loop perturbation theory and to all orders in the lattice spacing.

We apply the methodology for the renormalization to an ensemble of twisted mass fermions with N_f=2+1+1 dynamical quarks, and a pion mass of around 375 MeV.

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1 Introduction

Parton distribution functions (PDFs) describe the inner dynamics of partons inside a hadron [1]. They have a non-pertubative nature and, thus, they can not be computed in perturbation theory. Lattice QCD is an ideal formulation to study the PDFs from first principles, in large scale simulations. However, PDFs are usually defined on the light cone, which poses a problem for the standard Euclidean formulation. Hence, hadron structure calculations in lattice QCD are related to other quantities that are accessible in a Euclidean spacetime. This led to a long history of investigations of Mellin moments of PDFs and nucleon form factors (see [2, 3, 4, 5, 6] for recent reviews). In practice, there are severe limitations in the reconstruction of the PDFs mainly due to the small signal-to-noise ratio for the high moments. In addition, there are inevitable problems with power divergent mixings with lower dimensional operators. Therefore, the task of reconstructing the PDFs from their moments is practically unfeasible.

Ab initio evaluations of PDFs are of high importance as they would be a stringent test of non-perturbative aspects of QCD. The fact that a calculation of the PDFs from first principles is missing is a pressing problem that prevents a deeper understanding of the nucleon structure. Our current knowledge on the PDFs relies on phenomenological fits from experimental data using perturbation theory. These parameterized PDFs serve, for example, as input for the computation of cross sections used for presently running colliders, most notably the LHC, and also to plan future collider experiments which are themselves tests of the Standard Model. The parameterizations are, however, not without ambiguities [7]. In addition, there are kinematical regions that are not experimentally accessible. The large Bjorken x region is one of them, with large uncertainties most dramatically seen in the down quark distributions [8, 9]. The transversity PDF is yet another example of a PDF that is only poorly constrained by phenomenology.

A pioneering method for a direct computation has been suggested by X. Ji [10]. In this approach, instead of matrix elements defined on the light cone, one calculates matrix elements of fermion operators including a finite-length Wilson line, and whose Fourier transform defines the so-called quasi-PDFs. This is achieved by taking the Wilson line in a purely spatial direction, conventionally chosen to be the z-direction, instead of the $+$-direction on the light cone. In terms of hadron kinematics, the nucleon momentum is usually taken along this spatial direction. At large, but finite momenta, the quasi-PDFs can then be related to the light-front PDFs via a matching procedure [11, 12]. Ji's approach has already been tested in Refs. [13, 14, 15, 16, 17] and all these results are promising, i.e. they give a correct shape of the PDFs after the matching procedure. Certain properties of quasi-PDFs, like the nucleon mass dependence and target mass effects, have also been analyzed via their relation with transverse momentum dependent distribution functions (TMDs) [18, 19]. Note also the appearance of a related approach, pseudo-PDFs, which is a different generalization of light-cone PDFs to finite nucleon momenta [20, 21]. Refs. [22, 23] discussed the role of the Euclidean signature in the computation of quasi-PDFs, as compared to light-front PDFs, and the latter reference proved that matrix elements obtained from Euclidean lattice QCD are identical to those obtained using the LSZ reduction formula in Minkowski space. Nevertheless, the matrix elements of quasi-PDFs contain divergences that need to be eliminated via renormalization in order to obtain meaningful results that can be compared to the physical PDFs.

To date, all works on the quasi-PDFs only considered the bare matrix elements, as the renormalization process is highly non-trivial and was not addressed until recently. In particular, new complications arise in the renormalization of the Wilson line operators, compared to the local operators. For one, in addition to the logarithmic divergences there is a linear divergence [24] with respect to the lattice regulator, a, that prohibits one to take the continuum limit prior to its elimination. To one-loop level in perturbation theory the divergence is manifesting itself as a linear divergence, computed in Ref. [25] for a variety of fermion and gluon actions. However, it is of utmost importance to extract the power divergence non-perturbatively, which is one of the goals of this work. Another feature of these operators that brings in new complications is the fact that certain choices of the Dirac structure exhibit mixing [25].

To show that these matrix elements can be renormalized, in particular that the linear divergence associated with the Wilson line can be eliminated, is of paramount importance. Without renormalization, the whole quasi-PDF strategy is incomplete and unable to provide any useful information to the theoretical and experimental community. Some suggestions for the elimination of the linear divergence via the static potential were proposed in Refs. [26, 27]. In Ref. [12] a one loop calculation of the linear divergence has been made, and that motivated the definition of an improved quasi-PDF that is free of power divergences. One has, nevertheless, to show that such a procedure can be done non-perturbatively. Another method to extract the coefficient of the linear divergence using the nucleon matrix elements of the quasi-PDFs was also presented in Ref. [25]. An alternative technique to suppress the linear divergence was discussed in Ref. [28] utilizing the gradient flow. Very recently, two papers discussed the employment of the auxiliary field formalism for the renormalization of quasi-PDFs [29, 30], where the Wilson line is replaced by a Green's function of the introduced auxiliary field. The renormalizability of quasi-PDFs to all orders in perturbation theory was addressed in Ref. [31].

In this paper we propose, for the first time[1], a concrete renormalization method of the quasi-PDFs in a fully non-perturbative manner. We provide the prescription of the method and show examples of the renormalized matrix elements. We also discuss the elimination of the mixing between the unpolarized quasi-PDF and the twist-3 scalar operator. We employ the RI$'$ renormalization scheme [33] and we convert the results to the $\overline{\text{MS}}$ scheme at a reference scale, $\bar{\mu}$=2 GeV, using the one-loop conversion factor computed in Ref. [25]. As a test case, we focus on the helicity quasi-PDF to demonstrate results, as it is free of mixing.

The outline of the paper is as follows: In Section 2 we provide the theoretical setup related to the nucleon matrix elements and the renormalization prescription in the presence and absence of mixing, as well as the data on the conversion factor computed perturbatively. Section 3 includes results on the renormalization functions, a discussion on the systematic uncertainties in the Z-factors, renormalized matrix elements for the helicity case, as well as results of the matching to light-front PDFs. Finally we conclude and give our future directions.

2 Theoretical setup

In this section we briefly introduce the nucleon matrix elements for the quasi-PDFs that we aim to renormalize. We also explain the renormalization prescription for the three types of PDFs: unpolarized, helicity and transversity. For details on the computation of the nucleon matrix elements, we refer to Refs. [15, 17].

[1]After the submission of our work, the proposed renormalization programme was also applied in Ref. [32].

2.1 Nucleon matrix elements

We consider matrix elements of non-local fermion operators that contain a straight Wilson line, denoted by $h_\Gamma(P_3, z)$. The variable z is the length of the Wilson line, and P_3 is the nucleon momentum, which is taken in the same direction as the Wilson line. The quasi-PDFs can be computed from the Fourier transform of the following local matrix elements:

$$h_\Gamma(P_3, z) = \langle N|\bar{\psi}(0, z)\,\Gamma\, W_3(z)\psi(0, 0)|N\rangle, \tag{1}$$

where $|N\rangle$ is a nucleon state with spatial momentum $\vec{P}$=$(0, 0, P_3)$ along the 3-direction and $W_3(z)$ is a Wilson line of length z in the same direction. Γ denotes the Dirac structure of the operator insertion, which is γ_μ (unpolarized), $\gamma_\mu \cdot \gamma_5$ (helicity), $\sigma_{\mu\nu}$ (transversity). In the works appearing in the literature γ_μ is taken along the Wilson line. In principle, one may choose γ_μ orthogonal to the direction of the Wilson line. In this case, the unpolarized operator is free of mixing, while the helicity and transverity do mix. For example, choosing the γ-matrix in the temporal direction is important for a faster convergence to the physical PDFs, as discussed in Ref. [19]. Our recent work [25] indicates that $\vec{P}_3 \perp \vec{z}$ $(\vec{P}_3 \| \vec{z})$ is ideal for the unpolarized (helicity and transversity) case.

To calculate the bare matrix elements, we use the setup of Ref. [17]. We consider one ensemble of dynamical N_f=2+1+1 twisted mass fermions produced by ETMC [34], with volume $32^3{\times}64$, lattice spacing $a{\approx}0.082\,\mathrm{fm}$ [35] and a bare twisted mass of $a\mu$=0.0055, which corresponds to a pion mass of around 375 MeV. We performed our calculations on 1000 gauge configurations with 15 forward propagators and 2 stochastic propagators, i.e. 30000 measurements in total. We will present results for momentum $P_3 = \frac{6\pi}{L}$, which is around 1.4 GeV in physical units. Gaussian smearing has been employed on the nucleon interpolating fields in the calculation of the matrix elements [17].

2.2 Renormalization scheme

Here we discuss a fully non-perturbative renormalization prescription that will remove all divergences inherited in the matrix elements of the quasi-PDFs, as well as the mixing, as indicated in the perturbative analysis of Ref. [25]. In a nutshell, the proposed renormalization program:

1 removes the linear divergence that resums into a multiplicative exponential factor, $e^{-\delta m|z|/a+c|z|}$. The coefficient δm represents the strength of the divergence and is expected to be operator independent, as it is related only to the Wilson line. c is an arbitrary scale [36] that can be fixed by such a renormalization prescription;

2 takes away the logarithmic divergence with respect to the regulator, $\log(a\bar{\mu}_0)$, where $\bar{\mu}_0$ is the RI′ renormalization scale;

3 applies the necessary finite renormalization related to the lattice regularization;

4 eliminates the mixing that appears in the unpolarized operator, as the bare matrix element is a linear combination of the unpolarized quasi-PDF and the twist-3 scalar operator. The two may be disentangled by the construction of a 2×2 mixing matrix.

We adopt a renormalization scheme which is applicable non-perturbatively, that is, the RI′ scheme [33]. We compute vertex functions of the operators under study, between external quark states, with the setup being in momentum space, and the operator defined as:

$$\mathcal{O}_\Gamma = \overline{\psi}(x)\,\Gamma\,\mathcal{P}\,e^{i\,g\int_0^z A(\zeta)d\zeta}\,\psi(x + z\hat{\mu})\,, \tag{2}$$

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where $\Gamma = \gamma_\mu, \gamma_\mu \cdot \gamma_5, \sigma_{\mu\nu}$ ($\nu \neq \mu$). The path ordering of the exponential appearing in the above expression becomes, on the lattice, a series of path ordered gauge links. The renormalization functions (Z-factors) depend on the length of the Wilson line and, thus, we perform a separate calculation for each value of z. Typically, z goes up to half of the spatial extent of the lattice.

The renormalization prescription is along the lines of the program developed for local operators and the construction of the vertex functions is described in Ref. [37]. The difference between the renormalization of the local operators and the Wilson-line operators is the linear divergence that appears in the latter case. However, there is no need to separate this divergence from the multiplicative renormalization and, therefore, the technique described below may successfully extract both contributions at once.

Helicity and transversity quasi-PDFs

We first provide the methodology for a general operator with a Wilson line in the absence of any mixing. This is applicable for the helicity and transversity quasi-PDFs, provided that their Dirac structure is chosen along the Wilson line. The renormalization functions of the Wilson-line operators, $Z_{\mathcal{O}}$, are extracted by imposing the following conditions:

$$Z_q^{-1}\, Z_{\mathcal{O}}(z)\, \frac{1}{12}\mathrm{Tr}\left[\mathcal{V}(p,z)\left(\mathcal{V}^{\mathrm{Born}}(p,z)\right)^{-1}\right]\Big|_{p^2=\bar{\mu}_0^2} = 1\,, \tag{3}$$

where Z_q is the renormalization function of the quark field obtained via

$$Z_q = \frac{1}{12}\mathrm{Tr}\left[(S(p))^{-1}\, S^{\mathrm{Born}}(p)\right]\Big|_{p^2=\bar{\mu}_0^2}. \tag{4}$$

The trace is taken over spin and color indices, and the momentum p entering the vertex function is set to the RI$'$ renormalization scale $\bar{\mu}_0$. In Eq. (3) $\mathcal{V}(p,z)$ is the amputated vertex function of the operator and $\mathcal{V}^{\mathrm{Born}}$ is its tree-level value, i.e. $\mathcal{V}^{\mathrm{Born}}(p,z){=}i\gamma_3\gamma_5\, e^{ipz}$ for the helicity operator. Also, $S(p)$ is the fermion propagator and $S^{\mathrm{Born}}(p)$ is its tree-level. The RI$'$ scale $\bar{\mu}_0$ is chosen such that its z-component is the same as the momentum of the nucleon. Such a choice serves as a suppression of discretization effects, as different classes of spatial components have different discretization effects, and scales of the form $(n_t, 3, 3, 3)$ have small discretization effects [38]. We test both diagonal (democratic) and parallel momenta to the Wilson line (in the spatial direction), that is $a\vec{\mu}_0{=}\frac{2\pi}{L}\,(P_3, P_3, P_3)$ and $a\vec{\mu}_0{=}\frac{2\pi}{L}\,(0, 0, P_3)$, respectively. We will refer to these choices as "diagonal" and "parallel". The latter are expected to have larger lattice artifacts, as the ratio

$$\hat{P} {\equiv} \frac{\sum_\rho \bar{\mu}_{0_\rho}^4}{\left(\sum_\rho \bar{\mu}_{0_\rho}^2\right)^2} \tag{5}$$

is higher than for diagonal momenta. Using renormalization scales leading to a small value for such a ratio has been successful for the local fermion operators [39, 37].

The vertex functions $\mathcal{V}(p)$ contain the same linear divergence as the nucleon matrix elements. This is crucial as it allows the extraction of the exponential together with the multiplicative Z-factor, through the renormalization condition of Eq. (3). Regarding the renormalizability of quasi-PDFs, it was proven to be multiplicative to all orders in Refs. [27, 31]. The authors consider quasi-PDFs defined in coordinate space and prove the multiplicative renormalizability, and the same holds for the definition in Eq. (1). This is due to the fact

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that the former require, in addition, a consistent subtraction scheme to remove all terms divergent in the limit $\xi_z \to 0$. Moreover, the renormalization of bilocal composite operators is also studied in Ref. [27, 31]. Based on the above, $Z_\mathcal{O}$ can be factorized as

$$Z_\mathcal{O}(z) = \overline{Z}_\mathcal{O}\; e^{+\delta m|z|/a - c|z|}\,, \tag{6}$$

where $\overline{Z}_\mathcal{O}$ is the multiplicative Z-factor of the operator and δm is the strength of the linear divergence. The exponential includes a term with an arbitrary scale c that could be of the form $c|z|$ [36]. Note that the exponential comes with a different sign compared to the nucleon matrix element, as $Z_\mathcal{O}$ is related to the inverse of the vertex function,

$$Z_\mathcal{O} = \frac{Z_q}{\frac{1}{12}\mathrm{Tr}\left[\mathcal{V}(p)\left(\mathcal{V}^{\rm Born}(p)\right)^{-1}\right]\Big|_{p=\bar\mu}}\,. \tag{7}$$

Such a construction of the Z-factor justifies the reason why the elimination of the power divergence in the nucleon matrix elements is successful by multiplying with $Z_\mathcal{O}$, provided that it has been calculated on the same ensemble. Consequently, the above renormalization condition handles all the divergences which are present in the matrix element under consideration. Note that in the absence of a Wilson line (z=0), the Z-factors reduce to the ones for local currents, which are free of any power divergence. For example, for the helicity operator, $Z_A(z{=}0)$ is the standard Z-factor of the axial current.

We would like to point out that knowledge of the coefficient δm provides insight on the strength of the power divergence. One can pursue this direction via the static potential [27] or the technique proposed in Ref. [25]. If the linear divergence is extracted, one may apply the matching of Ref. [12], which includes the coefficient δm. However, there is still a necessity to compute the multiplicative Z-factor to cure any logarithmic divergences, apply finite renormalization, as well as fix the arbitrary scale c.

Unpolarized quasi-PDF

The case of the unpolarized quasi-PDF requires special treatment, if the Dirac structure is in the same direction as the Wilson line. As demonstrated in Ref. [25], for such a choice there is a mixing with the twist-3 scalar operator[2], that must be eliminated. This mixing appears in some lattice regularizations (in particular, Wilson and twisted mass fermions) due to the breaking of chiral symmetry, and is found to be finite. We establish notation by using the subscripts S and V for the scalar and vector (unpolarized) operators, respectively. The corresponding operators are:

$$\mathcal{O}_S \;=\; \overline{\psi}(x)\,\hat{1}\,\mathcal{P}\,e^{i\,g\int_0^z A(\zeta)d\zeta}\,\psi(x+z\hat\mu)\,, \tag{8}$$

$$\mathcal{O}_V \;=\; \overline{\psi}(x)\,\gamma_\mu\,\mathcal{P}\,e^{i\,g\int_0^z A(\zeta)d\zeta}\,\psi(x+z\hat\mu)\,, \tag{9}$$

and we represent their nucleon matrix elements as $h_S(P_3,z)$ and $h_V(P_3,z)$. To disentangle the two contributions from their bare matrix elements, one must compute the multiplicative renormalization and mixing coefficients from the following 2×2 matrix:

$$\begin{pmatrix}\mathcal{O}_V^R(P_3,z)\\ \mathcal{O}_S^R(P_3,z)\end{pmatrix} = \hat{Z}(z)\cdot\begin{pmatrix}\mathcal{O}_V(P_3,z)\\ \mathcal{O}_S(P_3,z)\end{pmatrix}, \tag{10}$$

[2]For twisted mass fermions the mixing is between the vector and pseudoscalar currents.

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where $\hat{Z}$ is the matrix of the mixing between the scalar and vector operators, that is

$$\hat{Z}(z) = \begin{pmatrix} Z_{VV}(z) & Z_{VS}(z) \\ Z_{SV}(z) & Z_{SS}(z) \end{pmatrix} . \tag{11}$$

According to the above mixing, the renormalized unpolarized quasi-PDF, $h_V^R(P_3, z)$, is related to the bare scalar and unpolarized via:

$$h_V^R(P_3, z) = Z_{VV}(z)\; h_V(P_3, z) + Z_{VS}(z)\; h_S(P_3, z)\,, \tag{12}$$

where Z_{VV} and Z_{VS} are computed in a certain scheme. In the present work we employ the RI′ scheme and then convert to the $\overline{\rm MS}$ scheme, at an energy scale $\bar{\mu}$. The Z_{ii} factors can be computed following a prescription similar to Eq. (3), that is:

$$Z_q^{-1}\, \hat{Z}(z)\, \hat{\mathcal{V}}(p, z)\Big|_{p=\bar{\mu}} = \hat{1}\,, \tag{13}$$

where the elements of the vertex function matrix $\hat{\mathcal{V}}$ are given by the trace

$$\left(\hat{\mathcal{V}}(z)\right)_{ij} = \frac{1}{12}\mathrm{Tr}\left[\mathcal{V}_i(p,z)\left(\mathcal{V}_j^{\rm Born}(p,z)\right)^{-1}\right], \quad i,j = S, V\,. \tag{14}$$

In the above equation $\mathcal{V}_i^{\rm Born}$ is the tree-level expression of the operator $\mathcal{O}_i$. Thus, all matrix elements of $\hat{Z}$ can be extracted by a set of linear equations, which can be written in the following matrix form:

$$Z_q^{-1} \begin{pmatrix} Z_{VV}(z) & Z_{VS}(z) \\ Z_{SV}(z) & Z_{SS}(z) \end{pmatrix} \cdot \begin{pmatrix} \left(\hat{\mathcal{V}}(z)\right)_{VV} & \left(\hat{\mathcal{V}}(z)\right)_{SV} \\ \left(\hat{\mathcal{V}}(z)\right)_{VS} & \left(\hat{\mathcal{V}}(z)\right)_{SS} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} . \tag{15}$$

As can be seen from Eq. (12), a major component in the renormalization of the unpolarized quasi-PDF is knowledge of the scalar nucleon matrix element $h_S(P_3, z)$. To date, no lattice calculation is available for $h_S(P_3, z)$, as a mixing was not anticipated prior the work of Ref. [25]. Therefore, a proper renormalization of the unpolarized quasi-PDF is still pending. However, the mixing appears to be greatly suppressed in the presence of a clover term in the fermionic action. This is of high importance, as we are currently computing the quasi-PDFs on an ensemble of twisted mass clover improved fermions at the physical pion mass [40, 41].

2.3 Conversion to the $\overline{\rm MS}$-scheme

The fermion operators with a finite-length Wilson line are scheme and scale dependent. As a result, having obtained the Z-factors in the RI′ scheme as depicted in Eqs. (3) and (13) at the RI′ scale $\bar{\mu}_0$, we must convert them to the $\overline{\rm MS}$ scheme at a scale $\bar{\mu}$. This conversion factor has been computed in one-loop continuum perturbation theory in Ref. [25]. In addition, comparison with phenomenological estimates is done typically at $\bar{\mu}$=2 GeV. This defines the value of the $\overline{\rm MS}$ renormalization scale chosen in the expressions for the conversion factors. Alternatively, one can choose $\bar{\mu}$=$\bar{\mu}_0$, and then evolve the scale to 2 GeV, which requires the anomalous dimension of the operator:

$$\gamma_{\mathcal{O}} = -3\frac{g^2\, C_f}{16\pi^2}\,. \tag{16}$$

Note that to one-loop level, the anomalous dimension does not dependent on the Dirac structure of the operator and is the same in the RI′ and $\overline{\text{MS}}$ schemes. The evolution to the $\overline{\text{MS}}$ renormalization scale $\bar{\mu}$=2 GeV can be performed using the intermediate Renormalization Group Invariant scheme (RGI), as employed in Refs. [42, 38]. In the framework of this paper we tested both methods and their difference was found to be negligible.

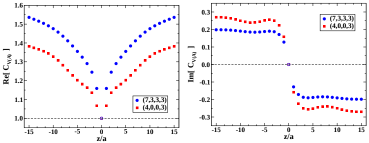

Figure 1: One-loop conversion factor from the RI′ to the $\overline{\text{MS}}$ scheme for $\bar{\mu}$=$\bar{\mu}_0$. In the left (right) plot we show the real (imaginary) part of the conversion factor as a function of the length of the Wilson line in lattice units. Two choices of the RI′ scale have been employed: $a\bar{\mu}_0$=$\frac{2\pi}{32}\,(\frac{7}{2}+\frac{1}{4}, 3, 3, 3)$ (blue circles) and $a\bar{\mu}_0$=$\frac{2\pi}{32}\,(\frac{4}{2}+\frac{1}{4}, 0, 0, 3)$ (red squares). We use the abbreviation (7,3,3,3) and (4,0,0,3) in the legends, respectively.

As discussed in the previous section, we employ two types of the RI′ renormalization scale regarding the spatial direction: one the same as in the nucleon matrix elements, $a\vec{\mu}_0$=$\frac{2\pi}{L}\,(0, 0, P_3)$, (parallel to the Wilson line direction), and one which is diagonal and each direction has a value of P_3, that is $a\vec{\mu}_0$=$\frac{2\pi}{L}\,(P_3, P_3, P_3)$. The conversion factor depends on both the RI′ and $\overline{\text{MS}}$ renormalization scales. While $\bar{\mu}$ is typically fixed to 2 GeV, $\bar{\mu}_0$ can change affecting the numerical values of the conversion factor. Such a case is illustrated in Fig. 1 for the unpolarized and helicity operators, which share the same conversion factor[3]. We focus on the renormalization of the matrix elements with the nucleon boosted by aP_3=$\frac{6\pi}{L}$, and we apply the same to the conversion factor, for the two cases of the renormalization scale. For the specific value of P_3, we choose the temporal direction of $\bar{\mu}_0$ such that the ratio $\hat{P}$ defined above, is as small as possible in order to suppress lattice artifacts. Nevertheless, for the "parallel" case the minimum value for the ratio is $\hat{P}$=0.54 which is already very high. The chosen values for $a\bar{\mu}_0$ are: $\frac{2\pi}{32}\,(\frac{7}{2}+\frac{1}{4}, 3, 3, 3)$ and $\frac{2\pi}{32}\,(\frac{4}{2}+\frac{1}{4}, 0, 0, 3)$ for the "diagonal" and "parallel" case, respectively.

The real and imaginary parts of the conversion factor are plotted as a function of the length of the Wilson line, z. We stress that the dependence of the conversion factor on the renormalization scales and the length of the Wilson line is highly non-trivial and is expressed in terms of integrals of modified Bessel functions. Consequently, the data points shown in Fig. 1 have been computed numerically for the specific scales, at each value of z separately. We observe that the real part of the conversion is an order of magnitude larger

[3]The one-loop calculation does not depend on the prescription which one adopts for extending γ_5 to D dimensions (see, e.g., Refs. [43, 44, 45, 46, 47, 48] for a discussion of four relevant prescriptions and some conversion factors among them).

than the imaginary part. The real part consistently increases for increasing values of z, while the imaginary part almost immediately stabilizes when z becomes non-zero. Also, the conversion factor at z=0 is equal to unity, as it corresponds to the local vector and axial currents. Note that the case z=0 is not extracted from the calculation of Ref. [25] as it is strictly for $z\neq0$: the appearance of contract terms beyond tree level renders the limit $z \to 0$ nonanalytic. On the contrary, the non-perturbative prescription of the previous section can be applied for any value of z, as the calculation is performed on each z separately. The values of Fig. 1 will be used in the following section to bring the RI′ non-perturbative Z-factors to the $\overline{\rm MS}$ scheme. To reliably extract the Z-factors we have extend the calculation including several values of $\bar\mu_0$ as explained in Section 3. The conversion factor was found to have the same qualitative behavior for all values of $\bar\mu_0$.

3 Results

In this section we apply the prescription suggested above and we present our results for the non-perturbative Z-factors both in the RI′ and the $\overline{\rm MS}$ schemes. For demonstration purposes we focus on the data with 5 steps of Hypercubic (HYP) smearing that suppress the power divergence and bring the results closer to renormalized nucleon matrix elements. We focus on the multiplicative renormalization for the helicity quasi-PDF, and only briefly discuss the case of the unpolarized operator.

3.1 RI′ scheme and conversion to the $\overline{\rm MS}$ scheme

As a starting point, we have applied the two values of the RI′ scale $\bar\mu_0$ used in Fig. 1 ("parallel" and "diagonal"). After converting both cases to the $\overline{\rm MS}$ scheme at $\bar\mu$=2 GeV, we can quantify the systematic uncertainties related to lattice artifacts and truncation of the conversion factor, as explained in the next subsection.

In Fig. 2, we show the extracted values for the helicity Z-factor, $Z_{\Delta h}$, that renormalizes the bare matrix element $\Delta h(P_3, z)$. In each plot we overlay the results for the RI′ (open symbols) and the $\overline{\rm MS}$ (filled symbols) schemes, for the real and imaginary part of the Z-factor. We employ the momentum source technique [49, 38] that offers high statistical accuracy with a small number of measurements, typically of $\mathcal{O}(10)$. As can be seen in the plots, the statistical uncertainties are almost invisible. The left (right) plot corresponds to the "parallel" ("diagonal") choices for $\bar\mu_0$. We find that the imaginary part of $Z_{\Delta h}^{\overline{\rm MS}}$ is reduced compared to its counterpart in $Z_{\Delta h}^{\rm RI'}$, and is also rather small for low values of z. This is more pronounced in the right plot of Fig. 2 for "diagonal" $\bar\mu_0$, for which ${\rm Im}\,[Z_{\Delta h}^{\overline{\rm MS}}]$ (blue circles) is smaller than the corresponding data from the "parallel" scale, especially for large values of z. It is worth mentioning that the perturbative Z-factor in dimensional regularization and in the $\overline{\rm MS}$ scheme is real to all orders in perturbation theory, as it is extracted only from the poles. Therefore, it is expected that the imaginary part of the non-perturbative estimates should be highly suppressed. The behavior of the "diagonal" scale is encouraging, as the imaginary part is very close to zero for $|z|$ up to $\sim 10\,a$.

As we have done in previous applications of the RI′ scheme to extract the renormalization functions of ultra-local operators (see, e.g., Ref. [38]), here we also use a range of values for the RI′ renormalization scale, $\bar\mu_0$. This allows us to study the scale dependence of the vertex functions with the external momentum and identify the window in which renormalization factors can be extracted as reliably as possible. Unlike the case of the local currents, the

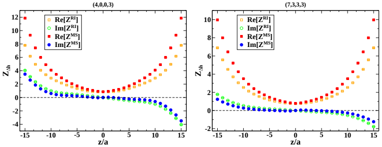

Figure 2: The z-dependent renormalization function for the matrix element $\Delta h(P_3, z)$ with $aP_3 = \frac{6\pi}{L}$. The "parallel" and "diagonal" choices for $\vec{\mu}_0$ are shown in the left and right plots, respectively. Open (filled) symbols correspond to the RI$'$ ($\overline{\mathrm{MS}}$) estimates.

computation of the lattice artifacts to $\mathcal{O}(g^2\, a^\infty)$ is extremely laborious and not available yet. Thus, one has to be careful with the choices for $\bar{\mu}_0$, as we want to avoid non-perturbative contaminations, and satisfy the criterion that $\hat{P}$ (Eq. (5)) is small, preferably below 0.3, to avoid enhanced cut-off effects. We compute the renormalization functions using the values reported in Table 1, from which we choose an optimal range for the fit using the diagonal choices. The parallel $\bar{\mu}_0$ have only been used to explore the systematic uncertainties in Subsection 3.3.

Label	(n_t, n_x, n_y, n_z)	$(a\bar{\mu}_0)^2$	$\bar{\mu}$ (GeV)	$\hat{P}$
		diagonal		
m1	(4,3,3,3)	1.236	2.671	0.261
m2	(5,3,3,3)	1.332	2.773	0.251
m3	(6,3,3,3)	1.448	2.891	0.251
m4	(7,3,3,3)	1.583	3.023	0.261
m5	(8,3,3,3)	1.737	3.167	0.280
m6	(9,3,3,3)	1.911	3.321	0.306
m7	(10,3,3,3)	2.104	3.484	0.339
m8	(11,3,3,3)	2.316	3.656	0.370
		parallel		
m9	(4,0,0,3)	0.542	1.769	0.539
m10	(9,0,0,3)	1.216	2.649	0.592
m11	(11,0,0,3)	1.622	3.097	0.664

Table 1: Values for the RI$'$ scale defined as $a\bar{\mu}_0 = \frac{2\pi}{L}(\frac{n_t}{2} + \frac{\pi}{4}, n_x, n_y, n_z)$, where L is the spatial extent of the lattice. The values are given in lattice and physical units. The last column corresponds to the ratio $\hat{P}$ defined in Eq. (5).

Since we present the renormalized matrix elements for the helicity PDF, we focus on $Z_{\Delta h}$ for this analysis. In Fig. 3, we show $Z^{\overline{\mathrm{MS}}}_{\Delta h}$ for selected values of z as a function of the

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RI′ scale, $(a\,\bar\mu_0)^2$, using the diagonal values labeled by m1−m8. The real (imaginary) part is shown in the left (right) panel. We find a residual dependence on $(a\,\bar\mu_0)^2$, mostly affecting the imaginary part. This is due to the fact that the vertex function depends not only on the magnitude and direction of the renormalization scale, especially the z-direction, as this is parallel to the Wilson line. Upon evolving to the same scale, the Z-factors should not depend on the initial scale if the evolution is known to higher loops in perturbation theory and discretization effects are sufficiently small. The nonzero slope of the plots shown in Fig. 3 is an indication of non-negligible lattice artifacts, as well as, a consequence of the truncation of the conversion factor to one-loop level. However, further investigation is required in order to attempt disentangling the two effects. We address this in Subsection 3.3.

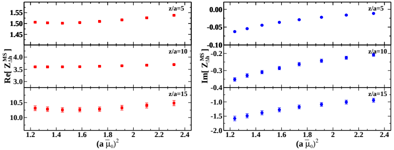

Figure 3: Left: Real part of $Z^{\overline{\rm MS}}_{\Delta h}$ for scales labeled by m1−m8 and z/a=5, 10, 15. Right: Same as left panel for the imaginary part of $Z^{\overline{\rm MS}}_{\Delta h}$.

To remove the residual dependence on $(a\,\bar\mu_0)^2$ we fit $Z^{\overline{\rm MS}}_{\Delta h}$ with the function

$$Z^{\overline{\rm MS}}_{\Delta h} = Z^{\overline{\rm MS}}_{0,\,\Delta h} + Z^{\overline{\rm MS}}_{1,\,\Delta h} \cdot (a\,\bar\mu_0)^2 \tag{17}$$

and we take $Z^{\overline{\rm MS}}_{0,\,\Delta h}$ as our final result. This process is done for each value of the length of the Wilson line z, and repeated for several fit ranges for the diagonal choices m1−m8 given in Table 1. For the extrapolated value $Z^{\overline{\rm MS}}_{0,\,\Delta h}$ at 2 GeV, we choose the one obtained in the range $(a\,\bar\mu_0)^2\,\epsilon\,[1.4,\,2.0]$. By excluding $(a\,\bar\mu_0)^2$ close to 1, we avoid contamination from non-perturbative effects. The choice for the above fit range also excludes the two higher scales that have $\hat{P}{>}0.3$, which may carry sizable lattice artifacts. As an additional check of the quality of the fit, we find that the $\chi^2/d.o.f$ is small for the chosen range. In Fig. 4, we plot the extrapolated value $Z^{\overline{\rm MS}}_{0,\,\Delta h}$ (for 5 steps of HYP smearing), which is applied on the bare nucleon matrix element of the helicity quasi-PDF in Subsection 3.3.

The prescription for obtaining Z_{VV} and Z_{VS} has also been applied on the same ensemble. We find that both the HYP smearing and the choice of "diagonal" scales suppresses the mixing coefficients Z_{VS} and Z_{SV}. In the left panel of Fig. 5, we plot $Z^{\rm RI'}_{VV}$ and $Z^{\rm RI'}_{VS}$ for the scale "(7,3,3,3)" and 5 steps of HYP smearing. The real and imaginary parts of $Z^{\rm RI'}_{VS}$ are of the same magnitude, but at least an order of magnitude smaller than the multiplicative factor $Z^{\rm RI'}_{VV}$. This can be seen from the right panel of Fig. 5, where we plot the ratios ${\rm Re}\,[Z^{\rm RI'}_{VS}]/{\rm Re}\,[Z^{\rm RI'}_{VV}]$ (red squares) and ${\rm Im}\,[Z^{\rm RI'}_{VS}]/{\rm Im}\,[Z^{\rm RI'}_{VV}]$ (blue circles). From the left plot

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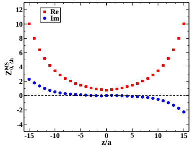

Figure 4: Extrapolated Z-factor for the helicity operator, $Z^{\overline{\mathrm{MS}}}_{0,\,\Delta h}$, for 5 steps of HYP smearing. The employed fit range for the RI′ scale is $(a\,\bar{\mu}_0)^2\,\epsilon\,[1.4{-}2.0]$.

one observes that the imaginary part of both $Z^{\mathrm{RI}'}_{VV}$ and $Z^{\mathrm{RI}'}_{VS}$ is compatible with zero for $|z/a| < 4$. Therefore, the large values of the blue points in the region $|z/a| < 4$ in the right plot are no indication of significant mixing.

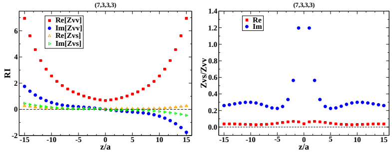

Figure 5: Left: multiplicative ($Z^{\mathrm{RI}'}_{VV}$) and mixing ($Z^{\mathrm{RI}'}_{VS}$) coefficients entering the renormalization of the unpolarized quasi-PDF, in the RI′ scheme for the "(7,3,3,3)" scale. Right: ratio of the real (red squares) and imaginary (blue circles) parts of $Z^{\mathrm{RI}'}_{VS}$ over $Z^{\mathrm{RI}'}_{VV}$.

3.2 Assessment of systematic uncertainties

In the framework of this study we have performed several investigations on the systematic uncertainties related to the truncation of the one-loop conversion factor, as well as, discretization effects. In this subsection we present the main conclusions of this study, and we give estimates of these effects.

Prior converting to the $\overline{\mathrm{MS}}$ scheme, we compute the Z-factors for different values of the RI′ scale $\bar{\mu}_0$. For example, $Z^{\mathrm{RI}'}_{\Delta h}$ shown in the two plots of Fig. 2 on the scales m4 and m9 is different. Upon evolving to the same scale, the extracted values should agree if the evolution

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has converged; typically this happens at two or three loops in perturbation theory. Thus, one should be able to compare the data extracted for $Z^{\overline{\rm MS}}_{\Delta h}$ at 2 GeV for the two scales presented in Fig. 2. The difference between the two, $\Delta Z(\mathrm{m4},\mathrm{m9})$, is an indication of the presence of lattice artifacts (mainly in the "parallel" case), coupled with the truncation of the conversion factor. This is demonstrated in Fig. 6 for both the real and imaginary parts, and it is interesting to see that the difference increases as z becomes larger. Based on this observation, we expect that such an increase of the lattice artifacts is also present for the "diagonal" case, but less severe. Another evidence of the presence of non-negligible systematics is the fact that the imaginary part of $Z^{\overline{\rm MS}}_{\Delta h}$ is nonzero. In Fig. 2, we find a small imaginary part for the "diagonal" scale, which has an increasing trend at large values of z.

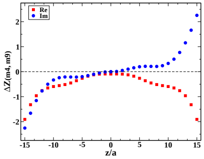

Figure 6: Difference of $Z^{\overline{\rm MS}}_{\Delta h}$ between the "parallel" (m4) and "diagonal" (m9) cases for $\bar{\mu}_0$. The real and imaginary parts are shown with red squares and blue circles, respectively.

Based on the above, we conclude that the systematic uncertainties related to lattice artifacts and the conversion factor must be addressed, in order to extract reliable estimates on the renormalization functions. It is our intention to reduce both effects in the near future, which will eliminate systematic uncertainties propagated to the estimates of the quasi-PDFs. Understanding the uncertainties dominating the large-z region is crucial, as the matrix element for these values also enters in the Fourier transform that yields the quasi-PDF. Preliminary explorations indicate that a likely magnitude of the two-loop contribution might suppress the imaginary part of $Z^{\overline{\rm MS}}_{\Delta h}$. Even though we are currently in no position to accurately quantify these systematics, we will estimate upper bounds. In particular, we try to estimate the effect of each one individually.

Lattice artifacts

The Z-factor for the helicity quasi-PDF at z=0 reduces to the renormalization function of the local axial current Z_A. Z_A is scheme- and scale-independent, and thus, the case z=0 allows us to study the lattice artifacts. Z_A has been already evaluated for this ensemble in Ref. [38]. In the latter calculation several diagonal scales have been employed and, together with a technique for the removal of the lattice artifacts, the extracted value was found to be Z_A=0.7556(5). In the present calculation at z=0 we find a value of Z_A=0.8620(15) using the scale m9 and Z_A=0.7727(2) for the scale m4. This is yet another indication that lattice artifacts are large for the "parallel" renormalization scale and less severe, but not negligible, for the "diagonal" case.

An additional test that may be performed for any value of z is the comparison of $Z^{\overline{\rm MS}}_{\Delta h}$ between scales with different components, but same $(a\,\bar\mu_0)^2$. For this, we utilize the values labeled as m10 and m11, which can be compared to m1 and m4, respectively. Since these pairs are approximately at the same value of $(a\,\bar\mu_0)^2$, we can compare them directly in the RI′ scheme without any evolution[4], and the difference can be interpreted as lattice artifacts. We observe similar behavior for the two pairs and as a demonstration we plot in Fig. 7 the following ratio for each value of z

$$D_{\rm Re\,(Im)}(\bar\mu_0,\bar\mu_0') \equiv \frac{Z^{\rm RI'}_{\rm Re\,(Im)}(\bar\mu_0) - Z^{\rm RI'}_{\rm Re\,(Im)}(\bar\mu_0')}{Z^{\rm RI'}_{\rm Re\,(Im)}(\bar\mu_0)}\,. \tag{18}$$

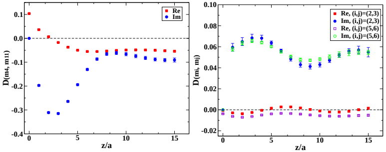

Figure 7: Left: The ratio of Eq. (18) using $(a\,\bar\mu_0)^2$=1.583 and $(a\,\bar\mu_0')^2$=1.622. Right: Similar to the left plot, for the pairs $((a\,\bar\mu_0)^2, (a\,\bar\mu_0')^2)$=(1.332, 1.448), (1.737, 1.911)

In the left panel we show the case where $\bar\mu_0$ and $\bar\mu_0'$ correspond to the values m4 and m11, respectively. The difference from zero can be attributed to lattice artifacts. We find that the Z-factor extracted from m11 deviates from the value extracted from m4 by 5-10% in the real part and up to 30% in the imaginary part. This deviation seems to stabilize after $z/a{\sim}8$ to 5% (10%) for the real (imaginary) part.

Note that the left plot of Fig. 7 gives the estimate of the discretization effects for the "parallel" case compared to the "diagonal" one. It would be interesting to compute $D_{\rm Re\,(Im)}$ between two neighboring "diagonal" momenta in order to understand the change in the artifacts. We choose 2 pairs (m2,m3) and (m5,m6) and we find that both $D_{\rm Re}$(m2, m3) and $D_{\rm Re}$(m5, m6) are less than 1%, while $D_{\rm Im}$(m2, m3) and $D_{\rm Im}$(m5, m6) are of the order of 6%. This is a confirmation that excluding "parallel" momenta in the fit of Eq. (17) reduces significantly the discretization effects.

[4] We confirmed numerically that the conversion factors from m1 to m10, and from m4 to m11 deviates from unity by less than 1‰, so we ignore it.

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Truncation effects

In order to assess quantitatively the influence of the conversion factor that is known to one-loop perturbation theory, we form the ratios

$$R^{\mathrm{RI}'\,(\overline{\mathrm{MS}})}_{\mathrm{Re}\,(\mathrm{Im})}(z,\bar{\mu}_0,\bar{\mu}_0';\bar{\mu}) \equiv \frac{Z^{\mathrm{RI}'\,(\overline{\mathrm{MS}})}_{\mathrm{Re}\,(\mathrm{Im})}(z,\bar{\mu}_0;\bar{\mu})}{Z^{\mathrm{RI}'\,(\overline{\mathrm{MS}})}_{\mathrm{Re}\,(\mathrm{Im})}(z,\bar{\mu}_0';\bar{\mu})} \tag{19}$$

both for the real and imaginary parts of the helicity Z-factor. In Fig. 8, we plot Eq. (19) for the RI' and $\overline{\mathrm{MS}}$ case. These have been extracted at different values of $(a\bar{\mu}_0)^2$ (m1−m8) and evolved perturbatively to the same scale of 2 GeV, using the results of Ref. [25]. The ratio is always taken with respect to the smallest "diagonal" scale, m1. $R^{\mathrm{RI}'}_{\mathrm{Re}\,(\mathrm{Im})}$ depend on the truncation effects in the one-loop evolution to 2 GeV, while $R^{\overline{\mathrm{MS}}}_{\mathrm{Re}\,(\mathrm{Im})}$ is affected by scheme conversion truncation effects. Without contamination from lattice artifacts and truncation effects, Eq. (19) should equal 1 in both schemes and for all values of $(a\bar{\mu}_0)^2$ and z/a. This realization allows for investigation of the truncation effects.

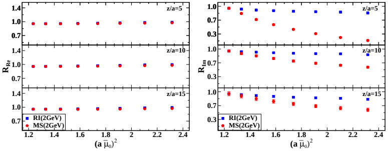

Figure 8: The ratio of Eq. (19) for z/a=5, 10, 15 as a function of the initial RI′ scale. All results have been evolved to 2 GeV and are normalized with the value of the Z-factor at $(a\bar{\mu}_0')^2$=1.236 (m1). Left: real part. Right: imaginary part.

One observes that the imaginary part is more sensitive to the change of the initial RI′ scale, which is given in the x-axis. To demonstrate this, we keep the range of y-axis the same for all plots of Fig. 8. The difference between the ratios of the real part extracted from different RI′ scales $(\bar{\mu}_0)^2$ is consistently small and reaches at most approx. 5% in the RI′ scheme and 3% in the $\overline{\mathrm{MS}}$ scheme, regardless of the Wilson line length, z. As we mentioned above, for z=0 we can compare to the highly reliable value found in Ref. [38]; our current value from the scale m1 is around 2% higher. Hence, we can estimate the typical size of discretization effects to be of order 2-5% in the real part of the Z-factors.

The differences in the $\overline{\mathrm{MS}}$ ratios are combinations of lattice artifacts and the conversion truncation. We observe that the conversion to the $\overline{\mathrm{MS}}$ scheme decreases the differences in the $\overline{\mathrm{MS}}$ scheme to at most 3%. Hence, truncation alone in the real part seems not to have an effect larger than 2%. Since we know that conversion truncation effects are very small for small values of z (in particular, they are zero for z=0, where Z_A is scheme-independent), we can take 0-2% as our estimate. In the end, given the fact that truncation effects seem to be opposite to the influence of lattice artifacts, we estimate that the total effects present in the real part of $Z^{\overline{\mathrm{MS}}}_{\Delta h}$ should not exceed 5%.

The situation is somewhat different in the ratio of the imaginary parts shown in the right panel of Fig. 8. We conclude that the uncertainty from lattice artifacts can be of the order of 10% in the RI′ scheme for intermediate and large z ($|z|/a \geq 5$[5]). From the right panel of Fig. 8, the total uncertainty in $R^{\overline{\mathrm{MS}}}_{\mathrm{Im}}$ can get enhanced to around 40% for large and up to 80% for intermediate Wilson line lengths. In addition, the total uncertainty in the imaginary part of $Z^{\overline{\mathrm{MS}}}_{\Delta h}$ is basically of the order of its magnitude, as it is expected to be real. Note that ignoring the imaginary part does not provide any solution, as they need to come out zero from the computation in order to claim that the computation is fully reliable. Thus, conversion truncation effects become the most major source of uncertainty of the imaginary parts of the Z-factors.

Based on the study presented in this subsection, we present in Table 2 a summary of our quantitative estimates on the systematic uncertainties present in $Z^{\overline{\mathrm{MS}}}_{\Delta h}$. The estimate of the total effect is not a simple sum of isolated effects, but takes into account their different signs discussed above. For the imaginary part, the estimates are for $|z/a| \geq 5$, since $\mathrm{Im}\,[Z^{\overline{\mathrm{MS}}}_{\Delta h}]{\sim}0$ for smaller values of z and the relative effects become meaningless.

We would like to stress that due to the complex multiplication of the Z-factors and the bare matrix elements, the large uncertainty in the imaginary part of the Z-factor implies that also real part of the renormalized matrix elements is affected. Furthermore, the uncertainties that we consider in the Z-factors translate differently to the uncertainties of the renormalized matrix elements for different Wilson line lengths and thus, we postpone the discussion of this influence to the next subsection.

Effect	$\mathrm{Re}\,[Z^{\overline{\mathrm{MS}}}_{\Delta h}]$	$\mathrm{Im}\,[Z^{\overline{\mathrm{MS}}}_{\Delta h}]$
Lattice artifacts	2-5%	$\lesssim 10\%$
Evolution truncation	1-2%	1-2%
Conversion truncation	$\lesssim 2\%$	$\lesssim 100\%$
Total	3-5%	$\lesssim 100\%$

Table 2: Quantitative estimates of systematic uncertainties in the real and imaginary parts of $Z^{\overline{\mathrm{MS}}}_{\Delta h}$.

3.3 Renormalized results

Once the Z-factors are obtained for the unpolarized, helicity and transversity quasi-PDFs, one may proceed with the application of the renormalization in the nucleon matrix elements. Here we mostly focus on the helicity case, as the renormalization is multiplicative. The case of the unpolarized quasi-PDF is briefly mentioned in the end of the subsection.

[5]For smaller values of z, the imaginary part is small in absolute terms and Eq. (19) becomes meaningless, i.e. even small absolute changes of the imaginary part of $Z^{\overline{\mathrm{MS}}}_{\Delta h}$ can imply large changes in R.

In Fig. 9, we show the renormalized helicity nucleon matrix elements, and compare them with the bare ones. This is a straightforward procedure as there is no mixing for the axial operator and, therefore, the renormalization is only multiplicative:

$$\Delta h^{\overline{\rm MS}}(z) = Z^{\overline{\rm MS}}_{\Delta h}(z) \cdot \Delta h^{bare}(z)\ . \tag{20}$$

The above formula involves complex quantities, and thus, $\Delta h^{\overline{\rm MS}}(z)$ is a mixture of the real and imaginary part of the bare matrix element. However, each value of z is renormalized independently. We use $Z^{\overline{\rm MS}}_{\Delta h}$ from the extrapolation of Eq. (17) in the range $(a\,\bar\mu_0)^2\,\epsilon\,[1.4,\,2.0]$, as shown in Fig. 4. One can see in Fig. 9 that for small values of z there is a slight suppression of the renormalized real part with respect to the bare one (${\rm Re}\,[Z^{\overline{\rm MS}}_{\Delta h} < 1]$). ${\rm Re}\,[\Delta h^{\overline{\rm MS}}]$ is compatible with zero for $|z/a|$>8, but with increased statistical uncertainties. The effect of the renormalization on the imaginary part of the matrix element is profound in the large z region, where we observe an amplification of its value and a shift of the maximum to larger z, as compared to the bare one.

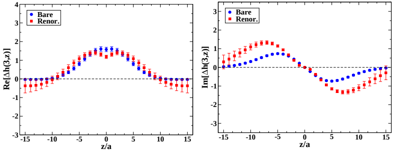

Figure 9: Renormalized matrix elements for the helicity quasi-PDF in the $\overline{\rm MS}$ scheme at $\bar\mu$=2 GeV using $Z^{\overline{\rm MS}}_{0,\Delta h}$ extracted from the fit range $(a\,\bar\mu_0)^2\,\epsilon\,[1.4,\,2.0]$.

The renormalized matrix elements of the helicity quasi-PDF are presented here for the first time, and they demonstrate the enormous progress in the field of the quasi-PDFs. However, before attempting to compare with the physical PDFs, we must understand the uncertainties that are inherited to $\Delta h^{\overline{\rm MS}}(z)$ from its renormalization function. As we argued in the previous subsection, a robust computation needs the subtraction of $\mathcal{O}(g^2\,a^\infty)$ lattice artifacts and a significant reduction of truncation uncertainties in the conversion between the RI$'$ and $\overline{\rm MS}$ schemes. We attempt setting bounds on these systematics, starting with the real part of renormalized matrix elements, which reads:

$${\rm Re}\,[\Delta h^{\overline{\rm MS}}] = {\rm Re}\,[Z^{\overline{\rm MS}}_{\Delta h}]\ {\rm Re}\,[\Delta h^{bare}] - {\rm Im}\,[Z^{\overline{\rm MS}}_{\Delta h}]\ {\rm Im}\,[\Delta h^{bare}]\,. \tag{21}$$

For small values of z, ${\rm Im}\,[Z^{\overline{\rm MS}}_{\Delta h}]$ is approximately zero and thus ${\rm Re}\,[\Delta h^{\overline{\rm MS}}]$ is approximately equal to the first term of Eq. (21). On the contrary, for $|z/a| \gtrsim 10$, ${\rm Re}\,[\Delta h^{\overline{\rm MS}}]$ receives significant contributions from the imaginary part, because ${\rm Re}\,[\Delta h^{bare}] < {\rm Im}\,[\Delta h^{bare}]$.

The uncertainty in the small-z region of renormalized matrix elements is dominated by uncertainties in the real part of the Z-factor, which, as argued in the previous subsection, should not exceed 5%. The local minimum observed at z=0 is likely to result from lattice artifacts and truncation effects. In the large-z region, the real part of the renormalized matrix elements receives negative contributions from the imaginary part of the bare matrix element if $Z_{\Delta h}^{\overline{\mathrm{MS}}}$ has a non-zero imaginary part and $\mathrm{Im}\,[\Delta h^{bare}]$ decays to zero more slowly than $\mathrm{Re}\,[\Delta h^{bare}]$, which is what we observe in the data. This effect is unphysical and is expected to be strongly suppressed or eliminated by extending our calculation to the two-loop order in the conversion and by subtracting lattice artifacts.

The imaginary part of renormalized matrix elements,

$$\mathrm{Im}\,[\Delta h^{\overline{\mathrm{MS}}}] = \mathrm{Re}\,[Z_{\Delta h}^{\overline{\mathrm{MS}}}]\ \mathrm{Im}\,[\Delta h^{bare}] + \mathrm{Im}\,[Z_{\Delta h}^{\overline{\mathrm{MS}}}]\ \mathrm{Re}\,[\Delta h^{bare}], \tag{22}$$

is enhanced in the intermediate-z regime, compared to the bare matrix elements. This results mostly from the fact that $\mathrm{Re}\,[Z_{\Delta h}^{\overline{\mathrm{MS}}}]$ increases with increasing z at a faster rate, than the decay of $\mathrm{Im}\,[\Delta h^{bare}]$. However, the obtained values receive also contributions from the second term of Eq. (22), where $\mathrm{Im}\,[Z_{\Delta h}^{\overline{\mathrm{MS}}}]$ is subject to a large uncertainty, as discussed in the previous subsection.

We want to stress that it is not possible to give a single number for the relative uncertainty of the real and imaginary parts of $\Delta h^{\overline{\mathrm{MS}}}$. According to Eqs. (21) - (22), different regions of z are influenced in a different way by the real and imaginary parts of the Z-factors. For small z, where $\mathrm{Im}\,[Z_{\Delta h}^{\overline{\mathrm{MS}}}]$ is small, the propagated uncertainty in the matrix elements is dominated by the uncertainty of $\mathrm{Re}\,[Z_{\Delta h}^{\overline{\mathrm{MS}}}]$, which is of the order of 5% and comparable to the currently attained statistical uncertainty. Thus, $\Delta h^{\overline{\mathrm{MS}}}$ is rather robust in this region. However, when z is increased, the uncertainty from $\mathrm{Im}\,[Z_{\Delta h}^{\overline{\mathrm{MS}}}]$ starts to dominate and reaches 100% for large z. In this way, the improvements expected by the perturbative subtraction of $\mathcal{O}(g^2\,a^\infty)$ lattice artifacts and the extension to the two-loop perturbative conversion to the $\overline{\mathrm{MS}}$ scheme are very important for obtaining meaningful values of renormalized matrix elements and hence, also quasi-PDFs.

The values of the multiplicative vector renormalization factor and the mixing coefficient can be used to properly renormalize the unpolarized quasi-PDF through Eq. (12). The successful renormalization requires the bare nucleon matrix elements for the scalar and vector operators, in order to extract the renormalized $h_V^{\mathrm{RI}'}$. Then, it must be multiplied by the conversion factor C_V to bring the results to the $\overline{\mathrm{MS}}$ scheme. Note that once the mixing is treated, the conversion factor is multiplicative and not a 2×2 matrix. This is due to the fact that no mixing is present in the continuum dimensional regularization.

3.4 Matching to light-front PDFs

Having the renormalized matrix elements, one can perform a Fourier transform and obtain the renormalized quasi-PDF, which represents the distribution of quark momenta for a finite-momentum nucleon moving in the chosen spatial (z) direction:

$$\tilde{q}(x,\mu,P_3) = \int_{-\infty}^{\infty} \frac{dz}{4\pi} e^{-izxP_3} \langle N|\bar{\psi}(0,z)\Gamma W(z)\psi(0,0)|N\rangle^{\overline{\mathrm{MS}},\mu}\,, \tag{23}$$

where x is the quark momentum fraction. The quasi-PDF, expressed in our case in the $\overline{\mathrm{MS}}$ scheme at μ=2 GeV, can then be connected to the light-front PDF, in the same scheme and at the same scale, using one-loop perturbative matching. The matching procedure uses the fact that only the ultraviolet physics is different in quasi- ($\tilde{q}$) and light-front (q) PDFs [11]. Hence, the one-loop difference between them is expressed as the difference between

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vertex corrections (denoted $Z^{(1)}$ below) and wave function corrections ($\delta Z^{(1)}$) for the finite momentum and infinite momentum cases. The generic matching formula for quasi-PDFs is: [50, 12]

$$\begin{aligned} q(x,\mu) &= \tilde{q}(x,\mu,P_3) - \frac{\alpha_s(\mu)}{2\pi}\tilde{q}(x,\mu,P_3)\,\delta Z^{(1)}\left(\frac{\mu}{P_3}\right) \\ &\quad -\frac{\alpha_s(\mu)}{2\pi}\int_{-\infty}^{\infty} Z^{(1)}\left(\xi,\frac{\mu}{P_3}\right)\tilde{q}\left(\frac{x}{\xi},\mu,P_3\right)\frac{d\xi}{|\xi|} + \mathcal{O}(\alpha_s^2), \end{aligned} \tag{24}$$

where $\alpha_s(\mu)$ is the strong coupling constant at the scale μ. In the integral, we exclude a small region around $\xi = 0$, such that the argument of $\tilde{q}$ is not too large, as we would then pick up contributions from the mirror images of $\tilde{q}$ resulting from the periodicity of the Fourier transform in Eq. (23). The linearly divergent terms $\propto \Lambda/P_3$ (Λ: transverse momentum cutoff) of the matching formulae from Ref. [11] are not present in our renormalized results.

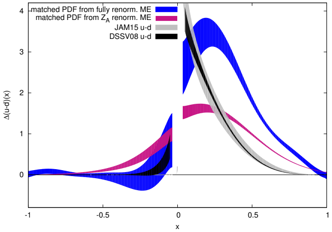

Figure 10: Comparison of matched helicity PDF obtained from quasi-PDF computed with either fully renormalized matrix elements (blue) or with bare matrix elements multiplied by the local (z=0) axial current Z-factor, Z_A (magenta). For purely orientational purposes, we also plot phenomenological PDFs (DSSV08 [51] and JAM15 [52]). However, we emphasize that no quantitative comparison with our results is aimed at, since careful consideration of a number of systematic effects is still needed. These include: cut-off effects, non-physical pion mass, finite volume effects, possible contamination by excited states, extrapolation to infinite nucleon boost, as well as the improvements in the computation of $\overline{\text{MS}}$ renormalization functions, postulated in the previous subsection.

In Fig. 10, we show the matched helicity PDF computed with either fully renormalized matrix elements obtained in this work (blue) or with bare matrix elements multiplied by the local (z=0) axial vector current renormalization function Z_A (magenta), corresponding to our results from Ref. [17]. We observe that the renormalized matrix elements from this work move towards the phenomenological PDFs, which is promising. In particular, the antiquark asymmetry is not overestimated any longer and actually this asymmetry becomes compatible with zero under current uncertainties. In the quark part, there is an enhancement of the matched PDF for all values of x. We emphasize again that the comparison with the phenomenological PDFs should be understood as only qualitative. For quantitative comparison, a careful investigation of a number of systematic effects is still needed. These include: cut-off effects, non-physical pion mass, finite volume effects, possible contamination by excited states, extrapolation to infinite nucleon boost, as well as the improvements in the computation of $\overline{\rm MS}$ renormalization functions, postulated in the previous subsection: subtraction of lattice artifacts computed in lattice perturbation theory and reduction of truncation effects in the perturbative conversion to the $\overline{\rm MS}$ scheme.

4 Conclusions and discussion

In this work we have presented a concrete prescription to renormalize non-perturbatively the matrix elements needed for the computation of quasi-PDFs. The employed scheme is RI′, which is then converted to the $\overline{\rm MS}$ scheme and evolved to 2 GeV; this is done perturbatively to one-loop. We have argued that the renormalization condition properly handles both kinds of divergences present in the matrix elements: the standard logarithmic divergence and the power divergence specific to non-local operators containing a Wilson line. Furthermore, we provide the renormalization conditions to eliminate the mixing in the case of the unpolarized quasi-PDF that mixes with the twist-3 scalar operator.

We have also demonstrated the implementation of the proposed prescription to the helicity quasi-PDF and presented the corresponding renormalized matrix elements. This has allowed us to draw conclusions how to make the computation more robust, which is the main outcome of this work.

- First, an essential ingredient of a computation with controlled systematic uncertainties is the subtraction of one-loop lattice artifacts in the framework of lattice perturbation theory. Following the ideas of Ref. [38], we are currently computing the $\mathcal{O}(g^2\, a^\infty)$ artifacts that will be subtracted from the non-perturbative estimates for the Z-factors. In this way, the presence of large cut-off effects in the renormalized functions (especially for "parallel" momenta) will be avoided to a large extent.
- Second, the conversion factor from the RI′ renormalization scheme to the $\overline{\rm MS}$ scheme is likely to have sizable higher order corrections that, among others, are responsible for the unphysical feature of the real part of the renormalized matrix element becoming negative for large Wilson line lengths. A two-loop computation of this conversion factor is expected to resolve this issue to a sufficient degree. We have performed numerical experiments that indicate that a natural change of the conversion factor by two-loop contributions, i.e. around 10-20% (which is approximately α_s at the considered scale), should be enough to suppress the unwanted effect. A perturbative calculation of the conversion factor to two loops is quite laborious and will be presented separately.

To summarize, the renormalization program presented in this work together with future improvements that are being pursued, will provide reliable estimates for the renormalization functions of the Wilson line fermion operators. In this fashion, the obtained renormalized matrix elements can be used as an input to calculate the quasi-PDFs and match them to light-front PDFs, which is the main aim of the whole approach. Apart from the helicity case discussed in this work, we will address the transversity PDF in an analogous manner. For the unpolarized case, one needs to take into account the mixing with the scalar operator, as explained and numerically demonstrated here. With this work, we have proposed and discussed a complete renormalization program of the quasi-PDFs, which has been a major uncertainty prior to this work and constitutes a crucial milestone in connecting lattice QCD results to the light-cone PDFs.

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P25：准部分子算符乘法可重正性

主题归属	非局域算符重整化与 hybrid 方案
底稿	arXiv:1809.01836
arXiv / DOI	1809.01836
完整原始 PDF	7 页；来源 PyQCD；SHA-256 90299e0614e9...
转排索引	EN 11 页；ZH 9 页（只作书目对照）
备注	索引未记录额外备注。

阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

页码范围：P25-p001 – P25-p007

Multiplicative renormalizability of quasi-parton operators

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(Dated: September 18, 2018)

Extracting parton distribution functions (PDFs) from lattice QCD calculation of quasi-PDFs has been actively pursued in recent years. We extend our proof of the multiplicative renormalizability of quasi-quark operators in Ref. [1] to quasi-gluon operators, and demonstrated that quasi-gluon operators could be multiplicatively renormalized to all orders in perturbation theory, without mixing with other operators. We find that using a gauge-invariant UV regulator is essential for achieving this proof. With the multiplicative renormalizability of both quasi-quark and quasi-gluon operators, and QCD collinear factorization of hadronic matrix elements of there operators into PDFs, extracting PDFs from lattice QCD calculated hadronic matrix elements of quasi-parton operators could have a solid theoretical foundation.

arXiv:1809.01836v2 [hep-ph] 17 Sep 2018

I. INTRODUCTION

The parton distribution functions (PDFs) encode important nonperturbative information of strong interactions. Based on QCD factorization [2], PDFs have been successfully extracted from high-energy scattering data with a good precision [3]. However, from both theoretical and practical points of view, extracting PDFs from first principle lattice QCD (LQCD) calculations must be done for testing non-perturbative sector of QCD, as well as needed for studying partonic structure of hadrons that could be difficult to do scattering experiments with.

Calculating PDFs in Euclidean-space LQCD *directly*, if not impossible, is difficult due to the time-dependence of the operators defining PDFs [3]. A novel approach was suggested by Ji [4], who introduced a set of LQCD-calculable quasi-PDFs and argued that the quasi-PDFs of hadron momentum P_z become corresponding PDFs when P_z is boosted to infinity. A number of other approaches to extract PDFs from LQCD calculations were also proposed [5–9]. In Refs. [10, 11], two of us proposed a QCD factorization based general approach to calculate PDFs in LQCD *indirectly*. Similar to the extraction of PDFs from experimental data of factorizable and measurable hadronic cross sections, we proposed to extract PDFs by the global analysis of data generated by LQCD calculations of good "lattice cross sections" (LCSs), which are defined as hadronic matrix elements that satisfy (1) calculable in Euclidean-space LQCD, (2) renormalizable for ultraviolet (UV) divergences to ensure a reliable continue limit, and (3) factorizable to PDFs with infrared-safe matching coefficients. It is the (3)-factorization that relates the desired PDFs to the LQCD calculable LCSs.

To extract the rich, precise and flavor separated information on PDFs, it is necessary to find as many good LCSs as possible, since different flavor PDFs are likely to contribute to the same LQCD calculated LCSs. For constructing good LCSs, we studied two types of operators in terms of (a) correlation of two gauge-dependent field operators with proper gauge links, which we call quasi-parton operators since they cover all operators defining quasi-PDFs and more, and (b) correlation of two gauge-invariant currents. In our approach, LQCD calculation of each of these good LCSs in coordinate space provides the needed information to *constrain* the PDFs, similar to the role of measuring various experimental cross sections to *constrain* the PDFs. For a comparison, in Ji's proposal [4], one focuses on reproducing PDFs by corresponding quasi-PDFs calculated in LQCD at a large enough P_z.

For LQCD calculations, it is relatively less expensive to calculate the type (a) operators comparing with type (b) operators, while the renormalization of the type (b) operators is much more simpler than that of the type (a) operators. Renormalization of the type (b) operators is almost trivial, for which one only needs to renormalize the gauge-invariant local currents, which is well-known. On the other hand, the renormalization of the type (a) operators is nontrivial due to the nonlocality of corresponding operators and potential power divergences. QCD factorization of both types of operators have been studied in Ref. [11], in which we found that multiplicative renormalizability of these operators is a necessary condition for the collinear factorization to be valid.

A lot of efforts have been devoted to explore the UV structure of the type (a) quasi-quark operators [12–19]. All-order multiplicative renormalizability of quasi-quark operators has been proved using two different methods: one relies on the auxiliary field technique [20, 21], and the other is based on diagrammatic expansion [1]. These proofs provided a firm theoretical basis for extracting the combination of quark distributions that are not sensitive to gluons from LQCD calculated hadronic matrix elements of quasi-quark operators [22–38].

In this paper, we study the UV divergences of quasi-gluon operators to all orders in QCD perturbation theory.

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The UV structure of quasi-gluon operators could be much more complicated, as we will explain. We define general bare quasi-gluon operators as

$$\mathcal{O}_{bg}^{\mu\nu\rho\sigma}(\xi) = F^{\mu\nu}(\xi)\,\Phi^{(a)}(\{\xi,0\})\,F^{\rho\sigma}(0)\,, \qquad (1)$$

where $\Phi^{(a)}(\xi,0) = \mathcal{P}e^{-ig_s\int_0^1 \xi\cdot A^{(a)}(r\xi)\,dr}$ is a path ordered gauge link in adjoint representation. To be definite, we assume ξ_μ along z-direction and introduce a unit vector $n^\mu = (0,0,0,1)$, defining $v\cdot n \equiv v_z$ for any vector v_μ. Due to the dimensional derivative operator in $F^{\mu\nu}$, superficial power counting tells us that the vertex between gluon field strength and gauge link could be linearly UV divergent. By using a cutoff regularization, one-loop calculation in Refs. [39, 40] indeed shows uncanceled linear divergences for this vertex, which would make the multiplicative renormalization of quasi-gluon operators almost impossible. For the comparison, the corresponding vertex of quasi-quark operators can be at most logarithmic UV divergent.

Another complication comes from operator mixing via UV renormalization. In principle, the general quasi-gluon operator in Eq. (1) could have 36 independent operators after taking into account the antisymmetry of gluon field strength, and all of them could be mixed under renormalization. In literature, quasi-gluon PDFs are constructed from linear combinations of $\mathcal{O}_{bg}^{\mu\nu\rho\sigma}$ by contracting them with some "tensors". In Refs. [4] and [40] and the first lattice simulation [41], different definitions of quasi-gluon PDFs were obtained by contracting $\mathcal{O}_{bg}^{\mu\nu\rho\sigma}$ by $n_\mu n_\rho g_{\nu i} g^i_\sigma$, $n_\mu n_\rho g_{\nu\sigma}$ and $g_{\mu 0} g_{\rho 0} g_{\nu\sigma}$, respectively. Renormalization of these quasi-gluon PDFs is non-trivial due to the mixing of these operators.

In this paper, we first perform an explicit one-loop calculation of quasi-gluon operators of an asymptotic gluon of momentum p, defined as $\langle g(p)|\mathcal{O}_{bg}^{\mu\nu\rho\sigma}(\xi)|g(p)\rangle$. We use dimensional regularization (DR) to regularize both logarithmic and linear UV divergences, which respectively appear as poles around $d = 4$ and $d = 4 - 1/n$ at n-loop order. We find that linear UV divergences of one-loop correction to the gluon-gauge-link vertex are canceled under DR, which makes the multiplicative renormalizability of quasi-gluon operators a possibility. We then explore all possible UV divergent topologies of higher order diagrams. Using gauge invariance, we find that all linear UV divergences from the gluon-gauge-link vertex are canceled to all-orders in perturbation theory. Then, we find that all of the 36 independent quasi-gluon operators can be multiplicatively renormalized without mixing with any other operators. Combining with our proof for quasi-quark operators in Ref. [1], our work presented in this paper for quasi-gluon operators completes the proof of multiplicative renormalization of the quasi-parton operators.

II. UV DIVERGENCES AT ONE LOOP

We present the relevant one-loop Feynman diagrams for quasi-gluon operators of an asymptotic gluon of momentum p in Fig. 1, where the bubble in the diagram (e) includes all one-loop self-energy diagrams of the active gluon. For the complete one-loop contribution, additional Feynman diagrams are needed. Some of them are mirror of diagrams (b), (c), (d), (e) and (g), while the rest can be obtained by replacing external momentum p to $-p$ in all these Feynman amplitudes. For the following one-loop calculation, we take the linearly combined quasi-gluon operator in Ref. [4] as an example, but our conclusion is true for any of the 36 independent operators.

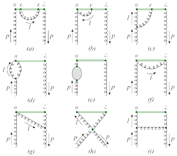

FIG. 1. Some typical Feynman diagrams for quasi-gluon PDFs of an asymptotic gluon of momentum p at one-loop order.

We choose Feynman gauge, and assume ξ_z to be positive for definiteness. The diagram (a) in Fig. 1 gives

$$M_{1a} = \frac{g_s^2\mu_r^{4-d}C_A}{i}\,e^{-ip_z\xi_z}\int_0^{\xi_z} dr_1\int_{r_1}^{\xi_z} dr_2\,\frac{d^dl}{(2\pi)^d}\,\frac{e^{il_z(r_2-r_1)}}{l^2}$$
$$\stackrel{\text{UV}}{=\!=}\ \frac{\alpha_s C_A}{\pi}e^{-ip_z\xi_z}\left(-\frac{\pi\mu_r\xi_z}{3-d}+\frac{1}{4-d}\right), \qquad (2)$$

where μ_r is a renormalization scale to compensate the mass dimension in DR. This diagram contributes to both linear and logarithmic UV divergences, as expected.

To understand where in this one-loop phase space the UV divergences in Eq. (2) come from, it is instructive to distinguish l_z - the z-component of the loop momentum l from $\bar{l}_\mu$ - the other components of l, as $l_\mu = \bar{l}_\mu - l_z n_\mu$ with $l^2 = \bar{l}^2 - l_z^2$ [1]. If $\bar{l}^2$ is constrained in a finite region in Eq. (2), integrating l_z, r_1 and r_2 cannot generate any UV divergence. Furthermore, there is no UV divergence if we do not include the region where $|r_2 - r_1|$ is very small, which can be demonstrated by introducing the following decomposition,

$$\frac{1}{(l+q)^2} = \frac{1}{(\bar{l}+\bar{q})^2} + \frac{(l_z+q_z)^2}{(l+q)^2(\bar{l}+\bar{q})^2}\,, \tag{3}$$

where q can be any unintegrated momentum. By applying this decomposition to $\frac{1}{l^2}$ in Eq. (2), the first term is free of l_z, and thus the integration of l_z gives $\delta(r_2 - r_1)$, while the second term is UV finite under the integration of $\bar{l}$. That is, the UV divergence in Eq. (2) can only come from the region of phase space where $\bar{l}$ are in UV region while $|r_2 - r_1|$ is very small. Therefore, we conclude that, with DR, all UV divergences of diagram (a) in Fig. 1 come from a region localized in spacetime.

By decomposing both $\frac{1}{l^2}$ and $\frac{1}{(p-l)^2}$ using Eq. (3), we obtain many terms for diagrams (b) and (c) and found that these terms are either free of l_z in denominator, which result in $\delta(r)$ or its derivatives, or UV finite under the integration of $\bar{l}$. Thus the UV divergences of these two diagrams are also localized in spacetime,

$$M_{1b} \overset{\mathrm{UV}}{=\!=} \frac{\alpha_s C_A}{\pi} e^{-ip_z\xi_z} \left(\frac{-i}{p_z\xi_z}\frac{\pi\mu_r\xi_z}{3-d} \right), \tag{4}$$

$$M_{1c} \overset{\mathrm{UV}}{=\!=} \frac{\alpha_s C_A}{\pi} e^{-ip_z\xi_z} \left(\frac{i}{p_z\xi_z}\frac{\pi\mu_r\xi_z}{3-d} + \frac{3}{4}\frac{1}{4-d} \right), \tag{5}$$

where diagram (b) has only linear UV divergence, while (c) has both linear and logarithmic UV divergence.

The diagrams (d) and (e) in Fig. 1 have only logarithmic UV divergences, which come from the region where all components of l^μ go to infinity, and thus, is localized in spacetime. All other diagrams in Fig. 1 are free of UV divergence, simply because the loop cannot be localized in spacetime due to finite ξ_z, same as the argument for quasi-quark operators [1]. In summary, we conclude that the UV divergences of quasi-gluon operators at one-loop can only be emerged from a region *localized* in coordinate spacetime. For a comparison, UV divergences of operators defining PDFs come from a region *nonlocal* along the "$-$" light-cone direction [1].

The linear UV divergence in Eq. (2) from the diagram (a) is harmless because it can be easily exponentiated to all-order as an overall phase factor and then multiplicatively renormalized, just like the case of quasi-quark operators [1, 42, 43]. However, the presence of linear UV divergence in Eqs. (4) and (5) from the diagrams (b) and (c), respectively, could challenge the multiplicative renormalizability. Fortunately, we find that with DR, the linear UV divergences from these two diagrams are canceled. On the contrary, the linear divergences from the diagrams (b) and (c) do not cancel if one uses a cutoff regularization that breaks the gauge symmetry [39, 40]. This implies that gauge invariance plays an important role to remove the linear UV divergences that may challenge the multiplicative renormalizability. In the following, we will use gauge invariance to show that all linear divergences, except that from self-energy of gauge links, are canceled by summing over all contributions, and quasi-gluon operators could be multiplicatively renormalized.

III. UV DIVERGENCES AT HIGH ORDERS

From the one-loop diagrams in Fig. 1, we can generate all high order loop diagrams by adding gluons, quark-antiquark pairs, or ghost-antighost pairs to them. Because of the isolation of z-component in the definition of quasi-parton operators, both 3-dimensional (3-D) and 4-D integration of loop momentum $\bar{l}$ and l could lead to UV divergence. In Ref. [1], we introduced the change of divergence index $\Delta\omega_3$ and $\Delta\omega_4$ for the 3-D and 4-D integration of loop momenta of higher order diagrams, respectively, and showed that it is sufficient, although it is not necessary, that quasi-parton operators are renormalizable if $\Delta\omega_3 \le 0$ and $\Delta\omega_4 \le 0$ are satisfied for all corresponding higher order diagrams. Based on the power counting rules derived in Ref. [1], we find that the only case that may increase superficial UV divergence of quasi-gluon operators at high orders is when we add a gluon with both ends of it attached to the gauge link, where the 3-D integration gets $\Delta\omega_3 = 1$. By applying the decomposition in Eq. (3) to the added gluon's momentum, it is straightforward to show that dimensional regularized UV divergences at any loop level are localized in spacetime, in the same way as quasi-quark operators shown in [1]. As a result, we find that $\Delta\omega_3$ is effectively irrelevant for studying UV divergences. Because $\Delta\omega_4 \le 0$ for all cases, there are only finite topologies of high order diagrams in Fig. 2 that have UV divergences.

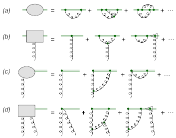

FIG. 2. Four topologies of diagrams which could give UV divergences to the quasi-gluon operators.

The blobs with topologies (a) and (c) in Fig. 2 denote one-particle-irreducible diagrams, and they both have linear superficial UV divergences. Because of the potential linear UV divergences, diagrams with one more gluon attached to the blobs can generate logarithmic UV divergences. Another possibility to produce logarithmic divergences is when a gluon is attached to the gauge link outside of, but very close to, the blobs, as shown in Fig. 2 (b) and (d), with the attachment denoted by a triangle. The blobs of topologies (b) and (d) include both kinds of logarithmic divergent diagrams mentioned here.

The topologies (a) and (b) in Fig. 2 are the same as that for quasi-quark operators, and their divergences can be renormalized similarly. Linear divergences from the diagrams of the topology (a) can be removed by an overall factor as the mass renormalization of a test particle moving along the gauge link [42], and its logarithmic divergences caused by end points of the gauge link can be removed by multiplying $Z_{wg}^{-1/2}$ - the "wave function" renormalization of the test particle [43]. The diagrams of the topology (b) has only logarithmic UV divergence, which can be taken care of by QCD renormalization [43].

The UV divergences from diagrams of topologies (c) and (d) in Fig. 2 are different from that of quasi-quark operators, and are studied in next two sections, respectively.

IV. RENORMALIZATION OF GLUON-GAUGE-LINK VERTEX

For the definiteness of following discussion, we assume that the gauge link in diagrams of the topology (c) in Fig. 2 starts at an arbitrary coordinate ξ_{1z} with the operator $F^{\mu\nu}(\xi_{1z})$, and ends at another arbitrary coordinate ξ_{2z} with no additional operators. With the "bare" coupling constant g_s, and "bare" field operators for the gluons, the Faddeev-Popov ghost and the antighost given by the symbols A, c and $\bar{c}$ respectively, a generalized Ward identity of the non-Abelian field relevant to this topology can be derived [44],

$$\begin{aligned}&\langle -i\partial_\lambda^y A_d^\lambda(y)[\Phi(\{\xi_{2z},\xi_{1z}\})]_{ab}\, F_b^{\mu\nu}(\xi_{1z})\rangle\\ &= \langle g_s\bar{c}_d(y)c_e(\xi_{2z})[t_e\Phi(\{\xi_{2z},\xi_{1z}\})]_{ab}\, F_b^{\mu\nu}(\xi_{1z})\rangle,\end{aligned} \tag{6}$$

where the t represents SU(3) generators of the adjoint representation. A pictorial representation of Eq. (6) is given in Fig. 3, where "1PR" denotes one-particle-reducible diagrams. The topology of the left-hand side of Fig. 3 is the same as that of the diagram (c) in Fig. 2, but is contracted with external gluon momentum and expressed in coordinate space. The topology of the first term on the right-hand side of Fig. 3 is nonlocal in space-time, and thus has no UV divergence. Furthermore, after the renormalization of QCD Lagrangian and gauge-link-related topologies (a) and (b) in Fig. 2, UV divergences of 1PR diagrams will be canceled. That is, the generalized Ward identity in Eq. (6) ensures that the topology (c) in Fig. 2 is free of UV divergence if it is contracted by external gluon momentum.

To understand the renormalization of the UV divergence of the topology (c) in Fig. 2, we represent the diagrams of this topology as $\Gamma^{\lambda\mu\nu}(p,n)$, which could be referred as the gluon-gauge-link vertex, where p and λ are the momentum and Lorentz index of the external gluon, respectively. Lorentz symmetry combined with antisymmetry between μ and ν enable us to do the general decomposition $\Gamma^{\lambda\mu\nu}(p,n) = \sum_{i=1}^4 c_i\Pi_i^{\lambda\mu\nu}$ with

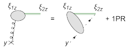

FIG. 3. Pictorial representation of the generalized Ward identity in Eq. (6) with dashed line represents the ghost field.

$$\begin{aligned}&\Pi_1^{\lambda\mu\nu} = g^{\mu\lambda}p^\nu - g^{\nu\lambda}p^\mu,\ \Pi_2^{\lambda\mu\nu} = (p^\mu n^\nu - p^\nu n^\mu)n^\lambda,\\ &\Pi_3^{\lambda\mu\nu} = (p^\mu n^\nu - p^\nu n^\mu)p^\lambda,\ \Pi_4^{\lambda\mu\nu} = g^{\mu\lambda}n^\nu - g^{\nu\lambda}n^\mu.\end{aligned} \tag{7}$$

Since $p_\lambda\Gamma^{\lambda\mu\nu} = 0$ for the UV divergent terms as discussed above, we obtain $c_2\, p\cdot n + c_3\, p^2 + c_4 = 0$, and consequently,

$$\begin{aligned}\Gamma^{\lambda\mu\nu}(p,n) =& c_1\Pi_1^{\lambda\mu\nu} + c_2(\Pi_2^{\lambda\mu\nu} - p\cdot n\Pi_4^{\lambda\mu\nu})\\ &+ c_3(\Pi_3^{\lambda\mu\nu} - p^2\Pi_4^{\lambda\mu\nu}).\end{aligned} \tag{8}$$

Since c_1 and c_2 have mass dimension 0, locality of UV divergences ensures that they can be at most logarithmic divergent, while c_3 is UV finite due to its mass dimension at -1. The only potential linearly UV divergent coefficient c_4 is removed by gauge invariance. We have therefore demonstrated that the cancellation of linear UV divergences of diagrams (b) and (c) in Fig. 1 at one-loop order can be generalized to all orders, which makes the multiplicative renormalizability of quasi-gluon operators a possibility.

At the lowest order in α_s, we have $\Gamma^{\lambda\mu\nu}(p,n) \propto \Pi_1^{\lambda\mu\nu}$. If we want $\Gamma^{\lambda\mu\nu}(p,n)$ not to mix with other operators under renormalization, we need its UV divergence to be proportional to $\Pi_1^{\lambda\mu\nu}$ to all orders. Fortunately, it is always true. For the case with μ (or ν) along z-direction or the case with both μ and ν not along z-direction, the coefficients of c_2 are proportional to $\Pi_1^{\lambda\mu\nu}$ or equal to zero, respectively. Therefore, the components of $\Gamma^{\lambda\mu\nu}(p,n)$ do not mix with each others at all, although two different renormalization constants are needed for the two different choices.

In summary, if we choose either $F^{z\bar{\nu}}$ or $F^{\bar{\mu}\bar{\nu}}$ for gluon-gauge-link vertex, $\Gamma^{\lambda\mu\nu}$, we can remove the UV divergences of the vertex by multiplying a corresponding renormalization factor $Z_{vg1}^{-1/2}$ or $Z_{vg2}^{-1/2}$, respectively.

V. RENORMALIZATION OF GLUON-GLUON-GAUGE-LINK VERTEX

Finally, we exam the renormalization of UV divergence of the topology (d) in Fig. 2, whose diagrams involve two gluons and a gauge link, and could be referred as the gluon-gluon-gauge-link vertex. Similar to the Ward identity in Eq. (6), we construct the following Ward identity for the "bare" fields and operators,

$$\langle\partial_\lambda^x A_d^\lambda(x)\partial_\rho^y A_e^\rho(y)[\Phi(\{\xi_{2z},\xi_{1z}\})]_{ab}\, F_b^{\mu\nu}(\xi_{1z})\rangle$$

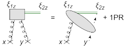

FIG. 4. Pictorial interpretation of the generalized Ward identity in Eq. (9) with dashed line represents the ghost field.

$$+i\delta^{(d)}(x-y)\delta_{de}\langle[\Phi(\{\xi_{2z},\xi_{1z}\})]_{ab}\,F_b^{\mu\nu}(\xi_{1z})\rangle \tag{9}$$
$$=g_s\langle f^{afg}\bar{c}_e(y)c_f(\xi_{2z})\partial^x_\lambda A^\lambda_d(x)[\Phi(\{\xi_{2z},\xi_{1z}\})]_{gb}\,F_b^{\mu\nu}(\xi_{1z})\rangle.$$

A pictorial interpretation of Eq. (9) is given in Fig. 4. Similar to Fig. 3, the topology of the left-hand side of Fig. 4 is the same as that of the diagram (d) in Fig. 2, except the external gluons of the diagrams are contracted with their respective momenta and expressed in coordinate space. The right-hand side of the equation in Fig. 4 is UV finite after all previously discussed renormalizations performed, including QCD Lagrangian, gauge links, and gluon-gauge-link vertex. That is, we find that the topology (d) in Fig. 2 is free of UV divergence if both external gluons are contracted by their respective momenta.

Similar to the discussion of the gluon-gauge-link vertex, the Ward identity helps reduce the superficial UV divergence of the topology (d) in Fig. 2. Since the diagrams of the topology (d) have only superficial logarithmic divergence, the additional reduction of the superficial UV divergence from the Ward identity makes the topology (d) in Fig. 2 UV finite. Therefore, after the renormalization of topology (c), the topology (d) requires no additional renormalization. This is similar to the case of renormalizing gluon vertexes of QCD Lagrangian, where gauge invariance guarantees that four-gluon vertex will be free of UV divergence once three-gluon vertex is renormalized.

VI. SUMMARY

We demonstrated that the UV divergences of quasi-gluon operators are actually similar to the UV divergences of quasi-quark operators if all divergences are regularized in a gauge invariant scheme. We proved that UV divergences of all 36 pure quasi-gluon operators are localized in spacetime, and could be multiplicatively renormalized without mixing with each others,

$$\mathcal{O}_g^{\mu\nu\rho\sigma}(\xi) = e^{-C_g|\xi_z|}Z_{wg}^{-1}Z_{vg1}^{-s/2}Z_{vg2}^{-(2-s)/2}\mathcal{O}_{bg}^{\mu\nu\rho\sigma}(\xi), \tag{10}$$

where s is the number of z components chosen for Lorentz indices $\{\mu,\nu,\rho,\sigma\}$, and C_g, Z_{wg}, Z_{vg1} and Z_{vg2} are renormalization constants. Like the quasi-quark operators, without mixing with each others, hadronic matrix elements of quasi-gluon operators could be examples of good "lattice cross sections", as defined in Refs. [10, 11], which could be calculated in LQCD and factorized into normal PDFs, from which PDFs could be extracted by QCD global analysis of the data of these good "lattice cross sections" generated by the first principle LQCD calculations.

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Note added: While our paper is being finalized, a preprint by Zhang *et al.* [45] appeared, in which these authors reached the similar conclusion although the approach is very different.

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P29：准光前关联函数的自重正化

主题归属	非局域算符重整化与 hybrid 方案
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阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

页码范围：P29-p001 – P29-p029

Self-Renormalization of Quasi-Light-Front Correlators on the Lattice

(Lattice Parton Collaboration (LPC))

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In applying large-momentum effective theory, renormalization of the Euclidean correlators in lattice regularization is a challenge due to linear divergences in the self-energy of Wilson lines. Based on lattice QCD matrix elements of the quasi-PDF operator at lattice spacing a= 0.03 fm $\sim$ 0.12 fm with clover and overlap valence quarks on staggered and domain-wall sea, we design a strategy to disentangle the divergent renormalization factors from finite physics matrix elements, which can be matched to a continuum scheme at short distance such as dimensional regularization and minimal subtraction. Our results indicate that the renormalization factors are universal in the hadron state matrix elements. Moreover, the physical matrix elements appear independent of the valence fermion formulations. These conclusions remain valid even with HYP smearing which reduces the statistical errors albeit reducing control of the renormalization procedure. Moreover, we find a large non-perturbative effect in the popular RI/MOM and ratio renormalization scheme, suggesting favor of the hybrid renormalization procedure proposed recently.

I. INTRODUCTION

Parton distribution functions (PDFs) provide an effective description of quarks and gluons inside a light-travelling nucleon [1, 2]. They play an essential role in calculating hardronic cross sections involving the nucleons [3], and their uncertainties from phenomenological extractions have been one of the major sources of errors in the theoretical predictions at Large Hadron Collider (LHC) and other hadron facilities. Thus, a precise knowledge of PDFs is very important both for the accurate tests of the Standard Model (SM) and for the search of new physics beyond the SM. On the other hand, the densities of partons in the nucleon provide direct information on its intrinsic properties such as the origin of the nucleon spin and mass, as well as the role of sea quarks for various physical quantities [4].

However, directly calculating parton physics from first principles of quantum chromodynamics (QCD) has been a difficult task. A review on various efforts of doing so can be found in Ref. [5]. The proposal of the large-momentum effective field theory (LaMET) has been an important step toward meeting the challenge [6–8]. So far, LaMET has been widely used in calculating quark isovector distribution functions [9–24], generalized parton distributions [25, 26], distribution amplitudes (DAs) [27–29], and transverse-momentum-dependent distributions [30–32].

LaMET suggests to calculate time-independent physical distributions in a finite momentum nucleon, which are Euclidean observables. Such finite-momentum quantities can then be matched to light-front (LF) parton properties using effective field theory techniques [33]. For example, for the collinear quark distributions, it has been suggested to first compute a Euclidean space correlation

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(or quasi-light-front (quasi-LF) correlation) on the lattice,

$$\tilde{h}(z, P_z) = \langle P|O_{\gamma_t}(z)|P\rangle, \quad (1)$$

where $|P\rangle$ is the nucleon state with a large momentum P and the non-local operator is

$$O_\Gamma(z) = \bar{\psi}(0)\Gamma U(0, z)\psi(z), \quad (2)$$

where ψ, $\bar{\psi}$ denote the bare quark field, Γ is a Dirac structure, and $U(0, z) = \exp(-ig\int_0^z dz' A_z(z'))$ is the Wilson link along the direction z, where A^μ is the gluon gauge potential. After renormalizing $\tilde{h}$ properly and matching it to some continuum scheme such as dimensional regularization and (modified) minimal subtraction ($\overline{\mathrm{MS}}$), one obtains the so-called quark quasi-PDF via a Fourier transform, from which the quark PDF can be extracted through perturbative QCD matching.

Renormalizing the quasi-LF correlation $\tilde{h}$ under lattice regularization is a non-trivial task. To see this, let us take, as a simple example, the matrix element of $O_\Gamma(z)$ in an off-shell quark state $|q\rangle$ with large Euclidean momentum p^2 (actually an off-shell truncated Green's function). It has the following form at 1-loop level [34],

$$\langle O_\Gamma(z)\rangle = \Gamma\left(1 + \gamma g^2 \log(z^2/a^2) - m_{-1}\frac{z}{a} + \ldots\ldots\right), \quad (3)$$

where g is the bare gauge coupling and a is the lattice spacing. Unlike the coefficient of the logarithm γ, the coefficient of the linear divergence m_{-1} can be sensitive to the details of the fermion and gauge actions on the lattice as those actions themselves define the lattice regularization. The linear-divergence term is proportional to the Wilson link length z/a in lattice units, and it can be exponentially large at large z or small a, when higher-order corrections are included. Thus, the linear divergence effect has to be removed before the lattice calculated quasi-PDF can be extrapolated to the continuum limit $a \to 0$ and eventually matched to the continuum-scheme PDF.

Theoretical studies so far have shown that the quasi-PDF operator is multiplicatively renormalizable [35–38] in a continuum theory. On the lattice, due to non-commutativity of the limit $z \to 0$ and $a \to 0$, it has been suggested recently [39] to use a hybrid scheme to renormalize the large and short distance correlations separately, which has the advantage of avoiding certain discretization effects and undesired infrared effects introduced in the renormalization stage. At small z where a short distance expansion is valid, one can use various ratio schemes, such as dividing by an off-shell quark matrix element of the quasi-PDF operator in the regularization-independent momentum subtraction (RI/MOM) scheme [34, 38, 40] or by a hadron matrix element in the rest frame $\tilde{h}(z, P_z = 0)$ [41, 42]. At large distance, one can directly remove the Wilson line self-energy effect by subtracting the power divergence [36–38, 43]. Apart from using the RI/MOM factor or the rest-frame hadron matrix element, other methods have been suggested to extract the linear divergence, including Wilson loop [27, 43], vacuum expectation values $\langle O_\Gamma(z)\rangle$ [44], and gauge fixed Wilson link [8], and so on [39].

However, numerically subtracting the linear divergence is an extremely delicate exercise. First, linear divergence could be sensitive to computational systematics in lattice calculations. In data, there may be slight differences between two matrix elements with the same linear divergence. These differences may lead to a failure of the suggested methods, especially for small lattice spacings. Second, some of the renormalization factors have other problems. For example, the leading contribution to the vacuum expectation value of $\langle O_\Gamma(z)\rangle$ at short distance vanishes and therefore, it is numerically very challenging to obtain the relevant linear divergence. Finally, as we shall see, linearly-divergent chiral symmetry breaking effects for Wilson fermions may render the linear divergence non-universal [45].

To understand better the effects of the linear divergence in LaMET applications, we study systematically the linear divergences of matrix elements for sets of lattice data, generated with different lattice actions. We propose a self-renormalization method to eliminate all divergences and discretization errors when data for several different lattice spacings are available. The idea of this method is to extract the renormalization factor and the residual contribution directly from the matrix element we want to renormalize, without using an additional matrix element for renormalization. We then match the empirically renormalized matrix elements to those in the continuum. Our method is largely model-independent, and each term of our fitting functions is motivated by physics considerations. Tests show that our method works well for all data sets considered, which include matrix elements calculated with the lattice spacings from 0.03 fm to 0.12 fm using the valence clover and overlap actions on the MILC [46] $N_f = 2 + 1 + 1$ and RBC [47] $N_f = 2 + 1$ configurations. We also find that the method works for matrix elements after applying smearing which is needed to improve the statistical precision.

The rest of the paper is organized as follows. We present the theory of perturbative renormalization in Sec. II. The definition of the matrix elements we calculate and the simulation setup can be found in Sec. III. In Sec. IV the linear divergence is analysed for the different ensembles. Sec. V presents our strategy to self-renormalize a matrix element and collects the results for the $O_{\gamma_t}(z)$ matrix elements in different states and calculated with different valence fermion actions. Sec. VI discusses other fitting options, which mainly differ in the treatment of higher-order terms. In Sec. IV, V, and VI, we only analyze matrix elements without HYP smearing. In Sec. VII, we extend our method to HYP smeared cases. In Sec. VIII, we test the linear divergence in some vacuum state matrix elements, including Wilson loop, quasi-PDF operator, and gauge fixed Wilson link, for the HYP smeared cases. In Sec. IX, we summarize the results and

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discuss issues to be addressed in further studies.

II. RENORMALIZATION IN PERTURBATION THEORY

According to the standard renormalization in local quantum field theories [48], the renormalization of the matrix elements of an operator does not depend on the external states but is only related to the short-distance property of the operator itself. Therefore, one can study the renormalization property of the operator in perturbative Green's functions, or off-shell quark and gluon matrix elements. For LaMET applications to PDFs, we are interested in the non-local operator $O_\Gamma(z) = \bar\psi(0)\Gamma U(0,z)\psi(z)$ in Eq. (2). There are two types of divergences: The linear divergence associated with the Wilson link (its self-energy) and the logarithmic divergences associated with the renormalization of the vertices involving the Wilson line and light quark. The renormalization is multiplicative, and only the linear divergence has a (linear) z dependence. Therefore the renormalized operator is [36–39]

$$O_\Gamma(z)_R = Z_O^{-1} e^{\delta\bar m z} O_\Gamma(z), \tag{4}$$

where $e^{\delta\bar m z}$ contains the linear divergence and Z_O the logarithmic ones. $\delta\bar m$ is not uniquely defined apart from the linear divergence, introducing a subtraction scheme dependence which affects the z-dependence of the renormalized operator [39].

The linearly-divergent mass renormalization can be calculated in perturbation theory. At one-loop order, the result is independent of lattice action [43],

$$\delta\bar m = \frac{2\pi}{3a}(\alpha_s + \mathcal{O}(\alpha_s^2) + ...)\ . \tag{5}$$

The energy scale of α_s can be chosen as $1/a$ to match the lattice results,

$$\alpha_s(1/a, \Lambda_{\rm QCD}) = \frac{2\pi}{b_0 \ln[1/(a\Lambda_{\rm QCD})]}\ , \tag{6}$$

where $b_0 = 11 - \frac{2}{3}n_f$ (we will take n_f=3) is the QCD β function at one-loop order with n_f species of fermions. $\Lambda_{\rm QCD}$ is the non-perturbative QCD scale. Including higher-orders in the β function will lead to a more complicated expression. Different choices of energy scale amount to including high-order corrections. $\Lambda_{\rm QCD}$ and higher-order α_s terms in $\delta\bar m$ also depend on the lattice action.

The perturbation theory does not converge due to infrared renormalons, see from example [49–52]. The existence of the renormalons signals a non-perturbative term in $\delta\bar m$ which is independent of a,

$$\delta\bar m = m_{-1}(a)/a - m_0\ , \tag{7}$$

where the minus sign is just a convention. The uncertainty in summing the perturbation series to get $m_{-1}(a)$ is compensated by the same uncertainty in the non-perturbative m_0, leaving the total independent of the renormalon uncertainty.

Additional uncertainty in m_0 comes from the subtraction scheme, or equivalently from the matching to the continuum scheme. To reduce the subtraction scheme dependence, we require the renormalized lattice correlation function to be consistent with the $\overline{\rm MS}$ result from continuum perturbation theory at short-distances z. To be more concrete, we determine m_0 by matching the lattice result to the $\overline{\rm MS}$ one within a window $a \ll z < z_S$, where $z_S < 1/\mu$ is the point beyond which perturbation theory ceases to work and μ is a perturbative scale. The condition $a \ll z$ ensures that the discretization effect on lattice results is small. For this special choice m_{0c}, the LaMET expansion does not have a linear term in $1/P^z$ [39].

At one-loop order and in dimensional regularization, the renormalization factor of the logarithimic divergence is [8, 53],

$$Z_O = 1 + \frac{3C_F\alpha_s}{2\pi}\frac{1}{4-d}, \tag{8}$$

where $C_F = 4/3$ is the Casimir operator for the fundamental representation of $SU(3)$ and d is the space-time dimension. One can resum the logarithimic divergence through solving the renormalization group equation,

$$\frac{dZ_O(\epsilon,\mu)}{d\ln(\mu)} = \gamma Z_O(\epsilon,\mu), \tag{9}$$

where $\epsilon = (4-d)/2$ and $\gamma = -\frac{3C_F}{2\pi}\alpha_s(\mu, \Lambda_{\rm QCD})$ is the leading anomalous dimension, which is independent of the regularization method. The form of the leading-order solution of Eq. (9) is independent of regularization scheme and, on the lattice, is

$$Z_O(1/a,\mu) = \left(\frac{\ln[1/(a\Lambda_{\rm QCD})]}{\ln[\mu/\Lambda_{\rm QCD}]}\right)^{\frac{3C_F}{b_0}}, \tag{10}$$

where the bare ultraviolet (UV) cut-off is $1/a$, and the renormalization scale is μ.

If one considers the contributions from sub-leading logarithms to the anomalous dimension γ when solving the renormalization group equation Eq. (9), we obtain [54]

$$Z_O(1/a,\mu) = \left(\frac{\ln[1/(a\Lambda_{\rm QCD})]}{\ln[\mu/\Lambda_{\rm QCD}]}\right)^{\frac{3C_F}{b_0}} \left(1 + \frac{d}{\ln[a\Lambda_{\rm QCD}]}\right), \tag{11}$$

where $\Lambda_{\rm QCD}$ and the constant d depend on the specific lattice action.

III. LATTICE MATRIX ELEMENTS AND SIMULATION SETUP

The standard non-perturbative renormalization of lattice matrix elements is through calculating some auxiliary matrix elements of the operator (which can also be computed in perturbation theory) and using them as the renormalization factor to approach the continuum limit. In this paper, we focus on the study of the auxiliary matrix elements potentially useful for the renormalization of quasi-LF correlations. We mainly consider the matrix elements of the quasi-PDF operators in the following cases: 1) in a large Euclidean momentum quark state, 2) in the physical zero-momentum state of a hadron such as the pion or nucleon. We will also study the matrix element in the vacuum [44] (case 3), and show that it is not a good choice for renormalization because at small z such a matrix element is suppressed as can be shown by operator product expansion (OPE). Moreover, since the z-dependent renormalization is mainly associated with the Wilson link, we shall also consider the matrix elements of Wilson loop as well as the Wilson link in a fixed gauge (case 4).

We calculate the pion and nucleon matrix elements $\tilde{h}^0_H(z) = \langle H|O_{\gamma_t}(z)|H\rangle_{\vec{P}=0}$ in the rest frame evaluating the following three-point function,

$$
\begin{aligned}
R_H(t_2,t,z) &\equiv \frac{\langle O_H(t_2)\sum_{\vec{x}} O_{\gamma_t}(z;(\vec{x},t))O^\dagger_H(0)\rangle}{\langle O_H(t_2)O^\dagger_H(0)\rangle} \\
&= \langle H|O_{\gamma_t}(z)|H\rangle \\
&+ \mathcal{O}(e^{-\delta m t}) + \mathcal{O}(e^{-\delta m(t_2-t)}) + \mathcal{O}(e^{-\delta m t_2}),
\end{aligned}
\tag{12}
$$

where $O_{\gamma_t}(z,(\vec{x},t)) = \bar{\psi}(\vec{x},t)\gamma_t U((\vec{x},t),(\vec{x}+\hat{n}_z z,t))\psi(\vec{x}+\hat{n}_z z,t)$, $\hat{n}_z$ is the unitary vector along the z direction, and O_H is the interpolation field of a hadron such as a pion and a nucleon. t_2 corresponds to the source-sink separation.

To obtain $\tilde{h}^0_H(z)$ accurately, we need both t and t_2-t to be large enough to suppress excited-state contaminations, or fit the three point function with a proper parametrization on the excited-state contaminations. We can use the first strategy for the pion case since the signal-to-noise ratio will not decay for the pion in the rest frame; while the second strategy is essential for the other cases where the statistical uncertainty increases exponentially with t and t_2. Due to (anti)periodic boundary conditions we can use at most $t_2 = T/2$ which is larger than 2 fm on most modern lattice ensembles. Since the mass gap $\delta m \sim 1$ GeV between the ground and first exited state of pion $\langle\tilde{h}^0_\pi(z)\rangle$ can be extracted with sufficient accuracy. At small perturbative z, $\langle\tilde{h}^0_\pi(z)\rangle^{-1}$ acts as a renormalization factor up to certain $\mathcal{O}(\Lambda^2_{\rm QCD}z^2)$ power corrections [41].

A common non-perturbative renormalization method for lattice QCD matrix elements with local operators is the RI/MOM scheme [55] and its modified versions [56]. One can calculate the bare matrix element with given operators in an off-shell quark state, for both the lattice and dimensional regularizations, and renormalize their difference in the $\overline{\rm MS}$ scheme to get the renormalization factor for the bare quantities. For the quasi-PDF, such a renormalization constant (the RI/MOM renormalization factor) is defined through the $O_{\gamma_t}(z)$ matrix element in the off-shell quark state with given momentum p^2 [34, 38, 40, 57]:

$$
Z^{RI}(z,\mu_R) = \frac{1}{4N_c}\mathrm{Tr}[\gamma_t\langle q|O_{\gamma^t}(z)|q\rangle]|_{p^2=-\mu_R^2, p_z=p_t=0},
\tag{13}
$$

where $N_c = 3$ is the number of colors, μ_R is the RI/MOM renormalization scale. We will take $\mu_R = 3$ GeV since the RI/MOM factor has little dependence on the renormalizaiton scale in a certain range [45]. We work with Landau gauge fixed configurations, and choose $p_z = p_t = 0$ to eliminate the difference between the projections [16].

Factor Z^{RI} can be calculated non-perturbatively for any lattice regularization, or equivalently for any quark and gluon action defined on the lattice. However, the current lattice QCD calculation is limited to the Landau gauge. The perturbative matching with off-shell states in Landau gauge can be complicated beyond one-loop level [58].

The z-dependent part of renormalization comes from the Wilson line, therefore one may extract the linear divergences from matrix elements of pure Wilson lines. First one can consider a Wilson loop [27, 28, 38, 43, 59],

$$
\mathcal{U}(r,t) = \langle U(\vec{r},t;\vec{r},0)U(\vec{r},0;\vec{0},0)U(\vec{0},0;\vec{0},t)U(\vec{0},t;\vec{r},t)\rangle,
\tag{14}
$$

Ref. [43] proposed this scheme to renormalize the pion DA, and the linear divergence coefficient was extracted precisely based on the calculation with several lattice spacings in the following Kaon DA study [27]. But most of subsequent lattice QCD calculations have switched to the RI/MOM scheme or used a hadronic matrix element.

One can also consider the matrix element of a single Wilson Link $\langle U(0,z)\rangle$ in a fixed gauge such as the Landau gauge. As the simplest choice without any external state, $\langle U(0,z)\rangle$ can provide a reference to identify whether the linearly divergent behavior is sensitive to the existence of the external state, as suggested in the multiplicative renormalizability studies of the quasi-PDF operator [35–38].

symbol	$6/g^2$	L	T	a(fm)
a12m310	3.60	24	64	0.1213(9)
a09m310	3.78	32	96	0.0882(8)
a06m310	4.03	48	144	0.0574(5)
a04m310	4.20	64	192	0.0425(4)
a03m310	4.37	96	288	0.0318(3)

TABLE I. Setup of the MILC ensembles, including the bare coupling constant g, lattice size $L^3 \times T$, and lattice spacing a.

Our lattice calculations are performed using the Chroma software suite [60] and QUDA [61–63] in the HIP programming model [64]. We use 2+1+1 flavors (degenerate up and down, strange, and charm degrees of freedom) of highly improved staggered quarks (HISQ) [65] ensembles generated by the MILC Collaboration [46] at five lattice spacings, and 2+1 flavor domain wall (DW) quarks and Iwasaki gauge ensembles from the

symbol	$6/g^2$	L	T	a(fm)
DW11	2.13	24	64	0.1105(3)
DW08	2.25	32	64	0.0828(3)
DW06	2.37	32	64	0.0627(3)

TABLE II. Setup of the RBC ensembles, including the bare coupling constant g, lattice size $L^3 \times T$, and lattice spacing a.

RBC/UKQCD collaboration [47] at three lattice spacings. The pion mass for all ensembles are tuned to be roughly 310 MeV based on the pion mass of the light sea quark mass on the corresponding ensemble. The lattice spacings for the MILC ensembles are determined using Wilson flow based on the parameters determined by Ref. [66].

We use the matrix elements without hyper-cubic (HYP) smearing in order to test the perturbative renormalization analysis (Sec. IV, V, and VI). However, since in practical calculations the results without smearing can be rather noisy, it is standard to use some type of smearing in lattice simulations. It is unclear to us how strongly the smearing will interfere with renormalization. Therefore, we regard our method as a purely phenomenological approach for the time being to analyze data with one step HYP smearing [67] in Sec. VII and VIII, hoping that the gain in statistical precision outweighs the additional systematic uncertainty from moderate smearing.

We use two kinds of fermion actions for the valence quarks: clover and overlap fermions. Clover fermions break chiral symmetry and the action is defined by

$$\begin{aligned} S_q^{clv} = \sum_x \bar{\psi}(x)\Big[&\frac{1-\gamma_\mu}{2a}U_\mu(x,x+\hat{n}_\mu)\psi(x+\hat{n}_\mu) \\ &+\frac{1+\gamma_\mu}{2a}U_\mu(x,x-\hat{n}_\mu)\psi(x-\hat{n}_\mu) \\ &-\left(\frac{4}{a}+c_{sw}\sigma_{\mu\nu}F_{\mu\nu}(x)a+m_q^0\right)\psi(x)\Big], \end{aligned} \tag{15}$$

where $\hat{n}_\mu$ is the unit vector along the μ direction, the clover coefficient c_{sw} is the tadpole improved tree level value, and $m_q^0 a$ is the bare quark mass which is determined by requiring the pion mass to be roughly 310 MeV. Parameters c_{sw} and $m_q^0 a$ should approach 1 and 0 respectively in the continuum, while the lattice spacing dependence is weaker than the $\mathcal{O}(a)$ discretization effect in the present range of a, and closer to that of the gauge coupling g^2 as predicted by lattice perturbative theory.

The overlap action preserves chiral symmetry but the simulation is rather expensive. We use it to test the dependence of renormalization on the fermion action. The overlap fermion action is defined by [68, 69]

$$S_q^{\text{ov}} = \sum_{x,y}\bar{\psi}(x)D_{\text{ov}}(x,y)\psi(y), \tag{16}$$

$$D_{\text{ov}} = \rho\Big(1+\frac{D_{\text{w}}(-\rho)}{\sqrt{D_{\text{w}}^\dagger(-\rho)D_{\text{w}}(-\rho)}}\Big),$$

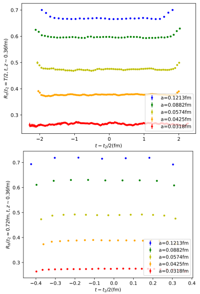


FIG. 1. The bare ratio $R_\pi(t_2 = T/2, t, z)$ for the pion (upper panel) and $R_N(t_2 \simeq 0.72\text{fm}, t, z)$ for the nucleon (lower panel) with $z \sim 0.36$ fm. For the pion case, we can see that the data in the region between $t - t_2/2 \in [-1, 1]$ fm are consistent with each other up to statistical fluctuations. For the nucleon case the ratio is also flat around $t \sim t_2/2$.

where

$$\begin{aligned} D_{\text{w}}(m_q^{\text{w}}; x, y) = &\tfrac{1-\gamma_\mu}{2a}U(x,x+\hat{n}_\mu)\delta_{x+\hat{n}_\mu,y} \\ &+\tfrac{1+\gamma_\mu}{2a}U(x,x-\hat{n}_\mu)\delta_{x-\hat{n}_\mu,y} - (\tfrac{4}{a}+m_q^{\text{w}})\delta_{x,y}, \end{aligned} \tag{17}$$

and $-\rho$ should be smaller than the bare quark mass for which vanishes the pion mass to make D_{ov} to be the same as the standard Dirac operator in the continuum limit. We choose $-\rho = 1.5$.

Although different active and sea fermion formulations will in general introduce non-unitarity, the effect shall vanish in the continuum limit. However, at finite a, the difference will generate systematic uncertainties which can affect the final results.

We start with nucleon and pion matrix elements for the MILC ensembles and clover action. For the pion matrix element, We average the $R_\pi(t_2, t, z)$ data with $t_2 = T/2$ and $t \in [T/8, 3T/8]$ to get a precise estimate of the ground state matrix element, and it should be also precise since T is between 7.7 and 9.1 fm in the ensembles we used, as shown in the upper panel of Fig. 1. Note

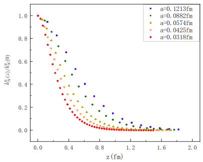


FIG. 2. The normalized bare nucleon matrix element in the rest frame, $\tilde{h}_N^0(z)/\tilde{h}_N^0(0)$ as the function of the Wilson link length z. The matrix element has a fast exponential decay at large z and the decay process accelerates with smaller lattice spacings.

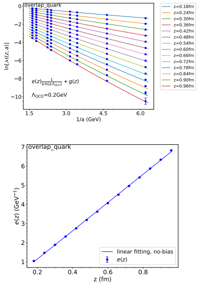


FIG. 3. Upper Panel: Using Eq. (18) to fit the bare $O_{\gamma_t}(z)$ matrix element in the off-shell quark state for overlap action without HYP smearing. Blue points are interpolated data and colorful curves are fitted curves for each z. $\Lambda_{\rm QCD}$ is fixed at 0.2 GeV. Lower Panel: $e(z)$ with respect to z. Blue points are the fitted parameters $e(z)$ from the upper panel. The blue curve is the no-bias linear fit for $e(z)$.

that $R_\pi(T/2, T/4, z)$ equals to $\langle\pi|O_{\gamma_t}(z)|\pi\rangle/2$ in such a case. The additional factor 1/2 corrects for the fact that there are forward and backward propagating states for (anti)periodic boundary conditions. The nucleon case is known to be much noisier especially at large t_2. Thus, we just consider $t_2 = 2t \simeq 0.72$ fm in all cases, and plot $R_N(t_2, t, z)$ in the lower panel of Fig. 1.

The normalized bare $\tilde{h}_N^0(z)/\tilde{h}_N^0(0)$ where $\tilde{h}_N^0(z)$ is approximated by $R_N(t_2, t, z)$ with $t_2 = 2t \simeq 0.72$ fm is plotted in Fig. 2, with a normalization factor $1/\tilde{h}_N^0(0)$ using jackknife resampling to make it to be exactly one at $z = 0$ at all lattice spacings. As in the figure, the linear divergence makes $\tilde{h}_N^0(z)/\tilde{h}_N^0(0)$ decay exponentially with both z and a, and thus one cannot obtain any meaningful continuum limit when $a \to 0$. A renormalization is necessary to recover a good continuum limit.

IV. SIMPLE TEST OF LINEAR DIVERGENCE

To study the renormalization properties of the quasi-PDF operator, we need to calculate matrix elements at several different lattice spacings. To ensure the data to be useful for a refined high-precision analysis, we start by testing whether they approximately show the linear divergence predicted by perturbation theory. In particular, one needs to show that the z-dependence of the linearly divergent term is linear.

To achieve this, we first extract the factor $e^{-\delta\bar{m}z}$ from the bare matrix element to test the $1/a$ dependence in $\delta\bar{m}$ in Eq. (5). Based on Eqs. (4), (5), and (6), if we take the natural log on a bare matrix element $\mathcal{M}$, we have,

$$\ln \mathcal{M}(z, a) = \frac{e(z)}{a \ln[a\Lambda_{\rm QCD}]} + g(z), \qquad (18)$$

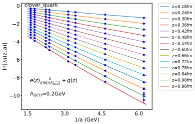
where the first term is linearly divergent and $g(z)$ is the residual. We have ignored the logarithmic divergence and discretization error here because these terms are much smaller and thus have little influence on this simple test of the linear divergence.

We use Eq. (18) to fit the bare matrix elements as a function of a. We treat $e(z)$ and $g(z)$ as unknown functions of z so that our fitting is performed at each value of z at which they are treated as free parameters. We need to do linear interpolation on the data $\ln \mathcal{M}(z, a)$ with respect to z. Here are the steps we perform in detail:

1. We do the linear interpolation with neighbourhood two data points to obtain the central value and uncertainty at $z = 0.06 \times n$ fm ($n = 3, 4, 5, ..., 16$).

2. For each z, we fit the dependence on a, treating $e(z)$, $g(z)$ as fitted parameters. For this initial test, we fix $\Lambda_{\rm QCD}$ at 0.2 GeV, which is a reasonable guess to start with [70].

3. We plot the fitted $e(z)$ with respect to z to see if $e(z)$ has a linear z dependence.



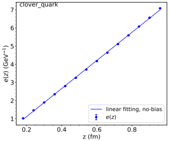


FIG. 4. Same as Fig. 3, except for the clover valence quark.

If we obtain a reasonable fit in step 2, we then see approximately the $1/a$ dependence of the divergences. In step 3, we can test the linear z dependence. The fitting results for the bare $O_{\gamma_t}(z)$ matrix element in the off-shell quark state for valence overlap and clover actions without HYP smearing are shown in Fig. 3 and Fig. 4, respectively. There are eight different lattice spacings. The range of z is taken from 0.18 fm to 0.96 fm and we analyse 14 different z values in this range.

The $1/a$ dependence in both cases is fitted very well, although due to the high precision of the data, chi-square is large. However, this is not our concern at this initial step. The z dependence of the divergent coefficients shows nicely the linear feature, approximately going through zero at $z = 0$. This is a strong indication that the linear divergence follows roughly the prediction of perturbation theory which gives us confidence to perform more refined analyses in the next sections.

V. SELF-RENORMALIZATION OF LATTICE MATRIX ELEMENTS

A. The Strategy

The non-perturbative matrix elements contain the following important contributions: a) the linear divergence, b) a finite term coming from non-perturbative renormalon physics, and c) the residual contribution encoding intrinsic non-perturbative physics. It is part c) that is required to extract the partonic structure information. Thus, it is important to separate out the latter from the former in a systematic way and with high precision.

Here we develop a self-renormalization method to extract the residual intrinsic non-perturbative physics and renormalization factor from a matrix element itself without using a different matrix element. The extraction of the residual is much harder than the extraction of the linear divergence term because after we take the logarithm of the matrix element (Eq. (18)), the linear divergence term is dominant and the residual is very sensitive to the subtraction. We need to properly take into account the fine features of the data, including logarithmic divergences and discretization errors.

Based on Eq. (11), we modify the fitting function Eq. (18) to be,

$$\ln \mathcal{M}(z,a) = \frac{kz}{a\ln[a\Lambda_{\rm QCD}]} + g(z) + f_{1,2}(z)a$$
$$+\frac{3C_F}{b_0}\ln\left[\frac{\ln[1/(a\Lambda_{\rm QCD})]}{\ln[\mu/\Lambda_{\rm QCD}]}\right] + \ln\left[1+\frac{d}{\ln(a\Lambda_{\rm QCD})}\right], \quad (19)$$

where the first term is linearly divergent. $g(z)$ is the residual, which contains the non-perturbative m_0 effect and the intrinsic non-perturbative physics. $f_{1,2}(z)a$ takes into account the discretization errors, which we allow to be different for the data calculated from MILC and RBC ensembles ($f_1(z)$ for MILC and $f_2(z)$ for RBC). The last two terms come from the resummation of leading and sub-leading logarithmic divergences, which only affect the overall normalization at different lattice spacings.

We use Eq. (19) to fit the logarithms to the data for the bare matrix elements for each z. The z points can be chosen in a large range where the lattice discretization error is small and, at the same time, the statistical error is limited. We fit the dependence on a, treating $g(z)$, $f_1(z)$, $f_2(z)$ as free parameters and fixing μ at 2 GeV, which is the renormalization scale we choose to use. We will discuss k and $\Lambda_{\rm QCD}$ in detail in Sec. V B and d in Sec. V C. We obtain the residual $g(z)$ from the fit.

The $g(z)$ obtained this way does not correspond to what is in the perturbative $\overline{\rm MS}$ scheme because it contains a non-perturbative m_0 effect. We need to eliminate this effect through matching the lattice result to the continuum scheme at short distance z. Based on Eq. (4) and Eq. (7), we use the following equation within a window

$a \ll z < 1/\mu$ to extract m_0,

$$g(z) - \ln[Z_{\overline{\rm MS}}(z, \mu, \Lambda_{\overline{\rm MS}})] = m_0 z, \tag{20}$$

where $Z_{\overline{\rm MS}}$ is the perturbative matrix element in the $\overline{\rm MS}$ scheme. For the $O_{\gamma_t}(z)$ matrix elements in the pion, nucleon, and off-shell quark state, we take $Z_{\overline{\rm MS}}$ at one loop [42],

$$Z_{\overline{\rm MS}}(z, \mu, \Lambda_{\overline{\rm MS}}) = 1 + \frac{\alpha_s(\mu, \Lambda_{\overline{\rm MS}}) C_F}{2\pi} \left[\frac{3}{2} \ln \frac{z^2 \mu^2}{4 e^{-2\gamma_E}} + \frac{5}{2} \right], \tag{21}$$

from operator product expansion, where we fix $\mu = 2$ GeV and $\Lambda_{\overline{\rm MS}} = 0.3$ GeV. The renormalized physical matrix element is then obtained from

$$\mathcal{M}(z)_R = \exp[g(z) - m_0 z], \tag{22}$$

and is expected to be consistent with $Z_{\overline{\rm MS}}$ at small z.

One could consider higher-order $\overline{\rm MS}$ matrix elements for matching, including summing over large logarithms. We choose not to do this here as it does not affect the procedure for the subsequent analysis, and therefore does not change the main conclusions of the paper. However, to obtain physical results at high precisions, one might need to examine this more carefully.

We thus define a renormalization factor

$$Z(z,a)_R = \exp\left[\frac{kz}{a \ln[a\Lambda_{\rm QCD}]} + m_0 z + f_{1,2}(z) a + \frac{3C_F}{b_0} \ln\left[\frac{\ln[1/(a\Lambda_{\rm QCD})]}{\ln[\mu/\Lambda_{\rm QCD}]} \right] + \ln[1 + \frac{d}{\ln(a\Lambda_{\rm QCD})}] \right], \tag{23}$$

which includes the discretization error as well. All the parameters here are either fixed, fitted or fine-tuned through the renormalization procedure from the matrix element $\mathcal{M}(z,a)$ itself. Dividing the bare matrix element $\mathcal{M}(z,a)$ by $Z(z,a)_R$, we get

$$\widetilde{\mathcal{M}}_R(z) = \mathcal{M}(z,a)/Z(z,a)_R. \tag{24}$$

If our procedure eliminates all the divergences and discretization errors, we should expect that $\widetilde{\mathcal{M}}_R(z)$ is a-independent and the same as $\mathcal{M}(z)_R$ (Eq. (22)). The degree to which this is the case can be used as a test of the self-renormalization procedure.

B. Renormalon Uncertainty

Now let us turn to the parameter k and $\Lambda_{\rm QCD}$ in Eq. (19). In Eq. (19), we have only considered the leading order perturbative term for the linear divergence and neglected higher-order ones. These can be included partially by a proper choice of $\Lambda_{\rm QCD}$. For example, the coupling constants with different choices of Λ are related to each other by perturbation theory [70],

$$\alpha_s(Q, \Lambda_{\rm QCD}) \sim \alpha_s(Q, \Lambda) + c_2 \alpha_s^2(Q, \Lambda) + c_3 \alpha_s^3(Q, \Lambda) + \ldots \tag{25}$$

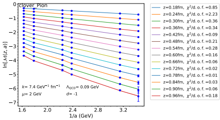


FIG. 5. Using Eq. (19) to fit the bare $O_{\gamma_t}(z)$ matrix element in the pion state for the clover action without HYP smearing. Blue points are interpolated data and colorful curves are fitted curves for each z. The fitted curves are broken lines because $f_1(z)$ and $f_2(z)$ are allowed to be different. k, $\Lambda_{\rm QCD}$, and d are fixed at 7.4 $\mathrm{GeV}^{-1}\mathrm{fm}^{-1}$ (1.46 if dimensionless), 0.09 GeV, and -1 respectively. The χ^2/d.o.f. of the fitting for each z is listed in the plot legend.

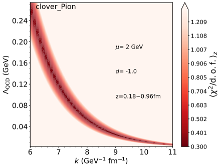


FIG. 6. χ^2 map with respect to k and $\Lambda_{\rm QCD}$. $\langle \chi^2/\text{d.o.f.} \rangle_z$ is the average of χ^2/d.o.f. among the fitting for each z. The "small-χ^2 band" shows that k and $\Lambda_{\rm QCD}$ are strongly correlated and uncertain for the chosen data.

In principle, $\Lambda_{\rm QCD}$ depends on the specific lattice action, and one needs to perform multi-loop calculations to relate them. In our analysis, we treat $\Lambda_{\rm QCD}$ as a fitting parameter which, therefore, can effectively take into account some higher-order effects.

Next we want to discuss the coefficient k of the linear divergence. QCD perturbation theory not only predicts the $1/a$ dependence and linear z dependence (which we have tested in Sec. IV), but also the value of k. Comparing Eq. (5) and Eq. (19), we get the one-loop perturbative value of k:

$$k = \frac{(2\pi)^2}{3b_0} = 1.46 = 7.4 \mathrm{GeV}^{-1}\mathrm{fm}^{-1}. \tag{26}$$

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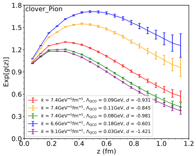

FIG. 7. The fitted residual Exp$[g(z)]$ for different sets of (k, $\Lambda_{\rm QCD}$) along the "small-χ^2 band".

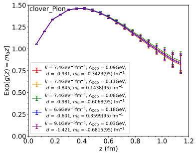

FIG. 8. The renormalized matrix element Exp$[g(z) - m_0 z]$ for different sets of (k, $\Lambda_{\rm QCD}$) along the "small-χ^2 band".

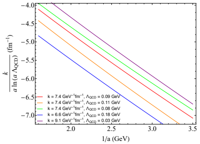

FIG. 9. The linear divergent term in the a range of our data for different sets of (k, $\Lambda_{\rm QCD}$) along the "small-χ^2 band".

Next, we check whether this value is consistent with our data.

We use Eq. (19) to fit the bare $O_{\gamma_t}(z)$ matrix element in the pion state for the clover action without HYP smearing, treating $g(z)$, $f_1(z)$, $f_2(z)$ as fitted parameters. If we take (k, $\Lambda_{\rm QCD}$) to be one set of values, e.g., (7.4 GeV^{-1}fm^{-1}, 0.09 GeV), we can perform a fitting for each z, as seen in Fig. 5. There is a χ^2/d.o.f. for the fit at each z and we calculate the average of them, denoted as $\langle\chi^2/\text{d.o.f.}\rangle_z$. Varying ($k$, $\Lambda_{\rm QCD}$) generates a χ^2 map, as seen in Fig. 6.

The sets of (k, $\Lambda_{\rm QCD}$) of small χ^2 lie in a band, which we call the "small-χ^2 band". That means that k and $\Lambda_{\rm QCD}$ are strongly correlated, while there is a large uncertainty for k and $\Lambda_{\rm QCD}$ separately. If we choose some sets of values from the "small-χ^2 band" to do the fitting, we find the fitted residuals Exp$[g(z)]$ are very different, as seen in Fig. 7. However, after eliminating the non-perturbative m_0 effect, the renormalized matrix elements Exp$[g(z) - m_0 z]$ are all the same (see Fig. 8). Therefore, the uncertainty in k and $\Lambda_{\rm QCD}$ will not significantly influence the final physical result.

The reason for this uncertainty lies in the behavior of $\frac{k}{a\ln[a\Lambda_{\rm QCD}]}$ (the term related to the linear divergence in Eq. (19)) in the a range of our data. Varying along the "small-χ^2 band" effectively shifts $\frac{k}{a\ln[a\Lambda_{\rm QCD}]}$ by a constant C, see Fig. 9. The Cz term is absorbed into $g(z)$ automatically during fitting so there is a large uncertainty for k and $\Lambda_{\rm QCD}$. However, when we eliminate the m_0 effect, the Cz term has been taken into account, and the final result does not change.

Another way to interpret this is that the perturbative value of k is consistent with the "small-χ^2" band and therefore is consistent with the fit. As a consequence, we can fix k at the one-loop perturbative value, 7.4 GeV^{-1}fm^{-1} (or 1.46 if dimensionless) in the subsequent analysis. Even with this choice of k, $\Lambda_{\rm QCD}$ can still vary within a reasonable range of χ^2 while the physical result is stabilized by the choice of m_0. We call this correlation between $\Lambda_{\rm QCD}$ and m_0 the "phenomenological renormalon uncertainty", in the sense that $\Lambda_{\rm QCD}$ is characteristic for the different truncation/resummation schemes for the perturbation series, but that the non-perturbative m_0 compensates the effects of the differences [49–52].

C. Tuning Parameter-d to Match the Continuum Scheme

In principle, the parameter d in Eq. (19) is determined by the lattice action. In our data, there are two different dynamical lattice ensembles (MILC and RBC) and two different valence fermion actions (overlap and clover). Here we treat d as a fine-tuning parameter to make sure that $(g(z) - \ln[Z_{\overline{\rm MS}}(z,\mu,\Lambda_{\overline{\rm MS}})])$ in Eq. (20) is propor-

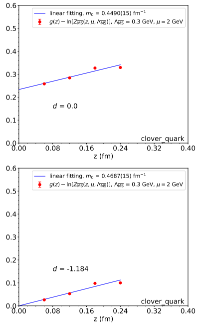

FIG. 10. The fitting to extract m_0 based on Eq. (20) for the bare $O_{\gamma_t}(z)$ matrix element in the off-shell quark state for the clover action without HYP smearing. Red points are $g(z) - \ln[Z_{\overline{\rm MS}}]$, where $g(z)$ is extracted through fitting the data for the bare matrix element using Eq. (19) and $Z_{\overline{\rm MS}}$ given in Eq. (21). The blue line is the linear fit of the red points (where the fitting function is $m_0 z + C_0$). The fitted m_0 is given in the plot legend. k is fixed at 7.4 GeV^{-1}fm^{-1} (1.46 if dimensionless) and $\Lambda_{\rm QCD}$ is chosen as the best fit value 0.118 GeV. In the upper panel, d is zero and $g(z) - \ln[Z_{\overline{\rm MS}}]$ is in a linear relationship with z but not proportional to z. (It does not go through the origin). In the lower panel, we tune d to make sure that $g(z) - \ln[Z_{\overline{\rm MS}}]$ is proportional to z.

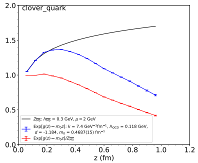

FIG. 11. Renormalized $O_{\gamma_t}(z)$ matrix element in the off-shell quark state for the clover action without HYP smearing. Blue points are the renormalized matrix element $\mathrm{Exp}[g(z) - m_0 z]$. The black curve denotes $Z_{\overline{\rm MS}}$ and the red points are their ratio. k is fixed at 7.4 GeV^{-1}fm^{-1} (1.46 if dimensionless) and $\Lambda_{\rm QCD}$ is chosen as the best fitted value 0.118 GeV. The parameter d is fine-tuned to be -1.184.

tional to z within a window $a \ll z < 1/\mu$. As shown in Fig. 10, this is quite effective. We have also tested that d has little influence on χ^2 since the effect of tuning d shifts $g(z)$ simply by a constant.

Fig. 11 shows that the renormalized matrix element $\mathrm{Exp}[g(z) - m_0 z]$ (blue) for clover quarks is consistent with $Z_{\overline{\rm MS}}$ (black) at small z. However, at large z, there is a significant discrepancy between $\mathrm{Exp}[g(z) - m_0 z]$ and $Z_{\overline{\rm MS}}$. That means that there is a significant non-perturbative effect at large z in the popular RI/MOM and ratio renormalization schemes used previously. To our knowledge, this is the first time that this difference has been studied in the literature. This finding supports the usage of the hybrid renormalization procedure proposed recently [39].

D. Self-Renormalization Procedure

The procedure developed in the previous subsections forms our self-renormalization strategy. This procedure can overcome some of the problems encountered in the renormalization of the linear divergence due to the numerical uncertainties associated with exponentially-amplified small errors. There are three steps in this process:

1. Use Eq. (19) to fit the data of the bare matrix elements $\mathcal{M}$. For each z (here z can be chosen in a large range and linear interpolation with respect to z might be needed for the data in $\ln\mathcal{M}(z, a)$), fit the dependence on a, treating $\Lambda_{\rm QCD}$ as a global parameter, namely the same for all z, and $g(z)$, $f_1(z)$, $f_2(z)$ as fit parameters, with fixing k=7.4 GeV^{-1}fm^{-1} (or 1.46 if dimensionless), and μ=2 GeV. Fine-tune d to make sure that $g(z) - \ln[Z_{\overline{\rm MS}}]$ is proportional to z within a window $a \ll z < 1/\mu$. We obtain the residual $g(z)$ through fitting;

2. Use Eq. (20) to fit the dependence on z (within a window $a \ll z < 1/\mu$) to extract m_0. Then calculate the renormalized matrix element $\mathrm{Exp}[g(z) - m_0 z]$ (Eq. (22)), which is considered valid for a large range of z;

3. Calculate $\mathcal{M}(z, a)/Z(z, a)_R$ (Eq. (24)) to see if showing any dependence on the lattice constant a and compare with $\mathrm{Exp}[g(z) - m_0 z]$.

In step 1 and 2, one can use another way to get d and m_0: Modify Eq. (19) through replacing $g(z)$ with $(\ln[Z_{\overline{\rm MS}}(z,\mu,\Lambda_{\overline{\rm MS}})]+m_0z)$, and then use the modified fitting function to fit the bare matrix element at small z to extract d and m_0. This way should be equivalent to the procedure outlined above and we will not follow it here.

In step 3, we need to calculate $Z(z,a)_R$ as well as estimate its error. We shall not propagate the fitting error of $\Lambda_{\rm QCD}$ to $Z(z,a)_R$ because the effect of slight variation of $\Lambda_{\rm QCD}$ can be compensated by m_0 and will not influence our result as we have discussed in Sec. V B.

Since the lattice data we use do not provide systematic errors, we will add a dummy systematic error $\delta_{\rm sys}$ to the data during our fitting [45]

$$(\sigma_{\ln\mathcal{M}})_{\rm new}=\sqrt{(\sigma_{\ln\mathcal{M}})^2_{\rm old}+(\delta_{\rm sys}a\mu)^2}, \qquad (27)$$

where we assume that there is an $a\mu$ dependence as well. This choice is intended to give more weight to the small lattice spacing data in the fitting.

E. Comparing Results from Different Actions and Matrix Elements

In this subsection, we apply the self-renormalization method to the $O_{\gamma_t}(z)$ matrix elements in the pion, nucleon, and off-shell quark state with valence clover and overlap actions without HYP smearing. We compare the renormalization factors and physics results. We find that apart from the case of the RI/MOM matrix element with clover valence fermion, all other renormalization factors are similar. Despite this difference in renormalization factor, the physical results are independent of fermion actions.

Figs. 12 through 16 show the detailed results of applying the self-renormalization method for different cases. For the RI/MOM factor with overlap and clover fermions, we analyse eight different lattice spacings from two different types of gauge ensembles. For the pion matrix element calculated with the overlap fermion action, we analyse three different lattice spacings from MILC ensembles. In the clover pion case, we use six different lattice spacings from two different types of gauge ensembles. For the nucleon matrix element with clover fermion action, we analyse five different lattice spacings from MILC ensembles.

Subfigure (a) in each figure is used to estimate the best fitted value of the global parameter $\Lambda_{\rm QCD}$ as well as its error. For each chosen $\Lambda_{\rm QCD}$, we can use Eq. (19) to fit the bare matrix element. The fitting for each z gives us a separate χ^2. We can calculate the averaged χ^2 for different z, denoted as $\langle\chi^2\rangle_z$. Subfigure (a) shows $\langle\chi^2\rangle_z-[\langle\chi^2\rangle_z]_{\rm min}$ with respect to $\Lambda_{\rm QCD}$. $\langle\chi^2\rangle_z-[\langle\chi^2\rangle_z]_{\rm min}$=0 gives us the best fitted $\Lambda_{\rm QCD}$ and $\langle\chi^2\rangle_z-[\langle\chi^2\rangle_z]_{\rm min}$=1 gives us its error. If the data points for different z were independent, we should use the total χ^2 for the fits at different z to estimate the error of $\Lambda_{\rm QCD}$. However, since we have done linear interpolations of the original data points, some of the points are correlated to each other. So here we just use $\langle\chi^2\rangle_z$ to estimate the error of $\Lambda_{\rm QCD}$. The range of z starts from 0.18 fm here but not 0.06 fm because the χ^2 for $z=0.06$ or 0.12 fm is much larger than others. d is fixed to be -1 since it has little influence on χ^2.

For $\delta_{\rm sys}=0.002$, the best fitted values of $\Lambda_{\rm QCD}$ are 0.1086(17), 0.1350(21), 0.093(10), 0.086(14), 0.0926(61) GeV for the correlations in the overlap quark, clover quark, overlap pion, clover pion, and clover nucleon, respectively. As one can see, they are quite similar except for the correlation in the clover quark state. While the size of the systematic error has some effects for the correlation in the quark case, it has little influence on the correlation in the physical states.

In subfigure (b) in each figure, after taking $\Lambda_{\rm QCD}$ to be the best fitted value, we use Eq. (19) to fit the bare matrix elements to extract $g(z)$. The range of z is taken from 0.06 fm to 1.02 fm with 17 different z values for correlations in the overlap quark case, from 0.06 fm to 1.14 fm, with 19 different z values for correlations in the overlap pion and clover pion cases, from 0.06 fm to 0.96 fm, with 16 different z in the clover quark case, from 0.06 fm to 0.9 fm, and with 15 different z values in the clover nucleon case. We fine-tune d to make sure that $g(z)-\ln[Z_{\overline{\rm MS}}]$ is proportional to z within the window 0.06 fm $\le z \le$ 0.24 fm. The fine-tuned d values are $-1.29,-1.35,-1.17,-0.92,-0.97$ in the five cases, respectively. They are all close to -1.

Subfigure (c) in each figure shows the fitting to extract m_0. For $\delta_{\rm sys}=0.002$, the fitted values of m_0 are 0.2339(81), 0.7667(83), $-0.181(24)$, $-0.363(25)$, and $-0.257(20)$ fm^{-1} in the five cases, respectively. They are all about the order of $\Lambda_{\rm QCD}\sim 0.1$ GeV (about 0.5 fm^{-1}), which supports the argument in Sec. II that m_0 originates from the non-perturbative renormalon effect.

In subfigure (d) in each figure, we show our result for the renormalized matrix element ${\rm Exp}[g(z)-m_0z]$ in blue points connected by lines. As z increases, ${\rm Exp}[g(z)-m_0z]$ increases first, following the perturbative result, and then decreases from about $z=0.25$ fm on. In all the cases, ${\rm Exp}[g(z)-m_0z]$ is consistent with $Z_{\overline{\rm MS}}$ at small z. However, at large z, there is a significant discrepancy between ${\rm Exp}[g(z)-m_0z]$ and $Z_{\overline{\rm MS}}$. This means that there is a large non-perturbative effect at large z in the popular RI/MOM and ratio renormalization scheme used previously, which supports the hybrid renormalization procedure proposed recently [39]. We have already briefly discussed this in Sec. V C.

Subfigure (e) shows the renormalized matrix element ${\rm Exp}[g(z)-m_0z]$ for different $\delta_{\rm sys}$. A change in $\delta_{\rm sys}$ does not influence the central values of ${\rm Exp}[g(z)-m_0z]$ for most cases except for the clover pion. In the clover pion case, the increase of $\delta_{\rm sys}$ can slightly decrease the renormalized matrix element ${\rm Exp}[g(z)-m_0z]$.

Finally, subfigure (f) shows the ratio of bare matrix ele-

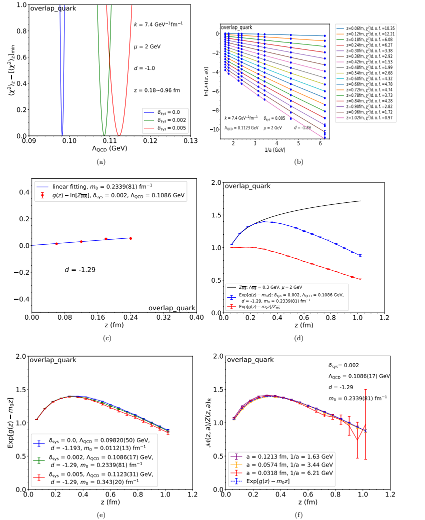

FIG. 12. Self-renormalizing the $O_{\gamma_t}(z)$ correlator in the off-shell quark state calculated for the overlap action without HYP smearing, using Eq. (19). (a) shows $\langle\chi^2\rangle_z - [\langle\chi^2\rangle_z]_{\rm min}$ with respect to $\Lambda_{\rm QCD}$. The best fitted $\Lambda_{\rm QCD}$ with its error can be estimated here. (b), (c) and (d) show the procedure to get the renormalized matrix element $\mathrm{Exp}[g(z) - m_0 z]$. (b) shows the fitting of the bare matrix element to extract $g(z)$. Blue points are interpolated data and colorful curves are fitted curves. (c) shows the fitting to extract m_0. (d) shows $\mathrm{Exp}[g(z) - m_0 z]$ (blue), compared with $Z_{\overline{\rm MS}}$ (black). (e) is the comparison of $\mathrm{Exp}[g(z) - m_0 z]$ for different $\delta_{\rm sys}$. (f) is the ratio of bare matrix element and renormalization factor (connected with solid lines), compared with $\mathrm{Exp}[g(z) - m_0 z]$ (connected with dashed lines).

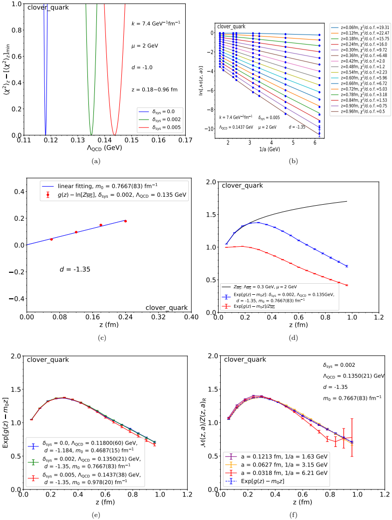

FIG. 13. Same as Fig. 12, with the $O_{\gamma_t}(z)$ correlator in the off-shell quark state calculated for the clover action.

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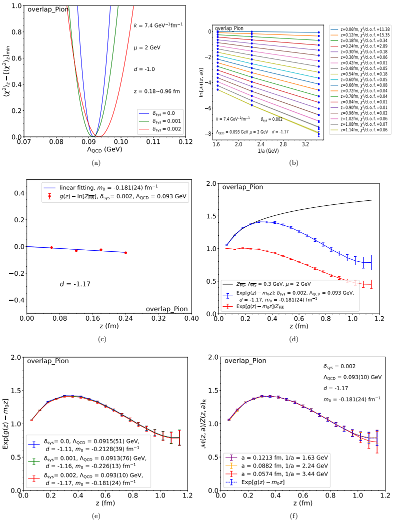

FIG. 14. Same as Fig. 12, with quasi-LF correlation in zero-momentum pion state calculated with the overlap fermion action.

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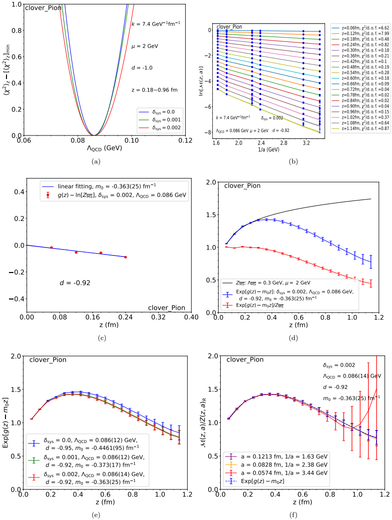

FIG. 15. Same as Fig. 12, with quasi-LF correlation in zero-momentum pion state calculated with the clover fermion action.

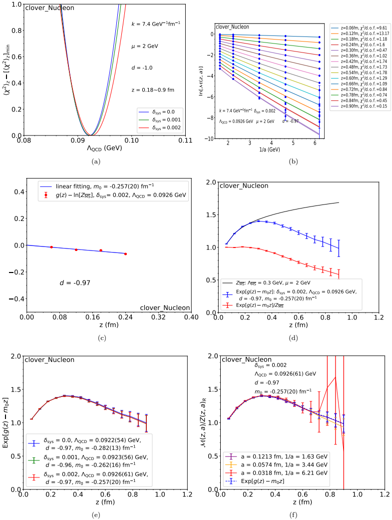

FIG. 16. Same as Fig. 12, with quasi-LF correlation in zero-momentum nucleon state calculated with the clover fermion action.

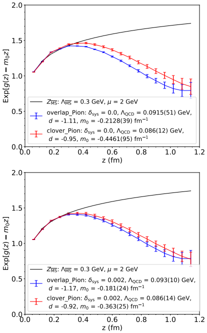

FIG. 17. Comparison of renormalized $O_{\gamma_t}(z)$ correlator in the zero-momentum pion state between overlap and clover actions without HYP smearing. Points connected by solid lines are renormalized matrix elements $\mathrm{Exp}[g(z)-m_0 z]$. Black curve is $Z_{\overline{\mathrm{MS}}}$. In the upper panel, $\delta_{\mathrm{sys}}=0$. In the lower panel, $\delta_{\mathrm{sys}}=0.002$.

ment and renormalization factor $\mathcal{M}(z,a)/Z(z,a)_R$, compared with the renormalized matrix element $\mathrm{Exp}[g(z)-m_0 z]$. $Z(z,a)_R$ here is not extracted from a different matrix elements but from $\mathcal{M}(z,a)$ itself. This subfigure is a consistency check of our method. In all cases, $\mathcal{M}(z,a)/Z(z,a)_R$ has little dependence on a within error and is consistent with $\mathrm{Exp}[g(z)-m_0 z]$. Therefore, our self-renormalization method can isolate the residual intrinsic physics from the linear divergence, renormalon uncertainty, log divergence and discretization error with a reasonable precision.

Although the ratio $\mathcal{M}(z,a)/Z(z,a)_R$ has a good behavior for most of the lattice spacings, it may have large errors for small lattice spacings like 0.0318 fm or 0.0574 fm. Therefore we do not recommend using $\mathcal{M}(z,a)/Z(z,a)_R$ as the renormalized matrix element.

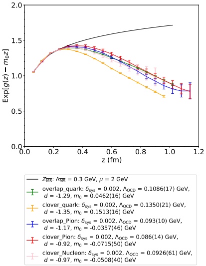

FIG. 18. Comparison of renormalized $O_{\gamma_t}(z)$ correlations for all the cases without HYP smearing. Points connected by solid lines are renormalized matrix elements $\mathrm{Exp}[g(z)-m_0 z]$. Black curve is $Z_{\overline{\mathrm{MS}}}$.

Since $\mathrm{Exp}[g(z)-m_0 z]$ is achieved through our fitting, the large-error data points have little influence on it. We take this as the renormalized matrix element, while using $\mathcal{M}(z,a)/Z(z,a)_R$ only to test consistency.

We show in Fig. 17 a comparison of renormalized $O_{\gamma_t}(z)$ correlation in the pion state between overlap and clover actions. Before we add the systematic error, both results show appreciable difference, which may indicate that two different valence fermions will lead to different results. After we add a small dummy systematic error during the fitting, the correlations in both cases lead to similar results. That means that the systematic error is important, and the residual intrinsic non-perturbative physics is independent of valence quark formulations if it is properly included.

Finally, for curiosity, we show in Fig. 18 the renormalized correlations as well as parameters related to the divergence for all cases we studied. As mentioned before, the fitted Λ_{QCD} for overlap quark, overlap pion, clover pion and clover nucleon are similar to each other. But for the clover quark, it is markedly different. We do not yet understand the reason, but it could be due to the combination of chiral symmetry breaking and the linearly-divergent quark mass. We plan to investigate this in the future, but for the time being, at least we understand why we cannot eliminate the linear divergence using RI/MOM for the clover action [45]. Quite surprisingly, the residual intrinsic non-perturbative correlations for overlap quark, pion and nucleon are very similar to each other, which we also do not fully understand.

VI. FITS WITH OTHER OPTIONS AND STABILITY

In a phenomenological analysis, the amount of information and the accuracy one can obtain depends, obviously, on the quality of the lattice data. Given the data we have, we would like to push the limit of the analysis by including more physics in the fit until the analysis no longer yields useful information.

The first question we would like to address is whether one can get more accurate information about the linear divergence. We have used so far the one-loop perturbative QCD prediction, but treated $\Lambda_{\rm QCD}$ as a free parameter to partially take into account higher order corrections. However, one might consider directly including at least the second-order corrections. Our strategy here is to fix k to its perturbative value, and to vary $\Lambda_{\rm QCD}$ and the explicit second order correction characterized by a new parameter λ

$$\ln \mathcal{M}(z,a) = \frac{k' z}{a}\tilde{\alpha_s}(1+\lambda\tilde{\alpha_s}) + g(z) + ... \qquad (28)$$

where the terms omitted are the same as those we had before in Eq. (19). k' is related to k by $-2\pi/b_0$ (since $-k'2\pi/b_0 = k$). On the other hand, one might use the λ parameter to include some higher order corrections [51],

$$\ln \mathcal{M}(z,a) = \frac{k' z}{a}\frac{\tilde{\alpha_s}}{1-\lambda\tilde{\alpha_s}} + g(z) + ... \qquad (29)$$

To be consistent, we now have to use the QCD running coupling constant $\tilde{\alpha_s}$ up to the second order β function $b_1 = 102 - \frac{38}{3}n_f$,

$$\tilde{\alpha_s} = \frac{4\pi}{b_0 \ln(\frac{1}{a^2\Lambda^2_{\rm QCD}})}\left(1 - \frac{b_1}{b_0^2}\times\frac{\ln[\ln(\frac{1}{a^2\Lambda^2_{\rm QCD}})]}{\ln(\frac{1}{a^2\Lambda^2_{\rm QCD}})}\right). \qquad (30)$$

Therefore, again we have two global parameters λ and $\Lambda_{\rm QCD}$ in the fit.

The fitting process with the above parametrization is similar to what is described in Sec. V D. The results are shown in Figs. 19 and 20. As one can see, the range of λ and $\Lambda_{\rm QCD}$ are rather large, and strongly correlated. After picking several correlated values of the pair, matching to the perturbative result, the resulting residual matrix elements are shown in the lower panel. One can hardly see any difference between the different choices. Moreover, the results are essentially the same as the fit without the extra λ parameter (Fig. 22). Therefore, we conclude that

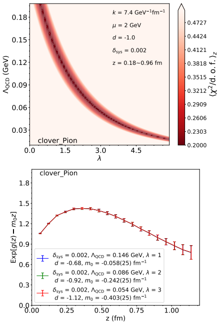

FIG. 19. Using Eq. (28) to renormalize the $O_{\gamma_t}(z)$ matrix element in the pion state for the clover action without HYP smearing. Upper Panel: χ^2 map with respect to λ and $\Lambda_{\rm QCD}$. $\langle\chi^2/\text{d.o.f.}\rangle_z$ is the average of $\chi^2/\text{d.o.f.}$ from fitting for each z. Lower Panel: renormalized matrix element for several sets of $(\Lambda_{\rm QCD}, \lambda)$ along the 'small-χ^2 band'.

with the data at hand, our result is stable with respect to the higher order corrections. Only with more accurate data, one might be able to see the difference of a higher-order fit.

Another possibility one can explore is that the higher-order corrections represented by $\Lambda_{\rm QCD}$ might be different for different ensembles. Therefore, one can discuss different $\Lambda_{\rm QCD}$ values for MILC and RBC data,

$$\ln \mathcal{M}(z,a) = \frac{kz}{a\ln[a\Lambda_{\rm QCD1,2}]} + g(z) + f_{1,2}(z)a$$
$$+\frac{3C_F}{b_0}\ln[\frac{\ln[1/(a\Lambda_{\rm QCD1,2})]}{\ln[\mu/\Lambda_{\rm QCD1,2}]}] + \ln[1+\frac{d}{\ln(a\Lambda_{\rm QCD1,2})}], \qquad (31)$$

where we fix the k parameter and obtain a two parameter fit, $\Lambda_{\rm QCD1}$ and $\Lambda_{\rm QCD2}$. The results of using the

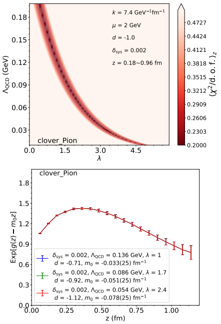

FIG. 20. Same as Fig. 19, fit using Eq. (29).

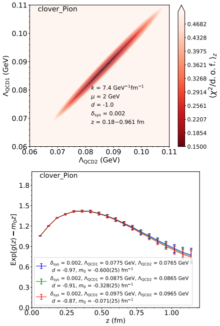

FIG. 21. Using Eq. (31) to renormalize the $O_{\gamma_t}(z)$ matrix element in the pion state for the clover action without HYP smearing. Upper Panel: χ^2 map with respect to $\Lambda_{\rm QCD1}$ and $\Lambda_{\rm QCD2}$. $\langle\chi^2/\text{d.o.f.}\rangle_z$ is the average of χ^2/d.o.f. from fitting for each z. Lower Panel: renormalized matrix element for several sets of ($\Lambda_{\rm QCD1}$, $\Lambda_{\rm QCD2}$) along the 'small-χ^2 band'.

above equation is shown in Fig. 21. As shown, the two parameters are strongly correlated and proportional to each other. The difference of the two is limited to a very small range. Furthermore, with different choices of the correlated pair ($\Lambda_{\rm QCD1}$, $\Lambda_{\rm QCD2}$), the final renormalized results do not change.

In Fig. 22, we show the results of renormalized matrix elements based on different fitting functions. All of these fitting functions lead to the same result. Thus, given the data, it is unnecessary to consider other high-order terms in the fitting function and Eq. (19) is good enough. This also supports the argument made in Sec. V B that we can use the fitted $\Lambda_{\rm QCD}$ in leading order to partially represent higher order corrections.

Finally, we consider $O(a^2)$ corrections to our fitting formula. These fits were unstable, which indicates that we introduce too many fit parameters for the given data quality. The conclusion is that higher precision data is needed to constrain $O(a^2)$ terms. One can also make fits by including $O(a^2)$ instead of the linear correction. The physics for doing it is weak in our cases.

VII. RENORMALIZATION OF LATTICE SMEARED MATRIX ELEMENTS

In many lattice calculations, the data can be very noisy which calls for more statistics. However, large statistics means more resources which are hard to come by sometimes. Therefore, lattice practitioners have invented phenomenological approaches to quench the short-distance fluctuations: smearing. However, smearing might lead to configurations that are not connected with fundamental theory. On the other hand, smearing in some sense just makes the effective UV cutoff (the inverse of the lattice spacing) smaller, and in the limit $a \to 0$, all smearing becomes local effects which can be taken into account properly through some kind of renormalization. In this section, we check if the procedure we discussed in the

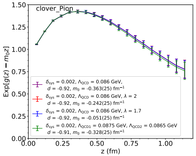

FIG. 22. Renormalized $O_{\gamma_t}(z)$ correlations in the pion state for the clover action without HYP smearing based on different fitting functions: Eq. (19) (purple), Eq. (28) (red), Eq. (29) (blue), Eq. (31) (green).

previous section still works for smeared matrix elements.

To ensure that the HYP smearing data are useful for refined analysis, we can do a simple test on whether the HYP smearing data show the properties of the linear divergence predicted by perturbation theory, as we have done in Sec. IV. Here we use the following function to fit the bare matrix element $\mathcal{M}$ to extract the linear divergence factor,

$$\ln \mathcal{M}(z,a) = \frac{e(z)}{a \ln[a\Lambda_{\rm QCD}]} + g(z) + f_{1,2}(z)a, \qquad (32)$$

where we allow for a discretization error to get a better fit. We fix $\Lambda_{\rm QCD}$ at 0.39 GeV in this simple test. Fig. 23 shows the fits for different actions and states which all look reasonable. We have not shown χ^2 which will be considered in the extraction of the renormalization factor.

Fig. 24 shows the extracted $e(z)$ with respect to z. The linear z dependence is approximately reproduced but some deviations are seen at large z. We can perform a linear fitting of $e(z)$ and compare the fitted slopes. Overall, the quality of the fits is not as good as in the unsmeared cases, although it is still rather impressive. The slope extracted for the linear divergence for the clover quark is totally different from the others, which explains why we cannot use the RI/MOM factor to eliminate it in the clover case for HYP smearing data [45]. Although the slopes for the overlap quark and pion, clover pion and nucleon are similar to each other, there are some small discrepancies. Using the RI/MOM factor for these cases may leave small residual linear divergences, and the self-renormalization method provides a better way to renormalize the matrix elements.

Next, we extend our self-renormalization method to the HYP smeared data. Our fitting function, Eq. (19) may have no clear physical meanings for the HYP smeared date, but it may be regarded as a phenomenological model.

We start with a fit function with two parameters in the linearly divergent part, k and $\Lambda_{\rm QCD}$. Fig. 25 shows the χ^2 map with respect to them. It is clear from the plots that in the "small-χ^2 band", we cannot find a solution for k with the one-loop perturbative value 7.4 GeV^{-1}fm^{-1} (1.46 if dimensionless, see Eq. (26)) any more. This is expected because the lattice space is enlarged by a factor of two, and therefore the values of k in the "small-χ^2 band" are expected to be at most half of this, named 3.7 GeV^{-1}fm^{-1} (0.73 if dimensionless). In fact, most probable k is now around 3.0 GeV^{-1}fm^{-1} (0.59 if dimensionless) or below. Another way of saying this is HYP smearing smooths out the short-range behaviors such as the linear divergence. The value of $\Lambda_{\rm QCD}$, on the other hand, has a larger range, possibly corresponding again to a larger effective lattice spacing.

So for the HYP smearing cases, we choose a set of (k, $\Lambda_{\rm QCD}$) near the center of the "small-χ^2 band". We choose $k = 2.5$ GeV^{-1}fm^{-1} (0.49 if dimensionless) for overlap quark and clover quark, and $k = 2.6$ GeV^{-1}fm^{-1} (0.51 if dimensionless) for overlap pion, clover pion and clover nucleon. The corresponding $\Lambda_{\rm QCD}$ shows a larger dispersion, with $\Lambda_{\rm QCD} = 0.43$ and 0.38 GeV for overlap quark and pion correlations respectively, and $\Lambda_{\rm QCD} = 0.61, 0.37$ and 0.35 GeV for clover quark, pion and nucleon correlations, respectively.

Once the (k, $\Lambda_{\rm QCD}$) parameters are chosen, we subtract the linear divergences, and match the result to the perturbative matrix elements to determine possible further subtractions for the non-perturbative mass effect. Fig. 26 shows that the renormalized matrix elements for HYP smearing in blue ticks and lines, in comparison with the cases without HYP smearing.

The large difference between smeared and unsmeared matrix elements is seen in the case of clover quark correlation at large z. Again, we speculate that due to the chiral symmetry breaking effects, the linear divergence has some curious behavior there. However, the difference is smaller in the pion matrix element where the error bars are also bigger. For the overlap case, the correlations for quark and pion show no substantial change due to smearing effects. On the other hand, the matrix element in the nucleon case shows a much smaller error bar in the smeared case, which demonstrates the power of smearing.

To conclude, our analysis method to renormalize the linear divergent matrix elements can successfully be used for smeared matrix elements as well. Smearing does not seem to change the behavior of renormalization qualitatively, but only quantitatively. The intrinsic physics does not seem to change under (moderate) smearing, consistent with the expectation that smearing is an alternative method to reduce the statistical noise in lattice calculations.

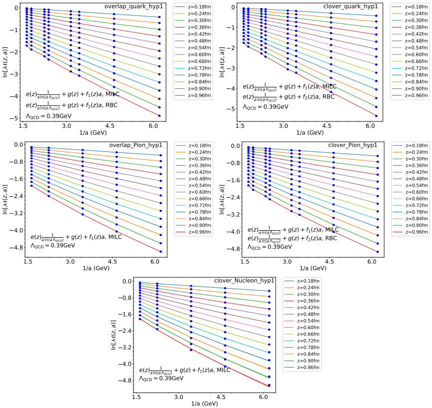

FIG. 23. Using Eq. (32) to fit $O_{\gamma_t}(z)$ matrix element for HYP smearing cases. Λ_{QCD} is fixed at 0.39 GeV.

VIII. LINEAR DIVERGENCE OF CORRELATIONS IN QCD VACUUM

In this section, we analyze the linear divergence of the Wilson line in the matrix elements taken in the QCD vacuum. We try to test if the divergence is universal and if the divergent mass can be extracted successfully from these matrix elements. We consider three new types of matrix elements: 1) large-size Wilson loop that has been used to extract heavy-quark potential, 2) the vacuum matrix element of the quasi-PDF operator, 3) the vacuum matrix element of Wilson link operator in a fixed gauge. The data are for MILC ensembles and, when applicable, with the clover action. According to the last section, smearing does not change the form of renormalization, and therefore, we consider here the higher-precision data with one step smearing. Since we just want to test the linear divergence, we can use the simple function Eq. (32) in our fitting.

A. Wilson Loop

To extract the linear divergence from lattice data, a Wilson loop is a good choice because it is easy to calcu-

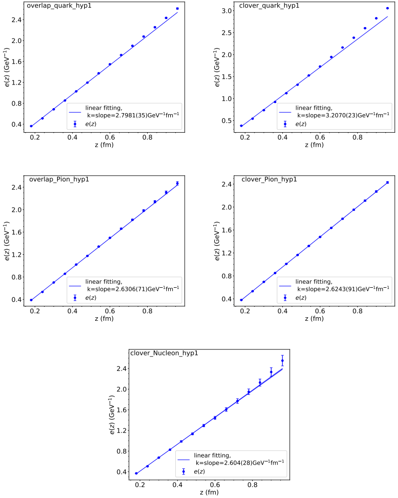

FIG. 24. $e(z)$ with respect to z. $e(z)$ is extracted from the fitting in Fig. 23. The slopes are labeled in the plot legend.

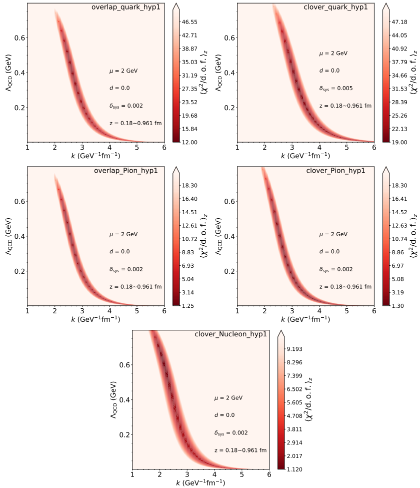

FIG. 25. Using Eq. (19) to fit the $O_{\gamma_t}(z)$ matrix element for HYP smearing. The plot shows the χ^2 map with respect to k and $\Lambda_{\rm QCD}$ for different actions and states. $\langle\chi^2/\text{d.o.f.}\rangle_z$ is the average of χ^2/d.o.f. from fitting for each z.

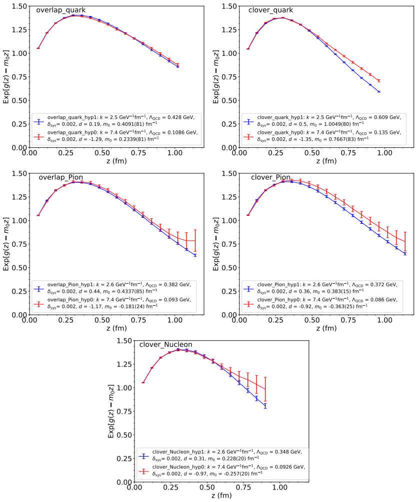

FIG. 26. Comparison between renormalized matrix elements $\mathrm{Exp}[g(z) - m_0 z]$ for unsmearing (hyp0, red) and HYP smearing (hyp1, blue) cases.

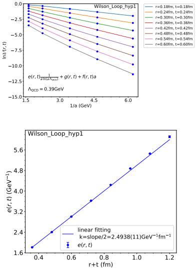

FIG. 27. Test of linear divergence for a Wilson Loop for HYP smearing cases. The slope is consistent approximately with hadronic matrix elements from the previous section.

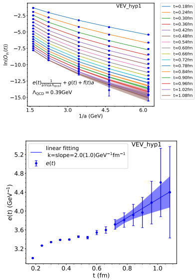

FIG. 28. Test of linear divergence for the vacuum matrix of the quasi-PDF operator. The matrix element is much suppressed and there is no leading twist contribution.

late, gauge invariant, and has high precision. It has been extensively studied in the literature [27, 28, 38, 43, 59], particularly in relation to the heavy quark potential.

In our analysis, we will not consider the heavy quark potential interpretation but simply consider a Wilson loop as a matrix element and fit its $1/a$ divergence by looking at the logarithm of the matrix element, as shown in Fig. 27. The lattice spacing dependence has been fitted well with our formula with a choice of $\Lambda_{\rm QCD}=0.39$ GeV, which is a value favored by our fitting in the previous section. After isolating the linear divergent coefficient, we plot it as a function of $r+t$, for which half of the slope gives us the divergent coefficient. Our value is about 2.5 GeV^{-1}fm^{-1} (0.49 if dimensionless). This value is very similar to the value we found in the previous section (Fig. 24), demonstrating that the linear divergence can be very well extracted from the Wilson loop.

B. Vacuum Matrix Element of Quasi-PDF Operator

Here we consider the vacuum matrix elements of a quasi-PDF operator which were suggested as choices for the renormalization factor in Ref. [44].

One can consider the vacuum expectation value (VEV) of $O_{\gamma_t}(t)$. As a gauge invariant choice proposed in Ref. [44]. $\langle O_\Gamma(t)\rangle$ can be obtained through the following stochastic estimation:

$$\begin{aligned}\langle O_\Gamma(t)\rangle &= \frac{1}{L_s^3}\langle\sum_{\vec{x}}\mathrm{Tr}[U(\vec{x},0;\vec{x},t)\Gamma S_w(\vec{x},t)]\rangle\\ &= \frac{1}{L_s^3}\langle\sum_{\vec{x}}\mathrm{Tr}[U(\vec{x},0;\vec{x},t)\Gamma S(\vec{x},t;\vec{x},0)]\rangle\\ &+\frac{1}{L_s^3}\sum_{\vec{x},\vec{y},\vec{y}\neq\vec{x}}\langle\mathrm{Tr}[U(\vec{x},0;\vec{x},t)\Gamma S(\vec{x},t;\vec{y},0)]\rangle \end{aligned}\quad (33)$$

where the second term on the right hand side of the second line vanishes as it is not gauge invariant, and $S_w(\vec{x},t)=\sum_{\vec{y}}S(\vec{x},t;\vec{y},0)$ is a wall source quark propagator without gauge fixing. Such a proposal can only be applied for the non-vanishing cases with $\Gamma=\gamma_t$ or $\mathcal{I}$. We will apply it to O_{γ_t}.

However, it is simple to see that the O_{γ_t} matrix element in the vacuum vanishes at small t using operator product expansion. Therefore the matrix element is susceptible to large O(a) correction. Only at very large t, one might see the correct slope.

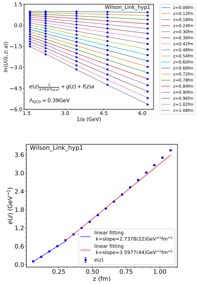

FIG. 29. Test of linear divergence for Landau gauge fixed Wilson link for HYP smearing cases. At small-z, it can be calculated in perturbation theory where the linear divergence works well. For large-z, the matrix element does not have a transfer-matrix interpretation in this gauge.

Results of a test of the linear divergence for the vacuum matrix element of the PDF operator is shown in Fig. 28. The slope at large t appears to be consistent with that from the Wilson loop case but only within uninterestingly large errors.

C. Landau-gauge-fixed Wilson link

For the gauge-dependent matrix elements, we consider the Wilson line in Landau gauge as the simplest choice.

The result of our test of the linear divergence is shown in Fig. 29. At small-z, where perturbation theory works well, the slope roughly agrees with the prediction from perturbation theory. For large-z, the matrix element does not have a transfer-matrix interpretation in this gauge.

Finally, Fig. 30 is a comparison of the linearly divergent term for Wilson Loop, VEV and Wilson link. At small z, the linear divergence terms of Wilson Loop and Wilson link are consistent while at large z they disagree strongly. The linear divergence term of VEV is different from the

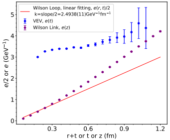

FIG. 30. Comparison of linear divergences in different vacuum matrix elements.

others because the large discretization errors make the extraction of the linear divergence very imprecise.

IX. SUMMARY

The proper renormalization of quasi-LF correlations calculated on the lattice has been a main challenge for the applications of LaMET to studying parton physics. Such correlations contain linear divergences associated with Wilson lines which have to be eliminated with high precision to extract the desired physics information. In this work, we propose a self-renormalization method to eliminate both the linear and the logarithmic divergence, as well as the discretization error, from a quasi-LF matrix element which can be matched to a continuum scheme at short distance. As a paradigmatic example, we show that our method works well for O_{γ_t} matrix elements in the pion, nucleon, and off-shell quark state. Our analysis shows that the renormalization factors are universal in the hadron state considered. Moreover, the renormalized correlations in the pion state for clover and overlap fermions are similar to each other. Besides, We find a large non-perturbative effect in the popular RI/MOM and ratio renormalization scheme used previously, which has to and can be avoided in the hybrid renormalization scheme proposed recently. It is certainly interesting to see that our method works also for HYP smeared matrix elements. This paves the way to do a phenomenological renormalization for smeared high-precision lattice data.

For the convenience of the reader, we collect the parameters related to renormalization for various matrix elements discussed in the previous sections, see Tables III and IV. For HYP smearing cases, there is much freedom for the choice of Λ_{QCD}. We choose $\Lambda_{QCD}=0.39$ GeV in Table IV. But we also find that $\Lambda_{QCD}=0.1$ GeV gives us smaller fitted discretization errors.

Cases	k	$\Lambda_{\rm QCD}$(GeV)	d	m_0(GeV)
overlap quark	1.46	0.109(02)	-1.29	0.0462(16)
clover quark	1.46	0.135(02)	-1.35	0.1513(16)
overlap pion	1.46	0.093(10)	-1.17	-0.0357(46)
clover pion	1.46	0.086(14)	-0.92	-0.0715(50)
clover nucleon	1.46	0.093(06)	-0.97	-0.0508(40)

TABLE III. Renormalization parameters based on the fitting functions Eq. (19) and Eq. (20) for the cases without HYP smearing for MILC and RBC ensembles, where $\delta_{\rm sys} = 0.002$. k is dimensionless here.

Cases	k	$\Lambda_{\rm QCD}$(GeV)
overlap quark	0.5521(07)	0.39
clover quark	0.6328(05)	0.39
overlap pion	0.5191(14)	0.39
clover pion	0.5178(18)	0.39
clover nucleon	0.5139(54)	0.39
Wilson Loop	0.4921(02)	0.39
VEV	0.39(21)	0.39
Wilson Link	0.5402(04), 0.7099(09)	0.39

TABLE IV. Renormalization parameters based on the fitting functions Eq. (32) for the cases with HYP smearing for MILC and RBC ensembles. k is the slope of $e(z)$ (half of the slope for Wilson Loop) and is dimensionless here.

For comparison purposes, we also show results based on 2+1 flavor clover quark and Luescher-Weisz (equivalent to Symanzik) gauge ensembles from CLS collaboration [71]. We choose valence fermion action to be clover action, the same as sea fermion action. The renormalization parameters for CLS ensembles are shown in Table V. The parameters k and $\Lambda_{\rm QCD}$ for pion matrix elements for CLS ensembles are similar to those for MILC and RBC ensembles, suggesting consistency between results using mixed actions or not.

Cases	k	$\Lambda_{\rm QCD}$(GeV)	d	m_0(GeV)
clover quark hyp0	1.46	0.170(06)	-0.68	0.1936(26)
clover pion hyp0	1.46	0.113(09)	-2.24	-0.0941(49)
clover quark hyp1	0.573(05)	0.39	0.53	0.2298(26)
clover pion hyp1	0.486(10)	0.39	-0.21	0.0312(29)

TABLE V. Renormalization parameters based on the fitting functions Eq. (19) and Eq. (20) for CLS ensembles, where $\delta_{\rm sys} = 0.002$. "hyp0" is the unsmearing cases and "hyp1" is the HYP smearing cases. k is dimensionless here.

We would like to stress that the application of our method is not limited to the matrix elements discussed in this paper. Instead, it can be applied to any matrix element of an operator in the form of Eq. (2). It can also be generalized, in principle, to any matrix element with a Wilson line in LaMET applications [8]. Of course, in these cases one also needs to consider the physical interpretations of the matrix elements and the details related to lattice calculations carefully.

ACKNOWLEDGEMENT

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P29-p029

P30：准光前关联的混合重正化方案

主题归属	非局域算符重整化与 hybrid 方案
底稿	arXiv:2008.03886
arXiv / DOI	2008.03886
完整原始 PDF	40 页；来源 PyQCD；SHA-256 56f633040d3b...
转排索引	EN 28 页；ZH 24 页（只作书目对照）
备注	索引未记录额外备注。

阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

页码范围：P30-p001 – P30-p040

A Hybrid Renormalization Scheme for Quasi Light-Front Correlations in Large-Momentum Effective Theory

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P30–p001

Abstract

In large-momentum effective theory (LaMET), calculating parton physics starts from calculating coordinate-space-z correlation functions $\tilde{h}(z, a, P^z)$ in a hadron of momentum P^z in lattice QCD. Such correlation functions involve both linear and logarithmic divergences in lattice spacing a, and thus need to be properly renormalized. We introduce a hybrid renormalization procedure to match these lattice correlations to those in the continuum $\overline{\mathrm{MS}}$ scheme, without introducing extra non-perturbative effects at large z. We analyze the effect of $\mathcal{O}(\Lambda_{\mathrm{QCD}})$ ambiguity in the Wilson line self-energy subtraction involved in this hybrid scheme. To obtain the momentum-space distributions, we recommend to extrapolate the lattice data to the asymptotic z-region using the generic properties of the coordinate space correlations at moderate and large P^z, respectively.

Keywords: Effective field theory, parton distribution function, lattice QCD, non-perturbative renormalization.

1. Introduction

Parton physics is important both for understanding the dynamics of high-energy collisions of hadrons and for studying their internal structure [1, 2]. The most familiar examples are quark and gluon parton distribution functions (PDFs) which, on one hand, provide the beam information for high-energy productions at colliders [3], and on the other hand, describe the bound-state physics of the colliding hadrons.

Despite its importance, calculating parton physics from first principles of quantum chromodynamics (QCD) has been a challenge. Recently, an effective field theory (EFT) approach–large momentum effective theory (LaMET)–has been proposed to extract parton physics from physical properties of hadrons moving at large momentum [4, 5], where the latter can be calculated from systematic approximations to Euclidean QCD such as lattice field theory. Since its proposal, LaMET has been widely used in calculating quark isovector distribution functions [6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21], distribution amplitudes (DAs) [22, 23, 24], generalized parton distributions [25, 26], and recently transverse-momentum-dependent distributions [27, 28, 29], and even higher-twist distributions [30]. Some recent reviews on LaMET can be found in Refs. [31, 32].

The key idea of LaMET is that partons in the infinite momentum frame (IMF) can be approximated by physical properties of a hadron at large but finite momentum. Due to the existence of ultraviolet (UV) divergences, this approximation is not completely straightforward. It requires using the standard EFT technology of matching and running. Detailed investigations have shown that the standard DGLAP evolution [33, 34, 35] has its origin in the momentum evolution of physical properties of the hadron [31].

In LaMET applications, one begins with lattice calculations of spatial correlation functions. Since lattice breaks the continuum symmetry, power divergences appear in bare correlation functions. They must be subtracted when matched to those in a continuum scheme such as dimensional regularization and (modified)-minimal subtraction, $\overline{\mathrm{MS}}$. In the past, the main approaches suggested in practical applications include the regularization-independent momentum subtraction method or RI/MOM [36, 37, 38, 39] and the ratio method [40, 41, 42, 43]. The latter relies on the validity of Euclidean operator product expansion (OPE) and can only be applied to correlations at short distances, and therefore cannot be used directly for LaMET applications. In contrast, the RI/MOM method appears to be applicable to large-z (z is the gauge-link length) distance at first glance. However, a detailed examination shows that this method introduces potential non-perturbative effects, for instance, through infrared (IR) logarithms such as $\ln(z^2\mu^2)$ (μ is a renormalization scale) in the scheme matching. Since UV divergences are supposed to be perturbative in asymptotically-free theories such as QCD, it shall be possible to find a renormalization procedure which does not introduce non-perturbative effects. It might be possible that RI/MOM does not introduce a large non-perturbative effect in the present precision of lattice-QCD calculations. However, a systematic effective-theory calculation with high precision cannot avoid addressing this issue.

To achieve this, we propose in this paper a hybrid renormalization procedure for lattice correlations in LaMET applications. At short distances where OPE is valid, the standard RI/MOM or ratio method is recommended. At large distances, we suggest to use the auxiliary field formalism [44, 45, 46, 47] which has been advocated in LaMET applications by a number of authors [48, 49, 50]. In this formalism, the Wilson line is replaced by two-point functions of the auxiliary field. The linear divergence in lattice correlation functions is then linked to the mass renormalization of the auxiliary field, whereas the logarithmic divergence appears in the renormalization of the "heavy"-light "currents" at the end of the Wilson line. Both divergences can be separately renormalized in a manner which is consistent with the $\overline{\mathrm{MS}}$ scheme [49, 50]. Although the mass subtraction of the Wilson line has been suggested before [51, 52], it has not been put into wide practical use because, to our knowledge, a reliable approach to calculate the non-perturbative mass has not been well-established in the literature. Here we suggest several ways to do so which shall be investigated through systematic lattice simulations in the future.

In addition, we also address several other issues that are important in extracting parton physics using LaMET, e.g., how to match appropriately to the continuum scheme near $z \sim 0$, and how to utilize the asymptotic behavior of relevant correlation functions at large light-front (LF) distance to remove the unphysical oscillations in the momentum distribution that arise from truncated Fourier transform.

2. Partons as quanta in infinite-momentum states and large-momentum expansion

Let us begin with a brief overview of the parton formalism. In the textbooks, PDFs are usually defined in terms of LF correlations in QCD [53, 54]. The LF is defined by $t-z=$ constant, if a massless particle is travelling along the z-direction, with variations of other coordinates, $t+z$ and transverse-space dimensions, defining a three-dimensional front surface. Introduce two independent LF four-vectors with dimension-one parameter Λ,

$$\begin{aligned} p^\mu &= (\Lambda, 0, 0, \Lambda) \ , \\ n^\mu &= (1/(2\Lambda), 0, 0, -1/(2\Lambda)) \ , \end{aligned} \tag{1}$$

then $p^2 = n^2 = 0$, and $p \cdot n = 1$. Different LFs are defined by different coordinate distance λ along the n-direction.

Consider now the quark PDFs in a state $|P\rangle$ with mass M and four-momentum $P^\mu = (P^0, 0, 0, P^z) = p^\mu + (M^2/2)n^\mu$, which can be used to solve for $\Lambda = (P^0 + P^z)/2$. Using ψ to denote a full-QCD quark field, the LF correlation function in coordinate space is,

$$h(\lambda) = \frac{1}{2P^+}\langle P|\overline{\psi}(\lambda n)\Gamma W(\lambda n, 0)\psi(0)|P\rangle \ , \tag{2}$$

where Γ is a Dirac matrix, W is a straight Wilson-line gauge link, and λ is the LF distance. All other coordinates have been taken to be zero. Due to the invariance of the LF under Lorentz boosts along the z-direction, the above correlation function is independent of the residual momentum P^z. Quite often, P^z is taken to be zero.

The quark PDF is just the Fourier transform of the above LF correlation [54],

$$f(x) = \int_{-\infty}^{\infty} \frac{d\lambda}{2\pi} e^{-ix\lambda} h(\lambda) \ . \tag{3}$$

In this way, partons can be studied without using the EFT machinery although they are effective degrees of freedom (dof's) to describe the LF collinear modes. The reason is that, the parton dof's are automatically projected out through the LF correlators applied to the full QCD state $|P\rangle$. On the other hand, these parton dof's can also be explicitly separated in the QCD Lagrangian, as is done in soft-collinear effective theory (SCET) where they are represented by LF collinear fields [55, 56, 57].

In the traditional parton formalism, the correlations are time-dependent, or in other words, the operators are in the Heisenberg picture. As such, we say that the formalism is Minkowskian and thus difficult for Monte Carlo simulations due to the famous "sign" problem. If one chooses $\xi^- = (t+z)/\sqrt{2}$ as the "new time" coordinate, and integrates it out, one obtains a Hamiltonian formalism for partons, which has been called LF quantization (LFQ) in the literature [58]. LFQ is also a very difficult formalism to work with, despite the fact that much progress has been made [59].

An alternative parton formalism can be obtained by adapting Feynman's original idea about partons to the context of a field theory [5, 31]. Feynman considered [60] the momentum distribution of a composite system, $f(k^z, P^z)$, where P^z is the center-of-mass momentum and k^z the longitudinal momentum carried by the parton whose transverse momentum $\vec{k}_\perp$ has been integrated over. The P^z-dependence of the momentum distribution is clearly a relativistic effect: According to Poincaré symmetry, the Hamiltonian of a system depends on the frame, and changes under Lorentz boosts according to,

$$[H, K^i] = iP^i \ , \tag{4}$$

where K^i $(i = 1, 2, 3)$ are the boost operators. Therefore, the wave functions are frame-dependent, leading to frame-dependent momentum distributions. Because K^i depends on interactions, the frame-dependence is a dynamical problem, and generally requires non-perturbative solutions.

An important feature of the momentum distribution of a system is that it is a static or time-independent quantity. In QCD, it is related to the following spatial correlation,

$$\tilde{h}(z, P^z) = \frac{1}{2N}\langle P|\overline{\psi}(z)\Gamma W(z,0)\psi(0)|P\rangle \ , \tag{5}$$

where $W(z,0)$ is a spacelike, straight-line gauge link, and N is normalization factor depending on the Dirac matrix. Feynman then considered the infinite-momentum limit, assuming that such a limit exists,

$$P^z \to \infty, \qquad z \to 0, \qquad \lambda = zP^z \text{ finite} \ , \tag{6}$$

i.e., the relevant correlation function for partons is

$$\tilde{h}(\lambda = zP^z) = \langle P^z = \infty|\overline{\psi}(z)\Gamma W(z,0)\psi(0)|P^z = \infty\rangle \ . \tag{7}$$

It is clear that in field theories this is a non-trivial limit. In fact, it can be shown that such a limit only exists in asymptotically-free theories, where the high-momentum modes are perturbative [31].

If one ignores the subtlety of the limit, the correlation in Eq. (7) is related to that in Eq. (2) by an infinite Lorentz transformation [4]. Our "new" form of parton formalism works with time-independent correlators and the IMF wave function. Since the operator is time-independent, it is the Schrödinger representation of parton physics if an analogy between time-translation and Lorentz boost is made [31].

In QCD, however, the correlations $\tilde{h}(\lambda)$ and $h(\lambda)$ are different. The difference arises from the presence of the UV cut-off. In the physical momentum distribution, the cut-off must always be much larger than the hadron momentum. As a result, the parton momentum is allowed to be larger than the hadron momentum, or $|x|$ can be larger than 1, without violating any laws of physics. On the other hand, the standard PDFs have support $|x| \leq 1$, corresponding to a UV cut-off smaller than the hadron momentum. Thanks to the asymptotic freedom, these two different UV limits can be connected to each other by perturbation theory in QCD. This makes it possible to extract LF parton physics defined in Eq. (2) from the Euclidean form in Eq. (7).

Eq. (7) is the starting point of the LaMET expansion, where we first compute the quasi-LF correlation functions at a finite, but large momentum $P^z \gg \Lambda_{\rm QCD}$. To make the expansion work, in principle one needs $\tilde{h}(z, P^z)$ with $-\infty < z < \infty$, or $\tilde{h}(\tilde{\lambda}, P^z)$ at all quasi-LF distances $\tilde{\lambda} = zP^z$. While in reality, of course, due to the finite volume, lattice data will always stop at some large $z(\tilde{\lambda})$ which we call $z_{\rm L}(\tilde{\lambda}_{\rm L})$. We will deal with issues of finite $\tilde{\lambda}_{\rm L}$ later. For the discussion in this section, we assume that $\tilde{h}(\tilde{\lambda}, P^z)$ is known in $[-\infty, \infty]$, i.e., in the whole $\tilde{\lambda}$ range at a large P^z.

With the above quasi-LF correlation, one can make a straightforward Fourier transformation

$$\tilde{f}(y, P^z) = \int_{-\infty}^{\infty} \frac{d\tilde{\lambda}}{2\pi}\, e^{i\tilde{\lambda}y}\, \tilde{h}(z, P^z)\ . \tag{8}$$

The physical interpretation of $\tilde{f}(y, P^z)$ hinges on the large momentum expansion [4, 61, 62, 63]

$$\begin{aligned}\tilde{f}(y, P^z) = &\int_{-1}^{1} dx\, C\Big(\frac{y}{x}, \frac{xP^z}{\mu}\Big) f(x, \mu)\\ &+ \mathcal{O}\Big(\frac{\Lambda^2_{\rm QCD}}{y^2(P^z)^2}, \frac{\Lambda^2_{\rm QCD}}{(1-y)^2(P^z)^2}\Big),\end{aligned} \tag{9}$$

where μ is a factorization scale, $\Lambda_{\rm QCD}$ is the hadronic scale, and $f(x,\mu)$ is the standard PDF that can be extracted from the above equation. The large scales $(yP^z)^2$ and $((1-y)P^z)^2$ are associated with the active quark and the spectator momenta, respectively. According to the standard EFT methodology, any large scale that is not forbidden shall be allowed in the expansion, and the linear dependence is absent in dimensional regularization due to space-time symmetry. Therefore, the validity of this expansion relies on the smallness of the expansion parameters $\Lambda^2_{\rm QCD}/[y^2(P^z)^2]$ and $\Lambda^2_{\rm QCD}/[(1-y)^2(P^z)^2]$.

The C factor in the above equation can be calculated perturbatively. At leading-order in α_s, we can identify $\tilde{f}(y, P^z)$ with $f(y,\mu)$ (ignoring the power corrections for the moment), thus they have the same asymptotic behavior as $y \to 0$ and $y \to 1$. Beyond leading-order, this will be changed by perturbative corrections. To see this, let us take the following simple form of $f(x,\mu)$ as an example

$$x^a(1-x)^b, \tag{10}$$

with a, b controlling the asymptotic behavior at $x \to 0$ and $x \to 1$, respectively. The perturbative one-loop corrections lead to the following change in the asymptotic behavior

$$\delta\tilde{f}(y, P^z) \sim \alpha_s y^a \ln y \quad \text{as} \quad y \to 0\,. \tag{11}$$

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When resummed to all orders in perturbation theory, this yields a power law behavior of the form $\tilde{f}(y, P^z) \sim y^{a+\gamma}$ with γ being associated with the anomalous dimension of the operator defining $\tilde{f}(y, P^z)$. Similar behavior also occurs as $y \to 1$ for realistic PDFs with $b > 0$. This can also be seen from the coordinate space analysis to be presented below.

Therefore, for a given large P^z, there is a range of y where high-order corrections as well as power corrections are small, and this range can be translated into a valid range x for the PDFs. Thus, one can systematically obtain the PDFs in an interval $[x_{\rm min}, x_{\rm max}]$ ($x_{\rm min}$ will approach 0 and $x_{\rm max}$ approach 1 as $P^z \to \infty$). In other words, the LaMET expansion provides a natural way to calculate parton distributions in an interval of the parton momentum x, similar to extracting parton distributions from experimental data at finite energies.

3. A hybrid renormalization procedure

As explained in the previous section, the LaMET expansion starts from calculating the coordinate-space correlation functions $\tilde{h}(z, P^z)$ at large momentum P^z and for the whole range of distance $-\infty < z < \infty$. On a discrete lattice with spacing a, the nonlocal quark bilinear operator that defines $\tilde{h}(z, P^z)$ in Eq. (5) can be multiplicatively renormalized as [48, 52, 49]

$$\begin{aligned}&\left[\bar{\psi}(z)\Gamma W(z,0)\psi(0)\right]_{\rm B}\\&= e^{\delta m|z|}Z(a)\left[\bar{\psi}(z)\Gamma W(z,0)\psi(0)\right]_{\rm R}\,,\end{aligned} \tag{12}$$

up to lattice artifacts [36, 64]. Here the operator on the l.h.s. is defined in terms of bare fields and couplings, denoted by the subscript "B", while the operator on the r.h.s. is renormalized and denoted by the subscript "R". Without an explicit statement, we always assume that the renormalized correlations are eventually defined in the $\overline{\rm MS}$ scheme before a Fourier transformation is made to the momentum space. There are both z-independent logarithmic and z-dependent linear divergences. The former arises from the renormalization of quark and gluon fields as well as the vertices at the endpoints of the Wilson line, which is included in the factor $Z(a)$, while the latter comes from the Wilson-line self-energy, which is factored into the exponential $e^{\delta m|z|}$ with δm being the "mass correction" .

A number of proposals have been made in the literature [65, 51, 66, 40, 36, 49, 37, 42, 43] to renormalize the above lattice correlation functions $\tilde{h}(z, a, P^z)$, among which the RI/MOM scheme has frequently been used [36, 37]. In this approach, one calculates the matrix elements (amputated Green's function) $Z(z,-p^2,a)$ of the bilocal operators $O(z,a)$ in a deep Euclidean state with momentum squared $-p^2 \gg \Lambda_{\rm QCD}^2$ in a fixed gauge, and then defines $\overline{\rm MS}$ operators as,

$$O_{\overline{\rm MS}}(z,\mu) \equiv Z_{\overline{\rm MS}}(z,-p^2,\mu)\frac{O(z,a)}{Z(z,-p^2,a)}, \tag{13}$$

where $Z_{\overline{\rm MS}}$ converts the RI/MOM renormalized result to the $\overline{\rm MS}$ scheme. The gauge and $-p^2$ dependences cancel between two Z-factors. The r.h.s. has a proper continuum limit $a \to 0$ without divergences.

However, while the RI/MOM approach is justified for local operators, it has potential problems when applied to nonlocal ones. For instance, when z becomes large, $Z_{\overline{\rm MS}}(z,-p^2,\mu^2)$ contains IR logarithms of z and the perturbative calculation of z-dependence is not reliable. Moreover, although the RI/MOM factor $Z(z,-p^2,a)$ helps to cancel the lattice UV divergences, the composite operator at large-z contains non-perturbative physics as well. Therefore, both Z-factors contain non-cancelling non-perturbative effects which alter the IR properties of $O(z)$. Thus, the RI/MOM renormalization scheme is not reliable at large-z. Moreover, when gluon distributions are involved, it requires external off-shell gluon states which bring in potential mixing with gauge-variant operators and make things much more complicated [67].

In addition to the renormalization issues at large distances, there are also subtleties for renormalization at short distances. While the standard renormalization of a bilocal operator makes it finite at any non-vanishing z, it becomes divergent in the $z \to 0$ limit. On the other hand, if one performs a resummation of the large logarithms at small z, the result vanishes at $z = 0$. However, the lattice result at $z = 0$ approaches to the matrix element of the vector current. Clearly, the two limits, $a \to 0$ and $z \to 0$, are not interchangeable.

To resolve these issues, we propose in this section a hybrid scheme to renormalize the correlation functions. The key point of this scheme is that we separate the correlations at short and long distances and renormalize them separately, and match both procedures at an intermediate distance z_S. The matching point must lie within $[0, z_{\rm LT}]$ where the leading-twist (LT) approximation for the correlation operator is valid. Discussions on the value of $z_{\rm LT}$ can be found in Sec. V.

3.1. Renormalization at short distance $0 \leq |z| \leq z_{\rm S}$

To renormalize $\tilde{h}(z, a, P^z)$ for $0 \leq |z| \leq z_{\rm S}$, particular attention shall be paid to the behavior of the correlation functions in the limit $z \to 0$.

In the continuum $\overline{\rm MS}$ sheme, the $z \to 0$ limit is not smooth and additional logarithmic UV divergences $\sim \ln z^2$ arise which when resummed yield zero. However, this is not the case for the lattice matrix element $\tilde{h}(z, a, P^z)$. For finite lattice spacing and non-vanishing z, $\tilde{h}(z, a, P^z)$ includes UV divergences related to the wave function renormalization of the bare fields, of the form $\alpha_s(a)\ln(z^2/a^2)$. At small z, particularly when $z = 0$ or a, $\tilde{h}(z, a, P^z)$ has discretization effects and is related to the lattice-regulated local matrix element $\bar{\psi}\Gamma\psi$. In particular, when $\Gamma = \gamma^\mu$, $\bar{\psi}\gamma^\mu\psi$ is conserved and its matrix element is finite in the $a \to 0$ limit. A function demonstrating this interesting interplay between lattice regulator and small physical distance is $\ln[(z^2 + a^2)/a^2]$.

The above discrepancy in the small-z regime can be removed through a perturbative conversion between lattice regularization and the continuum $\overline{\rm MS}$ scheme, which, however, is known to converge slowly. Instead, a more efficient strategy is to cancel the $\ln z^2$-dependences through lattice renormalization, which corresponds to a scheme "X" that is different from $\overline{\rm MS}$. As long as z is in the leading-twist region $|z| \leq z_{\rm S}$ where $z_{\rm S}$ is smaller than $z_{\rm LT}$, the difference between the X-scheme and $\overline{\rm MS}$ can be calculated in perturbation theory.

For example, the X-scheme can be implemented by forming the ratio of $\tilde{h}(z, a, P^z)$ and another matrix element of the same operator $O(z, a)$,

$$\frac{\tilde{h}(z, a, P^z)}{Z_X(z, a)}, \qquad \text{for } |z| \leq z_S\,, \tag{14}$$

where the renormalization factor Z_X corresponds to different choices of the matrix element. Possible choices for Z_X include

- Amputated Green's function of $O(z, a)$ in a single-particle deep Euclidean state, fixed in a particular gauge, e.g., Landau gauge, which defines the RI/MOM-type of schemes [36, 37, 38, 39].
- Matrix element of $O(z, a)$ in a hadron state with $P^z = 0$, depending on applications [40, 41].
- Vacuum ($|\Omega\rangle$) expectation value of $O(z, a)$ [42, 43].

In the second and third option, the matrix elements are gauge invariant, and therefore no gauge fixing is needed. For the third option, the quantum numbers of the operator must be the same as those of the vacuum. As discussed above, $z_{\rm S}$ has to be smaller than $z_{\rm LT}$, which is estimated in Sec. V to be about $0.25 \sim 0.33$ fm. Of course, the stability of the final result with respect to small variations of $z_{\rm S}$ shall be explicitly verified.

Due to the multiplicative renormalizability of the operator $O(z, a)$, all UV divergences cancel in the ratio in Eq. (14), thus allowing us to take the continuum limit,

$$\lim_{a\to 0} \frac{\tilde{h}(z,a,P^z)}{Z_X(z,a)} = \frac{\tilde{h}(z,\epsilon,P^z)}{Z_X(z,\epsilon)} \equiv \tilde{h}^X(z,P^z)\,, \tag{15}$$

where the term after the first equal sign refers to a $\overline{\rm MS}$ calculation of the same ratio with ϵ corresponding to dimensional regularization $d = 4-2\epsilon$ in the continuum theory. In the limit $|z| \to 0$, the $\ln z^2$-dependence is independent of the external state, so it cancels in the ratio, making the latter finite at $z = 0$. Moreover, in the leading-twist region $z \leq z_{\rm S} \leq z_{\rm LT}$, we can perturbtively match the ratio for any X-scheme to the LF correlation $h(\lambda, \mu)$ through the coordinate-space factorization formula [63, 68]

$$\begin{aligned}\tilde{h}^X(\lambda,P^z) = &\int_0^1 d\alpha\, \mathcal{C}^X\Big(\alpha, \frac{\lambda^2\mu^2}{(P^z)^2}\Big) h(\alpha\lambda,\mu) \\ &+ \mathcal{O}(z^2\Lambda^2_{\rm QCD})\,,\end{aligned} \tag{16}$$

where $\mathcal{C}^X$ is the matching coefficient, and we have suppressed its dependence on the renormalization scale in the X-scheme such as the RI/MOM. Also, higher-twist contributions have been suppressed in the above equation.

The one-loop matching coefficient for the second option for Z_X has been obtained in Refs. [69, 68, 63], and that for the RI/MOM scheme can be extracted from Ref. [36, 63]. The two-loop results for both the second and third option have been calculated as a series expansion in Ref. [43].

Eq. (16) also has an equivalent form in momentum space [63], with the matching coefficient calculated at one-loop [37, 13, 15] and two-loop [70] orders. The two-loop matching coefficient for Z_X defined by $P^z = 0$ matrix element can be extracted from Refs. [70, 43].

3.2. Renormalization at large distances $z \geq z_{\rm S}$

At large distance $z \geq z_{\rm S}$, UV renormalization needs a careful assessment because both the RI/MOM and the ratio scheme will introduce undesired non-perturbative effects. The only renormalization approach that will not introduce such extra non-perturbative physics is the explicit and separate subtraction of linear divergences (or δm) and logarithmic divergences [51, 48, 52, 49], which in principle can be done using the auxiliary field method [49, 50].

To calculate the mass renormalization δm of the Wilson line, there exist many suggestions in the literature. Here we provide probably an incomplete list:

- One can fit the hadron matrix element at large z, where the dominant decay is
$$\tilde{h}(z, a, P^z) \sim \exp(-\delta m|z|)\ . \tag{17}$$
δm can be obtained by fitting the ratio $\ln(\tilde{h}(z+a, a, P^z)/\tilde{h}(z, a, P^z))$ to a constant in z at large z. Of course, the result has to be independent of P^z, e.g., one can choose $P^z = 0$. Alternatively, one can fit $\ln \tilde{h}(z, a, P^z)$ to the $1/a$ dependence all z. This method has yet to be studied using real lattice data.

- One can use the single-quark Green's function as in the RI/MOM renormalization factor,
$$\begin{aligned} Z(z, -p^2, a) = \int & d^4x d^4y\ e^{ip\cdot(x-y)} \\ & \times \langle\Omega|T\psi(x)O(z, a)\bar{\psi}(y)|\Omega\rangle\,, \end{aligned} \tag{18}$$
which asymptotically goes like $Z(z) \sim \exp(-\delta m|z|)$. This matrix element needs a fixed gauge, and has been studied in Refs. [17, 71].

- On can also use the vacuum matrix element of $O(z, a)$
$$S(z) = \langle\Omega|O(z, a)|\Omega\rangle \tag{19}$$
which is gauge invariant. $S(z)$ again at very large z behaves like $S(z) \sim \exp(-\delta m|z|)$. This has been considered in Refs. [42, 43].

- Also the gauge-invariant Polyakov loop leads to the static potential between two heavy quarks. There exists a large number of references on this approach, see, e.g., [72, 73, 74, 75, 76].

- One can also calculate the vacuum expectation value of the Wilson line $W(z)$ directly in a fixed gauge, and again $\langle W(z)\rangle \sim \exp(-\delta m|z|)$ at large distance. This has been considered in Refs. [49, 50] using the auxiliary field method.

It is worth pointing out that, although all the proposals above work in principle, different practical issues may arise in their lattice realization such that some may work better than the others.

The mass renormalization δm is gauge-independent, just like the pole mass of a quark [77]. In the above suggestions where no gauge fixing is needed, this is obviously true. In the cases where a gauge-fixing is needed, one can demonstrate that the results in any other gauge are the same by constructing appropriate gauge-invariant operators [78]. Despite being gauge-independent, δm will depend on the specific action used in Monte Carlo simulations and on the definition of the matrix elements above. In the cases of vacuum matrix elements, δm may be interpreted as the non-perturbative pole mass in certain gauges [78].

The δm calculated from all the matrix elements above will have the following dependence on the lattice spacing a,

$$\delta m = m_{-1}(a)/a + m_0 \ , \tag{20}$$

where m_{-1}(a) is the coefficient of the power divergence, which is independent of the specific matrix element. The a-independent term m_0 has a more complicated origin. It can arise from various sources:

- Renormalon effect: In principle, $m_{-1}(a)$ can be calculated perturbatively, and is $2\pi\alpha_s(a)/3$ at leading order, similar to the mass counterterm in the Wilson formulation of fermions. However, the perturbation series is not convergent. When truncated at order $n \sim 1/\alpha_s(a)$, the perturbation series has an uncertainty of order $a\Lambda_{\rm QCD}$, which generates a contribution to m_0 [79, 80, 81, 82]. This means that in non-perturbative fitting, $m_{-1}(a)$ is determined only with an uncertainty of the order of $a\Lambda_{\rm QCD}$. Thus, an additional m_0 contribution of order $\Lambda_{\rm QCD}$ is expected.

- Pole mass: For certain matrix elements, like vacuum elements of bilocal operators, the z dependence can be viewed as originating from the pole mass of a meson consisting of an infinitely-heavy quark and a light one. In this case, δm is the pole mass apart from the linear divergence.

- Finite P^z effects: The correlation function at finite P^z has a long-range correlation, $\exp(-\lambda/\xi(P^z))$, where $\xi(P^z)$ is the correlation length. This contribution is included in m_0.

- Fitting effect. Since the data is always in finite z and a where the exponential decay cannot always be separated from an algebraic decay, there are fitting uncertainties contributing to m_0 as well as the separation between m_0 and m_{-1}.

To summarize, m_0 depends on the lattice matrix-element used and the fitting procedure [22, 23, 17, 71].

In Fig. 1 we show, as an example, the values of m_{-1} and m_0 determined from the quark RI/MOM renormalization factor calculated at the scale $\mu_R = 1.8$ GeV and $p_R^z = 0$, using the four ensembles with $a \approx \{0.045, 0.06, 0.09, 0.12\}$fm and 310 MeV pion mass from MILC collaboration [83]. Inspired by the asymptotic behavior at large z to be studied in Sec. 4, we use the following simplified form

$$e^{-(\frac{m_{-1}}{a}+m_0)|z|}\frac{c_1}{|z|^{d_1}} \tag{21}$$

to fit the renormalization factors at four different lattice spacings. It is worth pointing out that m_{-1} starts from $O(\alpha_s)$ and we therefore also include the dependence of the coupling on a in the fitting. For $a \approx 0.12$ fm, the fitted results for the coefficients m_{-1} and m_0 are

$$m_{-1} = 0.234 \pm 0.012, \quad m_0 = (350 \pm 60)\,\text{MeV}. \tag{22}$$

In principle, one can choose to subtract the power divergent piece only, namely m_{-1}/a. The less-well-determined m_0 term can be left in the lattice matrix elements, the momentum expansion will take care of the rest. Indeed, the difference between subtracting different m_0 is $O(1/P^z)$ effect, as has been demonstrated perturbatively in [61]. More precisely, assuming there are two quasi-LF correlations that define the quasi-PDFs and differ from each other by a factor $e^{-m|z|}$ with $m \sim \Lambda_{\text{QCD}}$,

$$\tilde{h}_1(z, P^z) = \tilde{h}_2(z, P^z)e^{-m|z|}\ , \tag{23}$$

then after Fourier transforming into momentum space, they are related by

$$\begin{aligned}&\tilde{f}_1(y, P^z) - \tilde{f}_2(y, P^z)\\ &= \frac{1}{\pi}\int_{-\infty}^{\infty} dy' \frac{\delta}{\delta^2 + (y-y')^2}\left[\tilde{f}_2(y', P^z) - \tilde{f}_2(y, P^z)\right]\ ,\end{aligned} \tag{24}$$

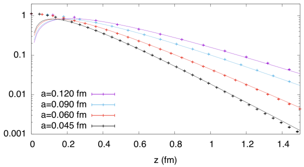

Figure 1: Fitting of the quark RI/MOM renormalization factor calculated using four ensembles with lattice spacings $a \approx \{0.045, 0.06, 0.09, 0.12\}$fm and 310 MeV pion mass from MILC collaboration [83].

with $\delta = m/P^z$. If the $\tilde{f}$'s are square integrable and their first order derivatives are continuous, one can show that as $\delta \to 0$,

$$|\tilde{f}_1(y, P^z) - \tilde{f}_2(y, P^z)| \sim \delta \ . \tag{25}$$

Therefore, the ambiguity of different schemes disappears in the large P^z limit.

However, this is still unsatisfactory because it appears that, due to non-perturbative effects from the linear divergence, the LaMET expansion will be an expansion in powers of M/P^z instead of $(M/P^z)^2$, which will significantly reduce the speed of convergence. Here we consider possible ways to overcome this deficiency. Recall that the LaMET expansion in $(M/P^z)^2$ is made in the $\overline{\text{MS}}$ scheme where no linear divergence exists, in a general scheme this expansion might contain odd powers in $1/P^z$. Therefore, there is a way to choose m_0 such that the condition of the $\overline{\text{MS}}$ scheme

$$\delta m_{\overline{\text{MS}}} = 0 \tag{26}$$

is met with non-perturbative calculations. We shall denote such a value of the subtracted mass by

$$\delta m_c = m_{-1}/a + m_{0c} \ . \tag{27}$$

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where m_{0c} can be determined by matching the matrix element $\tilde{h}_1(z, P^z, a)$ on lattice to the $\overline{\mathrm{MS}}$ result when z is small (around z_S) and QCD perturbative theory works. An alternative strategy is to vary m_0 in a certain range near Λ_{QCD}, and identify the value m_{0c} for which the linear term in $1/P^z$ in $\tilde{f}(y, P^z)$ vanishes. This is like searching for the critical value of κ_c for Wilson fermions for which a similar power divergent bare quark mass appears [84].

The mass-subtracted operator $O(z,a)e^{-\delta m(a)|z|}$ has no power divergence, but still has logarithmic dependence on a. These remnant logarithmic divergences are independent of z and can be renormalized, in principle, using the auxiliary field method [49, 50]. However, a more convenient option in practice is to fix the renormalization constant $Z_{\mathrm{hybrid}}(a)$ by directly matching the renormalized matrix elements of $O(z,a)$ at $z = z_{\mathrm{S}}$ from the short and long distance regimes, which is essentially a continuity condition,

$$Z_{\mathrm{hybrid}} e^{-\delta m |z_{\mathrm{S}}|} \langle P|O(z_{\mathrm{S}}, a)|P\rangle = \frac{\langle P|O(z_{\mathrm{S}}, a)|P\rangle}{Z_X(z_{\mathrm{S}}, a)}\,, \tag{28}$$

which leads to

$$Z_{\mathrm{hybrid}}(z_{\mathrm{S}}, a) = e^{\delta m |z_{\mathrm{S}}|} / Z_X(z_{\mathrm{S}}, a)\,. \tag{29}$$

In this way, one only has to calculate δm. Of course, one needs to vary z_{S} to check whether the final result is stable.

The matching coefficient $\mathcal{C}_{\mathrm{hybrid}}$ for the long distance regime is related to that for the X-scheme. For example, if one adopts the $P^z = 0$ matrix element for renormalization [69, 68, 63], then

$$\begin{aligned} \mathcal{C}_{\mathrm{hybrid}}(\alpha, z^2\mu^2, z^2/z_{\mathrm{S}}^2) &= \mathcal{C}_{\mathrm{ratio}}(\alpha, z^2\mu^2) \\ &\quad + \delta(1-\alpha)\frac{\alpha_s C_F}{2\pi}\frac{3}{2}\ln\frac{z^2}{z_{\mathrm{S}}^2}\ \theta(|z| - z_{\mathrm{S}})\,. \end{aligned} \tag{30}$$

However, due to the logarithms of $z^2\mu^2$ and z^2/z_{S}^2, the above matching coefficient is only valid for $|z| \ll \Lambda_{\mathrm{QCD}}^{-1}$, otherwise one has to resum the large logarithms for $|z| \sim \Lambda_{\mathrm{QCD}}^{-1}$ by evolving α_s to a highly nonperturabtive regime. Since our ultimate goal is to Fourier transform the final result to obtain the PDF, this will introduce uncontrolled sytematics.

To have a clearer way of separating the perturbative and non-perturbative regimes, we can perform the matching in momentum space, where nothing prevents using the correlations at large z, provided that yP^z is sufficiently

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large. In principle, we should first convert the hybrid scheme to the $\overline{\rm MS}$ scheme—where the factorization formula was proven [62, 31]—in coordinate space with the conversion factor

$$\mathcal{Z}(z, z_S, \mu) = \frac{Z_{\overline{\rm MS}}}{Z_{\rm hybrid}} = \frac{\theta(z_{\rm S}-|z|)}{Z_X^{\overline{\rm MS}}(z,\mu)} + \frac{\theta(|z|-z_{\rm S})}{Z_X^{\overline{\rm MS}}(z_S,\mu)}\,, \tag{31}$$

where $1/Z_X^{\overline{\rm MS}}$ converts the "X" scheme to $\overline{\rm MS}$, and for Eq. (30)

$$Z_{\rm ratio}^{\overline{\rm MS}} = 1 - \frac{\alpha_s C_F}{2\pi}\left[\frac{3}{2}\ln\frac{z^2\mu^2}{4e^{-2\gamma_E}} + \frac{5}{2}\right]. \tag{32}$$

The conversion factor $\mathcal{Z}$ is perturbative for all z as it does not include $\ln(z^2)$ at large distance $|z| > z_{\rm S}$. Then we can Fourier transform the $\overline{\rm MS}$ quasi-LF correlation and match it to the PDF in momentum space.

Since the scheme conversion is perturbative for all z, we can also do the Fourier transform first and directly match the hybrid scheme quasi-PDF to the PDF, and the matching coefficient $C_{\rm hybrid}$ is given by the double Fourier transform from Eq. (30),

$$\begin{aligned} C_{\rm hybrid}(\xi, \mu^2/(p^z)^2, z_{\rm S}^2\mu^2) &= C_{\rm ratio}(\xi, \mu^2/(p^z)^2) \\ &+ \frac{\alpha_s C_F}{2\pi}\frac{3}{2}\left[-\frac{1}{|1-\xi|_+} + \frac{2{\rm Si}((1-\xi)\lambda_{\rm S})}{\pi(1-\xi)}\right], \end{aligned} \tag{33}$$

where $C_{\rm ratio}$ can be found in [63], $\xi = y/x$, and $\lambda_{\rm S} = z_{\rm S}p^z$ with $p^z = xP^z$ being the parton momentum. The plus function is defined as

$$\frac{1}{|1-\xi|_+} \equiv \lim_{\beta\to 0^+}\left[\frac{\theta(|1-\xi|-\beta)}{|1-\xi|} + 2\delta(1-\xi)\ln\beta\right]. \tag{34}$$

We can also derive the corresponding scheme conversion factor and matching coefficient for using RI/MOM scheme in the short-distance renormalization. In the limit of $-z_{\rm S}^2p^2 \ll 1$, the RI/MOM renormalization factor $Z(z,-p^2)$ for $|z| < z_{\rm S}$ is equal to that of the zero-momentum matrix element, so the results are the same as those in Eqs. (31) and (33). However, if $-z_{\rm S}^2p^2$ is finite, then one needs to use the results from Refs. [36, 37] to derive the scheme conversion factor and matching coefficient. Finally, since the result of the PDF must be independent of the lattice renormalization scheme, we can try different short-distance schemes and check if they are consistent with each other.

In momentum space, the matching coefficient includes the logarithm of $\mu/(yP^z)$ which becomes non-perturbative for $y \sim \Lambda_{\rm QCD}/P^z$. This is consistent with the power counting parameter $\Lambda_{\rm QCD}^2/(yP^z)^2$. Therefore, the nature of the systematic uncertainties is clear, and we can only improve precision at small x by pushing to higher P^z.

4. Strategy of data analysis at asymptotic distances

For finite hadron momentum, lattice calculations of quasi-LF correlations always end up with data at finite $\lambda_{\rm L} = P^z z_{\rm L}$ where $z_{\rm L}$ is usually smaller than the lattice size due to increasing finite volume corrections and worse signal-to-noise ratios at large z. However, to reconstruct the full parton distribution, we need the correlations at all quasi-LF distances.

At finite momentum, the quasi-LF correlation in general has a finite correlation length (in the $\overline{\rm MS}$ or hybrid scheme) and exhibits an exponential decay at large z. This is similar to the case of density-density [85] or current-current correlation since the quasi-LF correlation can be viewed as the product of two heavy-to-light currents in the auxiliary field formalism [44, 45, 46, 47]. As a consequence, its Fourier transform converges fast at finite z or λ, as compared to that of the LF or twist-2 correlation which only decays algebraically at large λ due to the Regge behavior. If $z_{\rm L}$ is large enough such that the quasi-LF correlation falls close to zero, we can do a truncated Fourier transform up to $z_{\rm L}$ to obtain the quasi-PDF, and the resulting systematic uncertainty is negligible compared to other sources.

However, in practical lattice calculations, the choice of z_L is limited by fast-growing errors of quasi-LF correlations. This is particularly true for large hadron momentum. Thus, when we choose a z_L or $\lambda_{\rm L}$ with a target error, the quasi-LF correlation may still have a sizeable nonzero value at that point. In this case, a truncated Fourier transform will lead to an unphysical oscillation and inaccurate small-x result in the quasi-PDF, which can be formulated as an inversion problem [86]. Several strategies have been adopted in the literature to address this issue, e.g., the Backus-Gilbert method [86, 87], neural network and Bayesian reconstructions [86], the Gaussian reweighting method that suppresses the long-range correlations [88], the derivative method [89] which amounts to doing integration-by-parts and ignoring the boundary terms at the truncation point, or the Bayes-Gauss-Fourier transform which reconstructs a continuous form of the quasi-LF correlation over the whole domain by employing Gaussian process regression [90]. However, the assumptions employed in these strategies are mostly based on mathematical rather than physical reasons. Here we propose to use the knowledge of the asymptotic behavior of quasi-LF correlations and perform a physically motivated extrapolation to $\lambda = \infty$. After the extrapolation, one can perform a discrete Fourier transform for $|z| \leq z_{\rm L}$, where the discretization error can be studied with the lattice spacing dependence, while for the extrapolated part one can perform the Fourier transform analytically. Therefore, the mathematical inverse problem is solved by physics considerations. Although this extrapolation does not provide a first-principle prediction of the small-x PDF, it helps remove the unphysical oscillation and offers a reasonable way to estimate the systematic uncertainties in this region.

Depending on how large the momentum is, we propose to use either the exponential or algebraic decay form for the extrapolation. In the following, we discuss them in detail and describe how to estimate the corresponding uncertainty.

4.1. Exponential extrapolation at moderately large P^z

As mentioned above, for a moderately large momentum P^z, the quasi-LF correlations in general have a finite correlation length $\xi_z \sim 1/\Lambda_{\rm QCD}$ in the coordinate z space or $\xi_\lambda \sim P^z/\Lambda_{\rm QCD}$ in the LF distance λ space. This is due to the confinement property of non-perturbative QCD and spacelike nature of the correlation. The finite correlation length is associated with an exponential decay $\exp(-z/\xi_z)$, which becomes significant at large z. Before the quasi-LF correlation exhibits the exponential decay behavior, it is dominated by the leading-twist contribution which evolves slowly in P^z. In the λ space, the quasi-LF correlations at different P^z can be qualitatively described by Fig. 2. As P^z increases, the quasi-LF correlation evolves closer to leading-twist contribution, and starts to exhibit the exponential decay at larger λ values. In the limit of $P^z \to \infty$, ξ_λ approaches infinity and the quasi-LF correlation only includes the leading-twist contribution that decays algebraically at large λ.

Therefore, when P^z is not very large, e.g., about 2–5 GeV for the proton, we propose to use the exponential decay form $\sim e^{-\lambda/\xi_\lambda}$ to do the extrapolation (although some algebraic behavior can be added on the top to better represent the λ-dependence of the quasi-LF correlations). Note that to make the extrapolation under control, it is critical for the lattice data to exhibit the exponential decay before the error becomes too big. This shall be achieved with larger lattice volume and/or higher statistics of measurements, which

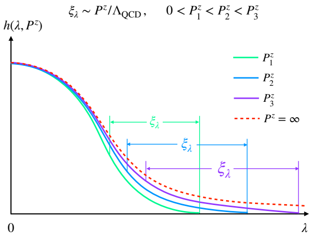

Figure 2: Qualitative behavior of the quasi-LF correlation in λ space at different P^z. For finite P^z, at short λ the correlation is approximated by the leading-twist contribution and evolves slowly in P^z. At large λ, the correlation starts to exhibit the exponential decay behavior, and both the starting point and correlation length ξ_λ increase with respect to P^z. In the $P^z \to \infty$ limit, ξ_λ approaches infinity and the quasi-LF correlation only includes leading-twist contribution which decays algebraically.

are feasible for contemporary computing resources. In the ideal case, at large enough z or λ, the quasi-LF correlation would become practically zero within the target error, and the extrapolation will barely affect the final result except for the extremely small-x region. In more practical scenarios, the lattice data shows the exponential drop, but still has a statistically significant nonzero value at $\lambda_{\rm L}$, then we can perform the exponential extrapolation to $\lambda = \infty$.

Note that although the exponential form is physically motivated, it remains unclear at what values of z or λ the lattice data should be included for the fitting, as the large-z data still includes leading-twist contribution which may obscure the result. Namely, one may fit to different values of the

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correlation length ξ_λ with different choices of the fitting range. Nevertheless, the variation in ξ_λ will mainly affect the region with very small x, which are anyway less predictive due to power corrections. Therefore, it is not a prerequisite to fit ξ_λ precisely. Instead, one should utilize this property by varying the fitting range, e.g., within $z_{\rm L} - 5a \leq z \leq z_{\rm L}$, and test the stability of the final result with different ξ_λ.

Last but not the least, the Fourier transform of an exponentially decaying correlation always leads to a finite quasi-PDF at $x = 0$, which is different from the Regge behavior of PDFs at small x. Besides, since the PDF at large x ($x \to 1$) is also sensitive to the long-range LF correlation which decays algebraically, the quasi-PDF shall deviate from the PDF in this region, too. These indicate the significance of power corrections in the end-point regions as $x \to 0$ and $x \to 1$, which gives us the hint on how to estimate the systematic uncertainty from the exponential extrapolation. To be specific, we can perform an algebraic extrapolation (see the section below) for the same range of data, which is essentially equivalent to ignoring all the power corrections at large z, and choose its difference to the exponential extrapolation as the error. One can anticipate that this estimate will lead to increasingly large systematic errors as x approaches the end points, which is consistent with the accuracy of the momentum-space expansion.

4.2. Algebraic extrapolation at very large P^z

When P^z becomes very large with future lattice resources, ξ_λ also becomes very large and it will take larger λ values to see the exponential decay in lattice data. We expect that the decay behavior follows more like an algebraic law rather than an exponential one as $\lambda \sim \lambda_{\rm L}$. In other words, the quasi-LF correlation is very close to the leading-twist correlation since the power corrections for $\lambda \leq \lambda_{\rm L}$ are expected to be well suppressed. In this case, we can use an algebraic form to extrapolate to $\lambda = \infty$.

The algebraic decay of leading-twist correlation is a consequence of its infinite correlation length ξ_λ, and is associated with the asymptotic Regge behavior [91]. At small x, it is well-known that parton distributions behave asymptotically like x^a, as suggested by Regge theory. For the non-singlet combination, the leading Regge trajectory indicates that $a \sim -1/2$. For the singlet combination, its mixing with gluon distributions under evolution makes things more subtle. In the so-called soft pomeron model, one has $a \sim -1$. However, scattering data at large momentum transfer indicate a more singular asymptotic behavior, reflecting the potential need for a contribution of the hard pomeron [92]. At large x, the asymptotic behavior is dictated by the quark counting rules [93]. As $x \to 1$, the hadron momentum is carried by the struck quark and no momentum is left for other spectator partons. The asymptotic behavior is then predicted to be $(1-x)^b$, where $b = 2n_s - 1 + 2|\Delta S^z|$ with n_s being the minimum number of spectator partons and ΔS^z the difference of the spin projections for the struck parton and the parent hadron [92, 94]. For example, for a valence quark in the proton $b = 3\,(5)$ if the struck quark has helicity parallel (antiparallel) to the proton as $n_s = 2$ and $|\Delta S^z| = 0\,(1)$, while for the pion one has $b = 2$ since $n_s = 1$ and $|\Delta S^z| = 1/2$. The above features have been widely used in global fits of PDFs, where one parameterizes the PDFs such that they behave as x^a for $x \to 0$ and $(1-x)^b$ for $x \to 1$ and fit the powers a, b to a large variety of experimental data. The role of such a power law behavior in global fits has been examined in detail in Refs. [95, 96].

When Fourier transformed to coordinate space, the asymptotic behavior described above implies that the correlation in the longitudinal space decays algebraically as $\lambda^{-\alpha}$ (α is a positive number related to a, b) rather than exponentially, and thus has an infinite correlation length. A similar algebraic decay behavior was also observed in a recent analysis of the LF wave functions [59] when Fourier transformed to conjugating coordinate space [97].

To see how the asymptotic behavior can help with the extrapolation of quasi-LF correlations at large momentum, let us begin with the following simple form of PDFs that incorporates the $x \sim 0, 1$ behavior,

$$x^a (1-x)^b \, . \tag{35}$$

The coordinate space matrix element can be defined as

$$h(\lambda) = \int_0^1 dx \, e^{ix\lambda} x^a (1-x)^b, \tag{36}$$

from which it follows that at large λ

$$h(\lambda) \sim \frac{\Gamma(1+a)}{(-i|\lambda|)^{a+1}} + e^{i\lambda} \frac{\Gamma(1+b)}{(i|\lambda|)^{b+1}} \, , \tag{37}$$

whose real (imaginary) part is even (odd) in λ, ensuring that parton distributions are real functions in momentum space. Therefore, the conjugate LF correlations behave at large λ as $\lambda^{-\alpha(a,b)}$ with

$$\alpha(a,b) = \min(a+1, b+1). \tag{38}$$

In most cases we are interested in, $\alpha(a,b) = a+1$. Applying the matching in Eq. (9) converts the light-cone correlations to quasi-LF correlations, and also induces logarithmic corrections to the asymptotic behavior. In regions where the factorization is valid, such corrections can be resummed as $\sim \exp(\gamma \ln z^2\mu^2) = (\lambda^2\mu^2/(p^z)^2)^\gamma$ to leading logarithmic (LL) accuracy, which modifies the asymptotic behavior of the quasi-LF correlation as

$$\tilde{h}(\lambda, z) \sim e^{\gamma \ln z^2\mu^2} \frac{1}{|\lambda|^{\alpha(a,b)}} \sim |\lambda|^{-\alpha(a,b)+2\gamma} . \tag{39}$$

This provides a useful approximation to the quasi-LF correlations at large λ with sufficiently large P^z, and is consistent with previous discussions based on the correlation length. Now we can use the following algebraic form to extrapolate the quasi-LF correlation to infinite λ (taking $\lambda > 0$ as an example)

$$\frac{c_1}{(-i\lambda)^{d_1}} + e^{i\lambda}\frac{c_2}{(i\lambda)^{d_2}}, \tag{40}$$

which accommodates the two different structures in Eq. (37). The parameters c_i, d_i can be fitted in the same way as that in the exponential extrapolation. Finally, we can use the uncertainty in these parameters to estimate the systematic error from extrapolation.

By supplementing lattice data with the above extrapolation strategy, we expect the final PDF result to be free of unphysical oscillation and converge better to the physical region $0 < x < 1$.

5. Large Momentum Vs. Short Distance Expansion

The Euclidean correlator in Eq. (5) introduced in Ref. [4] has also been considered in coordinate-space factorizaton (CSF) [40], which was introduced in an early work on meson DAs with current-current correlators [98] (see also [99]). The correlator can be factorized in terms of the LF correlations with expansion parameter $(z\Lambda_{\rm QCD})^2$. The formalism is naturally suited for calculating moments of PDFs or short-distance LF correlations. To obtain the full parton physics, however, one has to simultaneously consider the constraint on the external momentum $P^z \sim 1/z \gg \Lambda_{\rm QCD}$. This is identical to the observation in Ref. [5]: One must use large momenta to capture the full dynamical range of PDFs, which requires information on long-range correlations in λ. Despite their formal equivalence [100, 63, 68], some analytical matching calculations might more conveniently be done in coordinate space.

Not surprisingly, the same LaMET lattice data are needed for a CSF analysis to get the PDFs. Nominally, CSF can also admit data at small P^z, but the same information is contained already in large P^z data at smaller z.

The CSF expansion is formulated in terms of the Euclidean distance z, which is required to be small, i.e.,

$$z \ll 1/\Lambda_{\rm QCD} \ , \tag{41}$$

to ensure the validity of perturbation theory and leading-twist dominance. Assuming the largest z for the leading-twist approximation to be $z_{\rm LT}$ (say, the value of z for which the higher-twist contribution is at the level $\sim 20\%$), then the small expansion parameter is $(z_{\rm LT}\Lambda_{\rm QCD})^2 \ll 1$ when potential linear divergences are subtracted before the expansion is made. Therefore, only the matrix element of $O(z)$ within the range $[0, z_{\rm LT}]$ has a simple interpretation in terms of leading-twist parton physics.

An interesting question is then: What is the value of $z_{\rm LT}$? If we take $\Lambda_{\rm QCD} \sim 300$ MeV, and $z_{\rm LT}\Lambda_{\rm QCD} = 1/2 \sim 1/3$ as a small parameter, then the estimate is that $z_{\rm LT}$ is around $0.25 \sim 0.33$ fm. An upper limit is probably 0.4 fm. A good estimate of $z_{\rm LT}$ can be provided by comparing the matrix element $\langle P=0|O(z)|P=0\rangle$ or $\langle\Omega|O(z)|\Omega\rangle$, both of which have been proposed to renormalize the bare quasi-LF correlation [40, 42, 43], to the leading-twist contributions in their OPE.

Let us take the zero-momentum matrix element for the isovector case as an example. In the $\overline{\rm MS}$ scheme, it has a short distance expansion of the form [68, 63]

$$\begin{aligned}\tilde{h}(z,\mu,P^z\!=\!0) &= \frac{1}{2M}\langle P=0|\bar{\psi}(z)\gamma^0 W(z,0)\psi(0)|P=0\rangle \\ &= c_0(\mu^2 z^2)a_0 + \mathcal{O}(z^2\Lambda_{\rm QCD}^2)\,,\end{aligned} \tag{42}$$

where $W(z,0)$ is a spacelike straight gauge link. Here μ is the $\overline{\rm MS}$ renormalization scale, and $a_0 = 1$ is the conserved lowest moment of the correpsonding twist-2 PDF. The one-loop Wilson coefficient $c_0 = 1/Z^{\overline{\rm MS}}_{\rm ratio}$ with $Z^{\overline{\rm MS}}_{\rm ratio}$ given in Eq. (32) [63], and the two-loop result can be found in Ref. [43].

According to Eq. (12), the mass-subtracted matrix element includes logarithmic divergences that are independent of z and should not constitute significant corrections in lattice perturbation theory. Therefore, we can roughly approximate its OPE by replacing μ with $1/a$,

$$e^{-\delta m|z|}\tilde{f}(z,a,P^z\!=\!0)\!=\!c_0(z^2/a^2)\!+\!\mathcal{O}(z^2\Lambda_{\rm QCD}^2, a^2/z^2)\,, \tag{43}$$

where the lattice discretization effects are expected to be of $\mathcal{O}(a^2/z^2)$. Since the lattice matrix elements are convergent as $z \to 0$, which is contrary to the logarithmically divergent behavior in the $\overline{\mathrm{MS}}$ OPE, we expect the above approximation to be reliable within the range $a < |z| < z_{\rm LT}$ where the discretization and higher-twist effects are both suppressed. Note that lattice OPE is usually complicated by the broken Lorentz symmetry and operator mixings. Nevertheless, since for $P^z = 0$ the only leading-twist contribution comes from the conserved vector current, we can ignore such effects here.

In Fig. 3 we plot the mass-renormalized pion lattice matrix element. The bare lattice matrix element comes from a recent calculation of the pion valence PDF on an ensemble with $a = 0.06$ fm and pion mass $m_\pi = 300$ MeV [101]. On the same lattice ensemble, the Wilson-line mass correction δm was fitted from the quark-antiquark potential [17], and its value is given in lattice units as $a\delta m = -0.1568$. The leading-twist contribution is plotted with next-to-leading-order (NLO) corrections and NLO correction plus LL resummation for fixed α_s,

$$\left[1 + \frac{\alpha_s(1/a)C_F}{2\pi}\frac{5}{2}\right]\left(\frac{z^2}{4e^{-2\gamma_E}a^2}\right)^{\frac{\alpha_s(1/a)C_F}{2\pi}\frac{3}{2}} . \tag{44}$$

Here we choose α_s in the $\overline{\mathrm{MS}}$ scheme at scale $1/a$ as the input for OPE, which should allow for better convergence than the bare lattice coupling [84]. To estimate the uncertainty from the choice of α_s, we vary the $\overline{\mathrm{MS}}$ scale from $1/(2a)$ to $2/a$. Though a standard procedure of improvement shall be performed to define α_s on the lattice, we expect that it will not alter the following conclusion.

As one can see, for $z \leq a$, the lattice result is significantly different from the leading-twist approximations due to discretization effects. As z increases, the agreement becomes better. However, for $z \geq 0.3$ fm, the lattice result starts to deviate dramatically from the leading-twist approximations, showing that the higher-twist contributions become significant. Therefore, we can roughly estimate that $z_{\rm LT} \sim 0.3$ fm.

One can also look at the case of the better established heavy-quark potential. It is well-known that the static heavy-quark potential receives both perturbative and non-perturbative contributions. The perturbative static potential is known up to N^{3}LO level [74] and can be expressed in terms of the QCD running coupling constant. In Refs. [102, 103, 104], the running coupling constant has been extracted from lattice calculation of the static en-

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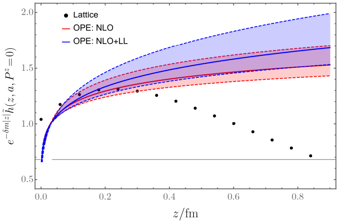

Figure 3: Comparison of the mass-renormalized pion lattice matrix element [101] and its leading-twist approximation with NLO and NLO+LL corrections for fixed α_s, respectively. The strong coupling is $\alpha_s(1/a) = 0.242$, and the error band was obtained by varying α_s from $\alpha_s(1/(2a)$ to $\alpha_s(2/a)$. This shows that the leading-twist approximation becomes unreliable at $z_{\rm LT} \sim 0.3$ fm.

ergy at short distances. The N^{3}LL perturbative result agrees well with lattice data up to $r \sim 0.2$ fm. However, it is well-known that the perturbative series for the static potential suffers from a renormalon ambiguity [105, 106] and breaks down at large distance. The non-perturbative heavy-quark potential can be simulated using lattice QCD, and is well-known to be dominated by the linear term of the form σr at large distance. Phenomenologically, the static potential can be well approximated by the linear+Coulumb QCD static potential, $V(r) = -e/r + \sigma r$ where $e = 0.25 \sim 0.5$ while $\sqrt{\sigma} \approx 477$ MeV [107, 108, 109]. When the perturbation theory is about to break down, the perturbative contribution $-e/r$ and the confining contribution σr should be of the same order of magnitude, which determines $r_c \approx \sqrt{e/\sigma}$ to be around $0.2 \sim 0.3$ fm. This is consistent with the result in Refs. [110, 111, 102]. The boundary $z_{\rm LT}$ should be of the same order of magnitude.

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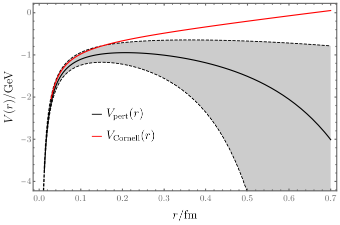

Figure 4: A comparison between the perturbative potential in α_s^3 order to the linear+Coulumb QCD static potential taken from [108, 109]. The shaded area corresponds to the perturbative prediction with the renormalization scale μ ranging from $r/2$ to $3r/2$. The black line corresponds to $\mu = 1/r$. From the figure, it is clear that beyond 0.2 to 0.3 fm, the perturbative potential starts to deviate from the full non-perturbative results and becomes unreliable.

With the estimated z_{LT} above, one can also define

$$\lambda_{\mathrm{LT}} = P^z z_{\mathrm{LT}}, \tag{45}$$

then the matrix element for the quasi-LF distance $[0, \lambda_{\mathrm{LT}}]$ can be used to extract parton distributions with the matching formula in Eq. (16) [40, 100, 63]. Thus, the coordinate-space approach is useful for extracting the LF correlation functions in a limited range, with the LF distance ranging between $[0, \lambda_{\mathrm{LT}}]$.

The CSF approach has also been used for products of currents made of quark bilinears [112, 98, 113, 114, 99, 115, 116, 117]. Renormalization of power divergences in the CSF expansion for the quark and gluon blinears with Wilson line is easier to handle. In particular, a version of the ratio method which divides by the matrix element at zero momentum, can be used to eliminate the power divergences in the lattice matrix elements [40, 41]. On the other hand, the current products can also be used in LaMET expansion after Fourier transforming into momentum space [31].

However, the CSF does not allow for directly calculating the x-distribution, because one needs the LF correlation at all LF distances. The requirement for $z \leq z_{\rm LT}$ makes it unfeasible to reach large λ values for the Fourier transform with the largest momentum on contemporary lattice resources. Thus, to reconstruct the PDF, one has to parameterize the functional form of the x dependence, just like that in the phenomenological fits, and then inverse Fourier transform it to the coordinate space and fit to a limited range of LF correlations [41, 118, 119, 120, 116, 121, 117, 87, 101, 122]. This process is hardly under control, because it is difficult to estimate the systematic uncertainty from the parameterization or assumptions of the PDF. As evident in the fits performed in literature so far [119, 120, 116, 121, 101, 122], either the errors in the end-point regions become smaller and smaller, or the errors shrink to almost zero for certain moderate values of x. These imply unaccounted systematics from the artifacts of the particular model used, which is also reflected by their inconsistency with global fits that use similar parameterizations.

From a different angle, the above practice amounts to postulating (or modeling) certain correlation between short- and long- distance behaviors of the LF correlations. Such a postulation has no first-principles foundation, and it can happen that the lattice data in the limited range of λ be fitted equally well by more than one parameterizations which have completely different asymptotic behaviors [117].

Despite the difficulty in providing a controlled calculation of the x-distribution, the CSF method allows for model-independent extraction of the Mellin moments using the OPE. Nevertheless, with limited range of λ, the LF correlations will be sensitive to only the lowest ones, which is related, but not in a direct one-to-one correspondence, to the predictive power in the x-space.

To make a more direct comparison between the momentum space and coordinate space approaches, let us consider the following example. Assume that the quasi-LF correlations defining the quasi-PDFs behave like

$$\tilde{h}(\lambda, z) = h(\lambda)e^{-m|z|} \ , \tag{46}$$

with $h(\lambda)$ being the light-cone correlator. The exponential $e^{-m|z|}$ with $m \sim \Lambda_{\rm QCD}$ is used to model higher-twist contributions. From this equation, it is clear that if one stays in position space, the CSF is accurate only when $z \ll 1/m$. The available range of λ is therefore much smaller than P^z/m, which indicates that the number of moments one can access is much less than P^z/m.

From the discussion above, it is clear that the momentum and coordinate space expansions are different expansion schemes. Even though they are equivalent in the infinite momentum limit, they are different at finite momentum P^z. In the latter, the information is filtered directly in coordinate space. One gets parton correlations in a finite range of LF distance which correspond to the number of moments controlled by $1/P^z$. In contrast, the former uses all the coordinate space information, filtering higher-twist physics in momentum space through $1/(y^2(P^z)^2) \ll 1$ and $1/((1-y)^2(P^z)^2) \ll 1$. Therefore, one gets partron distributions in an interval of x with systematic control of errors, which can be directly compared with experimental data.

Finally, we remark that the relative size of the perturbative correction to the quasi-PDF depends on x, where one usually observes larger corrections in the small- and large-x regions. On the other hand, although the size of the perturbative corrections in the coordinate space is usually small for finite λ, it can still lead to significant corrections in the end-point regions in momentum space.

6. Conclusion

To conclude, we have discussed some further subtleties in renormalization and matching of the quasi-LF correlations on lattice. We proposed a hybrid renormalization procedure to treat the short and long distance correlations separately. The short distance correlations can be renormalized by dividing the same correlator sandwiched in different external states, whereas the long distance ones are renormalized using the Wilson line mass renormalization with a continuity condition to match the short distance region. In this way, we avoid introducing extra non-perturbative effects at large distance in the renormalization stage. We also proposed how to extrapolate to large quasi-LF distance beyond the reach of lattice simulations by utilizing the asymptotic long-range behavior of the correlations, thus avoiding truncations in the ensuing Fourier transform. We finally compared the large-momentum expansion with the CSF approach when applied to LaMET data, showing that the former is a systematic expansion to extract the x-dependence of PDFs, whereas

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the latter is not. Our proposal here has the potential to greatly improve current computational strategies in lattice applications of LaMET.

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